

A Comprehensive Survey on 3D Deep Point Cloud Models

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Abstract

Recently, point cloud data has attracted the attention of researchers as a promising data representation model for a wide range of applications. As unlike 2D data, point clouds are unordered, irregular, and often large in scale, they might impose severe challenges when designing deep learning models. Over the past decade, substantial progress has been made in proposing architectures that address permutation invariance, geometric reasoning, scalability, and robustness, leading to rapid expansion across diverse 3D data oriented applications. The main aims of this paper are to present a comprehensive survey on existing literature and to analyze how different 3D representations have shaped the design and performance of deep learning models. In contrast to prior surveys that have emphasized on limited task subsets or specific model families, this survey reviews deep point cloud models through

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representation- and architecture-centric perspective. As such, beyond (1) core tasks such as classification, segmentation, detection, and tracking, this survey systematically provides insight into recent progress in broader directions, including (2) geometric modeling, alignment, and pose estimation, (3) foundation models and scene understanding, and (4) robustness, generalization, and reliability. Furthermore, this survey presents commonly used datasets and evaluation metrics, and finally summarizes challenges and future directions toward robustness, efficiency, and generalizability of 3D point cloud systems. Github page of the project: https://github.com/IPL-Sharif/3D_Survey

1 Introduction

In recent years, the use of 3D data has grown rapidly across various domains such as robotics, virtual reality, augmented reality (Wang et al., 2024h). The increasing availability of advanced sensors like LiDAR, RGB-D cameras, and multi-view stereo systems has made it possible to capture rich, detailed 3D information about real-world environments (Guo et al., 2020). Compared with 2D images, 3D data describes the real world in a more complete and natural way. A 2D image shows the scene on a flat surface, so information about depth and true spatial structure is lost. In contrast, 3D data keeps important geometric information such as the shape of objects, their size, and the distances between them. Because of this, 3D data is more suitable for tasks that need spatial understanding, such as recognizing objects, understanding scenes, reconstructing surfaces, and interacting with the physical world. In addition, geometric measurements are generally less dependent on illumination and surface texture than image appearance, although their quality remains affected by viewpoint, occlusion, sampling density, and sensor characteristics, which often cause problems in image-based methods. For these reasons, 3D representations are a strong and natural choice for understanding real-world environments.

Three-dimensional information can initially be encoded using point clouds, voxel grids, polygonal meshes, or multi-view projections. During model processing, these inputs may also be reorganized into spatially discretized, graph-structured, token-based, projected, or hybrid computational representations. Voxel-based approaches discretize space into regular 3D grids that are well suited to convolutional processing but often suffer from high memory consumption and limited resolution (Riegler et al., 2017). Meshes also come with challenges such as handling complex topology, managing high-resolution meshes with large memory and computation costs (Wang et al., 2024h). In contrast, a point cloud represents a scene or object as an unordered set of 3D points, each associated with attributes such as coordinates, color, intensity, or surface normals (Qi et al., 2017a). As a result, point clouds have become a preferred representation in many 3D tasks, including classification, detection, and segmentation (Qi et al., 2017a; Shi et al., 2019).

Despite these advantages, point clouds introduce unique challenges for data processing and learning. Unlike images or voxels, point clouds are inherently unordered, irregular, and unstructured, lacking a fixed grid or neighborhood definition (Bello et al., 2020). The density of points may vary significantly across regions, and large-scale point clouds can contain millions of points, posing serious computational and memory challenges. Additionally, real-world point cloud data is often noisy, incomplete, and dynamically acquired, further complicating robust feature extraction and learning (Quan et al., 2024). Designing models that are permutation-invariant, scalable, and capable of capturing both local geometric details and global context remains a central research problem.

Due to the rapid growth of research on 3D point cloud and its extensive use in a wide range of tasks and application domains, several survey papers have been published to review and categorize existing 3D data representations and methods from different perspectives. One of the early and influential works in the field, Guo et al. (2020) provided a foundational overview of deep learning approaches for point cloud processing, with a primary focus on fundamental tasks such as classification and segmentation. The survey categorized representative methods and highlighted the inherent challenges posed by the irregular structure and unordered nature of point cloud data.

Building upon earlier works, Lu & Shi (2020) expanded the scope of analysis beyond fundamental tasks by reviewing additional 3D point cloud understanding tasks such as flow estimation, registration, augmentation,

and completion, alongside core tasks like classification, segmentation, detection, and tracking. With the increasing adoption of attention mechanisms and large-scale models, Lu et al. (2022) delivered the first dedicated and systematic review of transformer-based architectures for 3D point cloud analysis.

More recently, Vinodkumar et al. (2023) presented a focused review of deep learning approaches for 3D object recognition, covering state-of-the-art models for classification, segmentation, and detection across multiple 3D data formats and benchmark datasets. In 2024, Sarker et al. (2024) provided a focused and detailed review of deep learning methods specifically for 3D shape classification and semantic segmentation. This work analyzed recent progress in these two core tasks and discussed commonly used benchmark datasets and evaluation metrics.

Another contemporary contribution, Wang et al. (2024h), explored the adoption of diffusion-based generative models in 3D vision, examining how iterative denoising frameworks originally developed for 2D image generation can be adapted to 3D tasks such as object generation, shape completion, point cloud reconstruction, and scene synthesis.

Despite the breadth and depth of these surveys at their respective times of publication, several limitations persist. Prior works often emphasize specific subsets of tasks and do not systematically integrate recent architectural advances, such as transformer-based frameworks, large-scale pretraining, and foundation models across the full spectrum of 3D point cloud analysis (Guo et al., 2020; Lu & Shi, 2020). Importantly, key research areas including point cloud compression, reconstruction, robustness, and scene understanding evaluation are not jointly examined within a single survey. In contrast, this survey covers a broader and more diverse set of 3D point cloud tasks, and it provides dedicated discussions of benchmark datasets, standardized evaluation metrics, and comparisons of classification, segmentation, and detection tasks. Beyond core tasks such as classification, segmentation, detection, and tracking, this survey covers several broader directions in deep point cloud learning. These include geometric modeling, alignment, and pose estimation, encompassing generation, reconstruction, registration, and 6DoF pose estimation; foundation models and scene understanding; and robustness. The literature is further organized along two complementary axes—representation choices and architectural paradigms—and their interactions, advantages, failure modes are analyzed. Together, these features provide a more comprehensive and up-to-date overview of the field, as summarized in Table 1.

This survey examines deep point cloud models through two central design dimensions: point cloud representations and architectural paradigms. A representation defines how 3D information is organized, while an architecture determines how that information is processed into task-specific features. By emphasizing their interaction, this survey shows how model behavior is shaped by representation-architecture combinations across different tasks. Following this perspective, we review core tasks, geometric modeling, spatial alignment and pose estimation, foundation models and scene understanding, and robustness.

From the **task perspective**, this survey organizes deep point cloud learning into several task families rather than treating individual tasks as isolated problems. Core tasks include classification, segmentation, detection, and tracking, while geometric and spatial tasks include compression, generation, reconstruction, registration, and 6DoF pose estimation. Beyond these classical settings, particular emphasis is placed on robustness and scene understanding, which are increasingly important for real-world 3D applications. Specifically, robustness-oriented studies address 3D ML pipeline security, noise and occlusion resilience, rotation invariance and equivariance, domain adaptation, and anomaly detection, alongside scene-level reasoning tasks such as 3D question answering, 3D grounding, 3D captioning, and 3D reasoning. Across these task families, the survey emphasizes how representation choices and architectural paradigms influence model design, performance, and generalization.

From a **methodological standpoint**, this survey structures deep point cloud models along two complementary axes: representation choices and architectural paradigms. The representation axis includes point-based, projection-based and multi-view, spatially discretized, token-based, graph-structured, and hybrid representations, while the architecture axis includes MLP-based, Convolution-based, Graph-based, and Transformer-based architectures. This organization clarifies how representation-architecture combinations shape feature extraction, computational behavior, failure modes, and task affinity, providing a more systematic taxonomy than task-only or model-family-specific surveys.

Table 1: Comparison of existing comprehensive survey papers on deep learning for 3D point cloud analysis.

Paper Name	Year	Application Tasks	Representation & Architecture Taxonomy	Datasets	Evaluation Metrics	Challenges & Open Problems
(Guo et al., 2020)	2020	Classification, Segmentation, Detection, Tracking	Point-Based(MLP-Based, CNN-Based, Graph-Based), Spatially Discretized-Voxel-Based, Projection-Based and Multi-View-Projection-Based, Hybrid	✓	✓	✓
(Lu & Shi, 2020)	2020	Classification, Segmentation, Detection, Tracking, Flow Estimation, Registration, Augmentation, Completion	Point-Based(MLP-Based, CNN-Based, Graph-Based), Spatially Discretized-Voxel-Based, Projection-Based and Multi-View-Projection-Based, Hybrid	✓	✓	✗
(Lu et al., 2022)	2022	Classification, Segmentation, Detection, Tracking, Registration, Video Understanding, Sampling, Denoising, Completion	Point-Based(Transformer-Based), Spatially Discretized-Voxel-Based(Transformer-Based)	✗	✗	✗
(Vinodkumar et al., 2023)	2023	Classification, Segmentation, Detection	Point-Based(MLP-Based, CNN-Based, Graph-Based, Transformer-Based), Spatially Discretized-Voxel-Based, Projection-Based and Multi-View-Projection-Based, Hybrid	✓	✗	✗
(Sarker et al., 2024)	2024	Classification, Semantic Segmentation	Point-Based(MLP-Based, CNN-Based, Graph-Based, Transformer-Based), Spatially Discretized-Voxel-Based, Projection-Based and Multi-View-Projection-Based, Hybrid	✓	✓	✓
(Wang et al., 2024h)	2024	Generation, Depth Estimation	Point-Based	✓	✓	✓
This Survey	2026	Classification, Segmentation, Detection, Tracking, Compression, Generation, Reconstruction, Registration, 6DoF Pose Estimation, Robustness, Scene Understanding	Point-Based(MLP-Based, Convolution-Based, Graph-Based, Transformer-Based), Projection-Based and Multi-View(Convolution-Based, Graph-Based, Transformer-Based), Spatially Discretized(Convolution-Based, Graph-Based, Transformer-Based), Token-Based(MLP-Based, Convolution-Based, Graph-Based, Transformer-Based), Graph-Structured(Graph-Based, Transformer-Based), Hybrid	✓	✓	✓

In addition, this survey introduces a dedicated **evaluation section** that reviews commonly used benchmark datasets and evaluation metrics across point cloud tasks, with performance analysis focused on classification, segmentation, and detection. Since point cloud tasks differ in datasets, evaluation protocols, and metrics, numerical comparisons are reported only when methods are evaluated under comparable settings. Classification, segmentation, and detection are selected for performance analysis because they have widely used benchmark datasets and many methods report results with compatible metrics. For other tasks, where benchmarks and metrics are less standardized, the discussion focuses on datasets, evaluation criteria, methodological trends, and remaining challenges rather than direct numerical ranking.

Figure 1 illustrates the overall organization of this survey. In summary, the main contributions of this work are as follows:

- A comprehensive survey of deep methods for 3D point clouds, covering a wide range of tasks, representations, and architectural paradigms within a unified framework.
- A representation-architecture taxonomy that distinguishes point-based, projection-based and multi-view, spatially discretized, token-based, graph-structured, and hybrid representations, and systematically analyzes their interactions with MLP-based, Convolution-based, Graph-based, and Transformer-based architectures.
- A task-family structure that organizes deep point cloud learning into core tasks; geometric modeling, alignment, and pose estimation; foundation models and scene understanding; and robustness, generalization, and reliability.
- A focused review of robustness and scene understanding in 3D point cloud learning, including studies on 3D ML pipeline security, noise and occlusion resilience, rotation invariance and equivariance, domain adaptation, anomaly detection, 3D question answering, grounding, captioning, and reasoning.

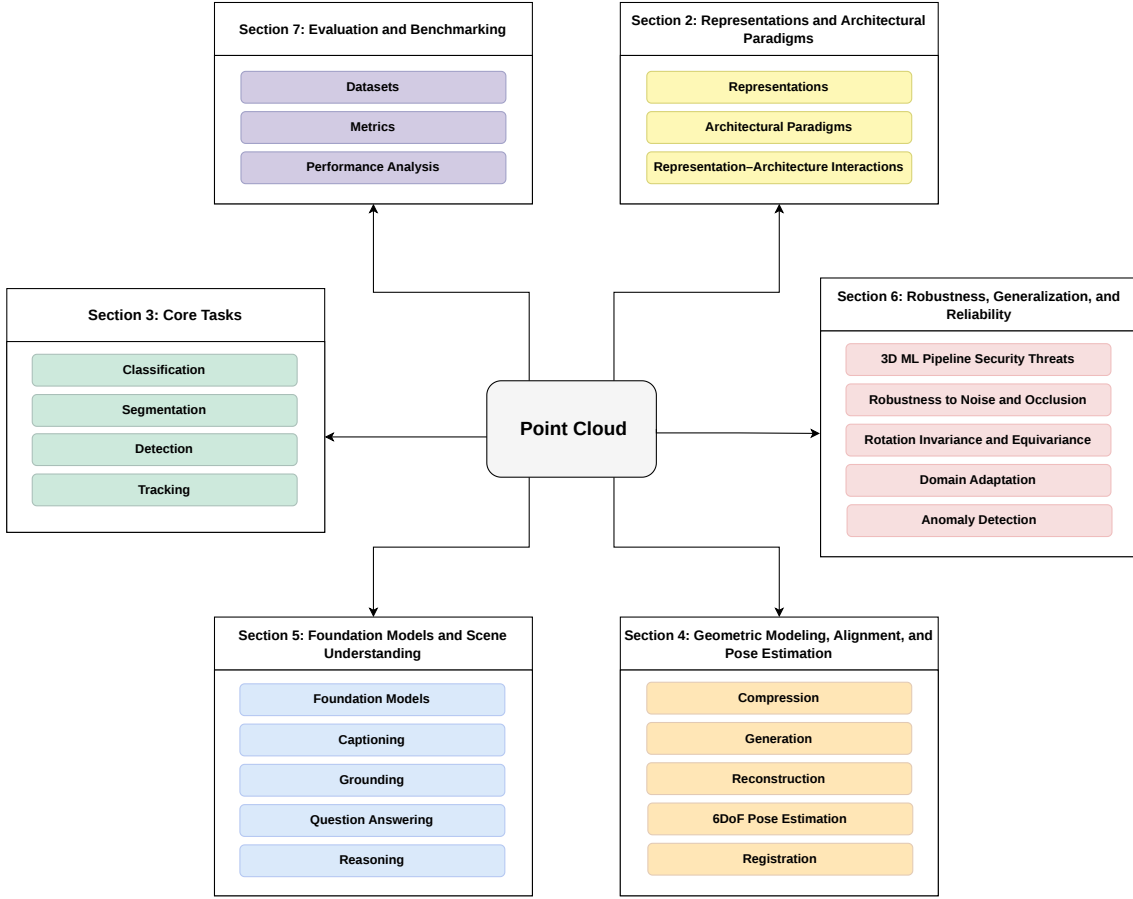


Figure 1: Overview of paper contents. This paper reviews 3D point cloud data across representations, architectural paradigms, and various tasks.

- A dedicated evaluation section that reviews commonly used datasets and metrics across point cloud tasks, with performance analysis focused on classification, segmentation, and detection where comparable benchmarks and compatible metrics are widely reported.

1.1 Literature Collection, Filtering, and Verification

A structured three-step process was followed to collect, screen, and verify the literature included in this survey.

Step 1. Initial literature search. Existing surveys were first examined to understand their scope, taxonomy, and organizational strategies. The selected point-cloud tasks were then distributed among the authors for focused investigation. For each task, representative studies were analyzed to identify relevant terminology, keywords, model families, datasets, and evaluation protocols. These terms were subsequently used to search multiple scholarly databases, including Google Scholar, Scopus, Semantic Scholar, Web of Science, IEEE Xplore, ScienceDirect, and SpringerLink. In addition, Google Scholar Lab was used with task-specific prompts to identify seminal and foundational works for each research topic.

Step 2. Expansion to recent works. After the foundational papers had been identified, the literature citing these works was explored, primarily through Google Scholar, to identify more recent developments. Given the large number of candidate papers, priority was given to highly cited studies and papers published in top-tier peer-reviewed venues to ensure broad coverage of influential and state-of-the-art methods.

Step 3. Final screening and selection. After a comprehensive collection of candidate papers had been assembled for each topic, a final screening was performed based on each paper’s technical contributions, relevance to the survey, and consistency with the categorization framework adopted in the corresponding section. This process ensured that the survey includes both foundational studies and representative recent advances while maintaining a focused and coherent presentation. The main search covered publications from 2013 to May 2026, while earlier foundational studies were included when necessary to explain the development of a task or methodology.

For the quantitative comparisons reported in Section 7.3, the reported performance values were independently extracted and verified by multiple members of the research team using the original primary sources. Each numerical result was traced to the corresponding table, appendix, or supplementary material of the published paper. In addition to the reported performance, the associated dataset, data split, evaluation metric, model configuration, supervision setting, and evaluation protocol were verified to ensure that the results were interpreted under the appropriate experimental conditions. The independently compiled records were then cross-checked, and any discrepancies were resolved through re-examination of the original source before inclusion in the manuscript. Consequently, direct quantitative comparisons were restricted to methods evaluated under sufficiently compatible experimental settings.

To further improve the reliability of the benchmark tables, the survey includes both foundational methods and representative recent approaches published in top-tier peer-reviewed venues. Such venues employ rigorous multi-reviewer peer-review processes, providing greater confidence in the validity of the reported results. Furthermore, priority was given to methods with publicly available implementations whenever possible, thereby improving transparency and facilitating reproducibility.

2 Point Cloud Representations and Architectural Paradigms

Despite the diverse tasks targeted by point cloud methods, they often share similar representations and architectural choices. Deep point cloud models are built upon two fundamental components: the representation used to encode 3D data and the architecture used to process it. This perspective is useful because the same 3D task can be approached through different representations, while a given representation can be paired with different neural architectures. This section reviews these common foundations and clarifies how different representation–architecture combinations shape model behavior across tasks.

2.1 Point Cloud Representations

A representation specifies how 3D information is organized and exposed to a model at a particular processing stage. It defines the elements operated on by the network—such as points, spatial cells, projected pixels, graph nodes, tokens, or neural primitives—and determines how geometric detail, attributes, neighborhood relations, and global structure are accessed. Representation differs from architecture, which defines the applied operators; learning paradigm, which determines how parameters are learned or adapted; modality, which identifies the input source; and task, which specifies the required output.

The categories below operate at related but distinct abstraction levels. Point-based, projection-based, and spatially discretized representations describe primary geometric organizations, whereas graph-structured and token-based representations are derived structures constructed from geometric or higher-level entities. Hybrid representations combine multiple structures within one system. Because models may transform data successively between these forms, the categories describe the principal computational structure at a particular stage rather than mutually exclusive end-to-end model families.

2.1.1 Point-Based Representations

Point-based representations retain individual points as the basic spatial units. A point cloud can be expressed as

$$\mathcal{P} = \{(\mathbf{x}_i, \mathbf{a}_i)\}_{i=1}^N, \quad (1)$$

where $\mathbf{x}_i \in \mathbb{R}^3$ denotes the coordinates of the i -th point and $\mathbf{a}_i$ contains optional attributes such as color, intensity, normals, timestamps, or learned features. Preserving individual coordinates avoids input-space discretization and projection collisions, making this representation suitable for tasks requiring fine geometry. However, sampling, pooling, neighborhood aggregation, and interpolation may still discard local information.

Because point clouds are unordered and lack fixed connectivity, point-based models require permutation-aware operators and often construct neighborhoods using k -nearest-neighbor search, radius queries, or hierarchical grouping. These operations can become bottlenecks in large or non-uniform scenes. Direct point processing is therefore most effective when point-level fidelity is important and the input size is manageable, whereas large-scale systems commonly use more scalable processing strategies. Point-based representations can be combined with different architectural paradigms, including MLPs, convolutions, graph neural networks, and transformers, depending on how local and global geometric information is extracted and aggregated.

2.1.2 Projection-Based and Multi-View Representations

Projection-based representations map point clouds into one or more structured 2D domains, enabling efficient 2D processing while introducing dependence on projection geometry. A distinction is made between sensor-centric range projections and object- or scene-centric multi-view renderings.

Sensor-centric projections, such as spherical range images, preserve the angular acquisition pattern of rotating LiDAR sensors and support efficient processing on a 2D lattice. However, projected proximity does not always reflect Euclidean proximity, especially near depth discontinuities. Projection may also create empty pixels, point collisions, boundary discontinuities, and reduced accuracy in sparse regions.

Object- or scene-centric multi-view representations render a point cloud from selected viewpoints as 2D images. They benefit from mature image backbones and complementary views, but depend on viewpoint coverage and fusion strategy. Occlusion or insufficient viewpoints may omit thin or hidden structures, whereas additional views increase computational cost.

Projection-based representations are therefore most effective when the acquisition geometry or available 2D priors provide a strong advantage. Nevertheless, they provide less direct access to complete metric geometry and viewpoint-independent 3D neighborhoods than native 3D representations. Consequently, applications requiring these properties often rely on back-projection, multi-view fusion, or subsequent 3D refinement.

2.1.3 Spatially Discretized Representations

Spatially discretized representations map continuous coordinates into predefined metric cells, providing explicit indexing, structured neighborhoods, efficient batching, and compatibility with optimized operators. Common forms include dense and sparse voxels, octrees, and pillars, whose features are often organized into metric bird’s-eye-view (BEV) grids. BEV representations may also be constructed through other aggregation or view-transformation mechanisms.

Voxel representations partition 3D space into volumetric cells and aggregate the enclosed points or attributes. Dense grids provide regular memory layouts and predictable receptive fields, but their cost grows rapidly with resolution. Fine voxels preserve boundaries and small structures at higher cost, whereas coarse voxels increase quantization and may merge nearby surfaces. Sparse voxels improve scalability by processing only occupied cells, although quantization, active-voxel growth, and indexing overhead remain.

Hierarchical spatial structures provide an alternative to uniform spatial discretization by recursively partitioning 3D space at multiple resolutions. Octree-based representations allocate finer cells to occupied or geometrically complex regions while retaining simpler regions at coarser levels, improving efficiency and effective resolution. However, irregular memory access, tree traversal, batching, and cross-level alignment increase implementation complexity. They are therefore most suitable for sparse or geometrically non-uniform data.

Pillar representations discretize the horizontal plane and aggregate points within vertical columns, whose features are typically arranged on a 2D BEV grid for efficient planar processing. Pillars are one mechanism

for constructing BEV features, whereas BEV representations may also be derived from voxels, point features, camera views, or multimodal inputs. These metric representations are well suited to large outdoor scenes, but pillar-based and height-collapsed BEV encodings may lose vertical detail and create ambiguity for overlapping or elevated structures.

Overall, spatially discretized representations trade geometric detail for computational structure and scalability. Dense voxels provide regular processing but are costly, sparse voxels and octrees improve efficiency through selective or adaptive allocation, and pillar-based BEV representations enable fast planar processing while reducing vertical detail. The appropriate representation therefore depends on the required balance among resolution, efficiency, and scene scale.

2.1.4 Token-Based Representations

Token-based representations are derived computational structures rather than primary geometric encodings, because they are constructed by grouping or aggregating points, voxels, projected regions, or higher-level entities into compact feature vectors called tokens. A common approach is patching, in which representative centers are sampled and neighboring points are grouped. Each patch is then encoded and pooled into a token that summarizes its local geometry and attributes. Tokens may also represent voxel regions, super-points, object proposals, or learned queries and commonly include positional and local geometric information. Tokenization design affects coverage, redundancy, and information preservation. Although commonly associated with transformers, the resulting tokens can also be processed by MLPs, graph networks, and other architectures.

Token granularity balances geometric fidelity and efficiency. Fine tokens preserve local detail but increase memory and computation. Coarse tokens reduce cost and facilitate broader interactions but may merge nearby structures or discard detail. In large, non-uniform point clouds, fixed grouping scales may produce uneven token quality. Hierarchical or adaptive tokenization can partly address this limitation by varying token size and resolution across processing stages.

Tokenization provides a compact interface for hierarchical processing and global context modeling by reducing or reorganizing the input sequence. Its effectiveness depends on sampling, grouping, and aggregation quality, as unstable neighborhoods or excessive pooling may weaken geometric consistency and point-level traceability.

2.1.5 Graph-Structured Representations

Graph-structured representations are derived relational structures rather than primary geometric encodings, because their connectivity is explicitly constructed over points, voxels, patches, superpoints, or objects rather than being inherent in the original point cloud. A graph can be expressed as

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}), \quad (2)$$

where the nodes in $\mathcal{V}$ correspond to geometric or semantic entities, and the edges in $\mathcal{E}$ encode geometric, spatial, or learned relationships between them. At a given processing stage, the nodes and their connectivity constitute a graph-structured representation, whereas the message-passing or graph-convolution operator applied to that structure belongs to the architectural level. This distinction is particularly important for dynamic models, in which connectivity may be repeatedly reconstructed during processing.

Graph construction commonly relies on k -nearest-neighbor (k -NN) search, radius-based neighborhoods, or dynamically recomputed feature-space neighborhoods. Each strategy introduces different geometric and computational assumptions. A fixed- k graph maintains a consistent number of neighbors, but its spatial extent varies with sampling density. Radius-based graphs preserve an explicit metric scale, but produce variable node degrees and require scale- and density-sensitive radius selection. Dynamic graphs can capture relationships beyond initial Euclidean proximity, but repeated construction increases computational cost and reduces geometric interpretability.

Explicit connectivity provides a useful inductive bias for modeling local geometry and structural relationships. However, its reliability depends on neighborhood stability under noise, outliers, density variation, and partial

observations, while graph construction and message passing can become computational bottlenecks in large-scale scenes.

2.1.6 Hybrid Representations

Hybrid geometric representations combine two or more model-facing encodings of the same 3D structure. Common examples include point-voxel, point-BEV, point-graph, and voxel-token representations. Their purpose is to combine complementary inductive biases: points retain coordinate precision, voxels and BEV provide structured spatial organization, graphs encode explicit relationships, and tokens support compact or hierarchical processing.

Hybridization can reduce the limitations of an individual representation. For example, a point-voxel model may use sparse voxels for scalable feature extraction and point features for boundary refinement. However, hybrid systems often require additional computation and memory, cross-representation alignment, and careful fusion design. Performance depends strongly on how the representations are synchronized and fused, and improvements may partly result from increased computational capacity.

Hybrid representations should be distinguished from multimodal systems. Combining point clouds with images, radar, or language changes the input modality, whereas a representation is hybrid only when the same 3D information is encoded through multiple geometric forms.

2.2 Architectural Paradigms

While a representation determines how 3D information is organized and exposed to a model, an architectural paradigm defines how features are transformed, aggregated, and propagated across the resulting computational elements. Architectural families differ primarily in their core operators, geometric inductive biases, receptive-field construction, and computational scaling. MLP-based architectures rely on shared feature transformations and symmetric aggregation, convolution-based methods use localized weight-shared kernels, GNNs propagate information through message passing, and transformers perform content-dependent aggregation through self-attention.

This subsection reviews the computational principles, major design developments, and intrinsic limitations of these architectural paradigms. Particular attention is given to how each family incorporates geometric information, expands its receptive field, and balances local feature extraction with broader contextual modeling. Because the same architecture can operate on different forms of 3D data, the benefits and limitations arising from specific representation-architecture pairings are discussed separately in Section 2.3.

2.2.1 MLP-Based Architectures

MLP-based architectures transform input features through shared multilayer perceptrons and commonly use symmetric aggregation to maintain permutation invariance. Their simple dense operations are parameter-efficient, hardware-friendly, and effective for hierarchical feature learning when combined with local grouping. However, MLPs do not inherently encode spatial locality or pairwise geometric relations, and independent element-wise processing may overlook local structure unless coordinates, neighborhoods, or relative features are explicitly incorporated.

These architectures have evolved from global point-wise processing toward hierarchical grouping, residual blocks, feature normalization, and multi-stage aggregation. They are most naturally suited to point-based and token-based representations, although MLP blocks can also transform features within voxel, graph, or hybrid models. Representative examples include PointNet, PointNet++, and PointMLP (Qi et al., 2017a;b; Ma et al., 2022).

2.2.2 Convolution-Based Architectures

Convolution-based architectures extract local patterns using spatially shared kernels. Weight sharing provides a strong locality bias, reduces the number of parameters, and supports hierarchical feature extraction through stacked layers, pooling, dilation, or strided operations. These properties make convolution effective

for learning local geometric structures, but its restricted receptive field limits direct long-range interaction and often requires deeper or multi-scale designs to capture broader context.

Convolutional variants include dense, sparse, continuous, kernel-point, and relation-dependent operators, which differ primarily in how kernels and neighborhoods are defined. Convolutions are particularly suitable for spatially discretized and projection-based representations with structured neighborhoods, while geometry-adaptive kernels also enable their application to point-based representations. Representative examples include VoxNet, SECOND, PointCNN, and KPConv (Maturana & Scherer, 2015; Yan et al., 2018; Li et al., 2018; Thomas et al., 2019).

2.2.3 GNN-Based Architectures

GNN-based architectures propagate information through message passing among connected entities. Each layer constructs messages from neighboring features and optional edge attributes, aggregates them using a permutation-invariant function, and updates the central features. Explicit connectivity provides a useful inductive bias for geometric and relational reasoning, but performance depends strongly on the quality of the graph, while repeated message passing can increase computational cost, propagate incorrect relations, and cause overly similar features in deep networks.

GNN variants differ in their message functions, aggregation rules, graph-pooling strategies, and use of fixed, hierarchical, or dynamically reconstructed connectivity. They are most naturally suited to graph-structured representations constructed over points, voxels, patches, superpoints, or objects. Representative examples include DGCNN and other graph-convolution or graph-attention models based on static or dynamic neighborhoods (Wang et al., 2019c).

2.2.4 Transformer-Based Architectures

Transformer-based architectures use self-attention to perform content-dependent feature aggregation. Attention supports flexible interaction across local and distant elements and is effective for combining fine geometric cues with broader contextual information. However, standard self-attention does not inherently encode three-dimensional geometry and has quadratic complexity with respect to sequence length, requiring positional encoding and scalable attention mechanisms for large point clouds.

Transformer designs commonly use geometric positional encoding together with global, local, windowed, sparse, grouped, or hierarchical attention. They are particularly suitable for token-based and point-based representations, while voxel, BEV, projection, or object features can also be processed after being organized as spatial tokens. Representative examples include PCT and the Point Transformer series (Guo et al., 2021; Zhao et al., 2021a; Wu et al., 2022; 2024d).

2.3 Representation–Architecture Interactions

A representation determines the computational elements over which features are extracted, whereas an architecture defines how they are transformed, aggregated, and propagated. Their compatibility therefore depends on whether the architecture’s structural assumptions match the organization provided by the representation.

The interactions are classified according to the dominant architecture responsible for feature extraction and aggregation. Because modern models frequently combine multiple and sometimes overlapping operators, these categories are operational rather than mutually exclusive. A **natural pairing** occurs when the representation directly provides the structure required by the dominant operator. A **conditional pairing** requires additional mechanisms such as neighborhood construction, grouping, positional encoding, tokenization, ordering, masking, or sparsification. A **weak pairing** requires substantial structural conversion or offers limited advantage over more compatible alternatives. Operators used only for feature projection, channel transformation, fusion, or prediction are treated as **auxiliary uses**. These categories describe structural compatibility rather than empirical performance; a conditional pairing may outperform a natural pairing when its adaptation mechanisms are well designed and appropriate for the target task.

Each interaction is examined in terms of structural compatibility, computational requirements, and common failure modes, while illustrative task affinities are summarized in Table 2. For conciseness, the task entries provide only representative applications of each interaction. They are non-exhaustive and do not imply that a representation–architecture pairing is limited to the listed tasks. Moreover, the representation categories are not always mutually exclusive, since some representations are derived from other underlying representations and may describe the computational elements used only at a particular processing stage. Consequently, the same model may be characterized by different representation–architecture pairings depending on whether it is examined from the perspective of its input representation or its internal feature-processing stages.

2.3.1 Point-Based Representation Interactions

Point-based representations retain individual points as the primary feature carriers and preserve coordinate-level geometry without input-space quantization or projection.

Natural pairings. Point-based representations form a natural pairing with shared point-wise MLP architectures because the same transformation is applied independently to each point, making point-wise features permutation equivariant, while symmetric aggregation produces permutation-invariant global representations without requiring grid construction, projection, or explicit connectivity. This pairing enables simple and efficient point-feature embedding, but point-wise transformations do not inherently encode spatial relations. Consequently, global pooling may discard local structure, whereas hierarchical variants depend on sampling, neighborhood grouping, interpolation, and downsampling quality.

Conditional pairings. Point-based representations can also be processed by Convolution-based, GNN-based, and Transformer-based architectures, but their dominant operators must first be adapted to the irregular and unordered structure of point sets. To apply convolution, local neighborhoods must be constructed and regular grid kernels replaced with geometry-aware kernels defined over relative point locations. This enables local surface modeling without voxelization, but makes performance sensitive to neighborhood scale, sampling density, and noise; applying standard convolution to an arbitrary point ordering would instead constitute a weak pairing. To use GNNs, explicit spatial or learned graphs must be constructed over the points so that message passing can model local dependencies and correspondences. This introduces graph-construction overhead and may propagate incorrect information when connectivity becomes unreliable under sparsity, density variation, or partial observations. Similarly, Transformer-based processing requires explicit access to three-dimensional geometric relationships, typically through absolute or relative positional encodings, coordinate-based attention biases, edge features, or other geometry-aware attention mechanisms. Local, sparse, grouped, or otherwise scalable attention strategies are also commonly employed to control the computational cost of point-wise attention. The effectiveness of this pairing therefore depends on how accurately geometric relations are encoded, how representative points or groups are selected, and how attention is partitioned across the point set.

2.3.2 Projection-Based and Multi-View Representation Interactions

Projection-based and multi-view representations organize point-cloud information on one or more structured 2D domains, enabling efficient feature extraction but introducing dependence on projection geometry, viewpoint coverage, and feature fusion.

Natural pairings. These representations form a natural pairing with Convolution-based architectures because range images, depth maps, and rendered views provide regular 2D lattices with explicit local neighborhoods that directly support standard convolution. This enables efficient weight-shared processing and the use of mature 2D backbones. However, projected proximity may not reflect three-dimensional proximity, while point collisions, empty pixels, depth discontinuities, occlusion, and limited view coverage can distort or remove geometric information.

Conditional pairings. Transformer-based and GNN-based architectures can also process projected or multi-view features, but require additional mechanisms to preserve spatial and cross-view relationships. Transformers typically require positional or geometric cues representing image coordinates, viewing geometry, or camera poses, together with scalable attention and cross-view fusion. GNNs require graphs constructed over projected regions or views; if back-projected 3D elements and their relations become a distinct

processing branch, the overall representation is more accurately regarded as graph-structured or hybrid. These adaptations increase attention, graph-construction, fusion, and alignment costs, and remain sensitive to viewpoint inconsistency, occlusion, and missing observations. When projected and native 3D features are jointly processed, the representation is more accurately classified as hybrid.

Auxiliary use. MLP-based operators are commonly used for feature embedding, channel transformation, view fusion, or prediction. Because they do not inherently exploit the spatial lattice or establish cross-view correspondences, they generally serve an auxiliary role rather than defining the dominant feature-extraction pairing.

2.3.3 Spatially Discretized Representation Interactions

Spatially discretized representations organize point clouds into dense or sparse voxels, hierarchical cells, pillars, or BEV grids. They provide explicit spatial indexing and structured neighborhoods, but introduce quantization and a trade-off between geometric resolution and computational cost.

Natural pairings. These representations form a natural pairing with Convolution-based architectures because their indexed cells directly support weight-shared local processing. Dense and sparse voxels commonly use 3D convolutions, whereas pillar-derived and height-collapsed BEV features enable efficient 2D convolution after vertical aggregation, while other BEV constructions similarly exploit a regular horizontal lattice. Sparse and hierarchical structures improve scalability by processing occupied or adaptively refined regions, but require specialized indexing. Performance remains resolution-dependent: coarse cells merge structures and weaken boundaries, while fine grids increase memory and computation. Pillar and height-collapsed BEV features may additionally lose vertical detail and confuse overlapping or elevated structures.

Conditional pairings. Transformer-based and GNN-based architectures can also process spatial cells, but require additional structural adaptation. Transformers organize cell or BEV features as spatial tokens and use positional encoding with sparse, local, or windowed attention, making performance dependent on token resolution, partitioning, and the number of active cells. GNNs require graphs over cells, hierarchical regions, or object features to capture irregular or nonlocal relations. This adds graph-construction overhead and offers limited benefit when it only reproduces the local adjacency already available to convolution.

Auxiliary use. MLP-based operators commonly embed points within cells, transform voxel or pillar features, and support channel mixing, fusion, or prediction. Because they do not independently exploit spatial connectivity, they usually serve an auxiliary role rather than defining the dominant feature-extraction pairing.

2.3.4 Token-Based Representation Interactions

Token-based representations encode or aggregate points, voxels, projected regions, patches, or higher-level entities as feature vectors processed as discrete computational units. They often reduce the number of elements through grouping or patching and support hierarchical or global processing, although their effectiveness depends on token construction, sampling, and granularity.

Natural pairings. Token-based representations form a natural pairing with Transformer-based architectures because tokens directly provide the elements processed by self-attention. Geometric positional encoding exposes spatial relationships to the attention mechanism, while attention supports adaptive interaction among local and distant tokens. This pairing enables broad contextual modeling and scalable pretraining, but remains sensitive to token quality: coarse or unstable tokenization may merge structures, weaken boundaries, or reduce correspondence with the original points, whereas fine tokenization increases attention cost.

Conditional pairings. MLP-based architectures constitute a conditional pairing only when explicit token-mixing operations provide the dominant cross-token aggregation. GNN-based and Convolution-based architectures can also process tokens, but require additional structural mechanisms: GNNs rely on graphs defined over geometric or learned token relations, whereas convolutions require tokens to be arranged into ordered grids, windows, or neighborhoods. These adaptations can improve efficiency or locality, but introduce sensitivity to token organization, connectivity, and neighborhood construction.

Auxiliary use. Independently applied MLPs commonly support token embedding, channel transformation, feed-forward processing, and prediction, but do not by themselves establish cross-token interactions.

2.3.5 Graph-Structured Representation Interactions

Graph-structured representations organize points, voxels, patches, superpoints, or objects as nodes connected by geometric, spatial, or learned edges. They expose relational structure explicitly, but make feature extraction dependent on graph construction and connectivity quality.

Natural pairings. Graph-structured representations form a natural pairing with GNN-based architectures because graph edges directly define the neighborhoods used for message computation, aggregation, and propagation. This provides a strong inductive bias for local geometry, structural relations, correspondences, and object interactions. However, graph construction and edge-wise processing can be costly, while unreliable connectivity under noise, outliers, density variation, or partial observations may propagate errors. Deep message passing may also lead to oversmoothed node features.

Conditional pairings. Transformer-based architectures can process graphs, but require graph-aware positional or edge encodings and often use connectivity masks, sparse attention, or neighborhood-restricted attention. These mechanisms enable content-dependent interaction beyond conventional message passing, but increase sensitivity to edge encoding and attention design. Convolution-based processing requires graph-specific kernels or conversion to an ordered local structure; when kernels operate directly over graph neighborhoods, the resulting operator may be interpreted as graph convolution and may therefore overlap with the GNN family, depending on whether kernel evaluation or message propagation constitutes the dominant mechanism.

Weak pairing. Standard grid convolution forms a weak pairing with graph-structured representations because graph neighborhoods have variable connectivity and no fixed ordering. Applying regular convolution therefore requires reordering, rasterization, or other structural conversion that may discard relational information.

Auxiliary use. MLP-based operators commonly embed node and edge attributes, construct messages, update features, or produce predictions. Because independent MLPs do not propagate information through graph connectivity, they generally serve an auxiliary role unless explicit cross-node mixing becomes the dominant operation.

2.3.6 Hybrid Representation Interactions

Hybrid representations encode the same 3D structure through complementary forms, such as point-voxel, point-BEV, point-graph, voxel-token, or point-projection combinations. They aim to combine geometric precision, structured processing, relational modeling, and contextual aggregation within one system.

No universal natural pairing.

Conditional pairings. Hybrid representations have no universal natural pairing because their components expose different computational structures. They are therefore typically processed by representation-specific encoders, such as point-wise operators for precise geometry, convolutions for voxel or BEV features, GNNs for explicit relations, and Transformers for token-level or long-range aggregation. Projection, alignment, and fusion are then required to exchange information across branches. These mechanisms can compensate for quantization, projection, or sampling losses, but increase memory, duplicated computation, and fusion complexity. Performance may degrade under spatial misalignment, unbalanced branches, or repeated information loss during conversion.

Weak pairing. Applying a single shared operator directly to heterogeneous and unaligned representations forms a weak pairing because the components differ in resolution, spatial organization, neighborhood structure, and feature semantics. Effective processing therefore requires representation-specific encoders or explicit alignment before joint aggregation.

Table 2: Representation–architecture interactions in terms of typical interaction mechanisms, principal advantages and failure modes, and representative task examples. The listed tasks and methods are illustrative and non-exhaustive.

Representation	MLP-based	Convolution-based	GNN-based	Transformer-based
Point-based	Natural pairing: Shared point-wise MLP + Efficient processing without input-space quantization – Limited explicit modeling of local geometry Tasks: Cls., Seg.	Conditional pairing: Geometry-aware point convolution + Detailed local surface modeling without voxelization – Sensitivity to sampling density and neighborhood scale Tasks: Cls., Seg.	Conditional pairing: Point-graph message passing + Explicit geometric and relational modeling – Graph-construction cost and unreliable connectivity Tasks: Cls., Det.	Conditional pairing: Local, grouped, or sparse point attention + Adaptive local and long-range contextual modeling – Sampling, geometric encoding, and attention cost Tasks: Cls., Det.
Projection-based and multi-view	Auxiliary use: Pixel, region, or view embedding + Efficient feature and channel transformation – No inherent spatial or cross-view aggregation Tasks: Various	Natural pairing: Range-view or multi-view 2D convolution + Efficient use of regular image lattices and mature 2D backbones – Projection distortion, occlusion, and depth discontinuities Tasks: Cls., Det.	Conditional pairing: Graph over views or projected regions + Explicit modeling of cross-view and regional relations – Graph construction, alignment, and fusion overhead Tasks: Cls., Pose.	Conditional pairing: Range-view or cross-view attention + Global aggregation within and across projected views – Resolution, viewpoint alignment, and attention cost Tasks: Gen.
Spatially discretized	Auxiliary use: Point-to-cell, voxel, or pillar embedding + Efficient encoding and aggregation of points within indexed cells – No independent spatial aggregation across cells Tasks: Various	Natural pairing: Dense/sparse 3D voxel, hierarchical-cell, or pillar/BEV 2D convolution + Direct weight-shared local processing over indexed spatial structures – Quantization, dense-grid memory, sparse-indexing cost, and possible vertical-detail loss Tasks: Seg., Det.	Conditional pairing: Graph over voxels, hierarchical cells, or spatial regions + Irregular and nonlocal relations beyond fixed grid adjacency – Graph overhead and limited benefit when local grid adjacency is duplicated Tasks: Seg., Rec.	Conditional pairing: Voxel, hierarchical-cell, or BEV sparse/window attention + Adaptive long-range context across occupied or selected spatial cells – Dependence on token resolution, active-cell count, and attention partitioning Tasks: Det.
Token-based	Conditional pairing or auxiliary use: Cross-token mixing or independent per-token MLP + Efficient token and channel transformation – Requires explicit token mixing; otherwise remains auxiliary Tasks: Cls., Seg.	Conditional pairing: Ordered, local, or windowed token convolution + Efficient aggregation among locally organized tokens – Dependence on token ordering, neighborhood definition, and window structure Tasks: Cls., Seg.	Conditional pairing: Token-graph message passing + Explicit relational modeling among geometric or learned tokens – Sensitivity to graph connectivity and edge-construction cost Tasks: Seg., Reg., Reas.	Natural pairing: Self-attention over geometric tokens + Adaptive global context and scalable representation pretraining – Tokenization loss, positional encoding, and attention cost Tasks: Cls., Cpl.
Graph-structured	Auxiliary use: Node, edge, or message embedding + Flexible transformation of node and relational attributes – No information propagation through connectivity when used independently Tasks: Various	Weak pairing: Rasterization followed by regular grid convolution + Efficient processing after conversion to a regular structure – Loss or distortion of variable graph connectivity Tasks: –	Natural pairing: Graph message computation, aggregation, and propagation + Direct use of explicit geometric and structural relations – Edge errors, graph-processing cost, and feature oversmoothing Tasks: Cls., Det.	Conditional pairing: Graph-aware or connectivity-restricted attention + Adaptive local and long-range interaction over graph nodes – Dependence on edge encoding, masking, and attention design Tasks: Seg., Reg.
Hybrid	Conditional pairing or auxiliary use: Cross-branch projection, channel alignment, or feature fusion + Mapping heterogeneous features into a shared feature space – Branch imbalance and limited spatial modeling when used alone Tasks: Various	Conditional pairing: Point-voxel or point-BEV convolutional processing + Point-level geometric precision with structured spatial efficiency – Cross-representation alignment and redundant computation Tasks: Det.	Conditional pairing: Cross-representation graph construction and fusion + Relational alignment among heterogeneous spatial elements – Graph-construction and branch-synchronization overhead Tasks: Cls.	Conditional pairing: Cross-representation or cross-modal attention + Flexible global interaction and feature fusion across branches – Memory cost, spatial misalignment, and feature-resolution mismatch Tasks: Det.

Abbreviations: Cls.: Classification; Seg.: Segmentation; Det.: Detection; Gen.: Generation; Rec.: Reconstruction; Cpl.: Completion; Pose.: 6DoF Pose Estimation; Reg.: Registration; Reas.: Reasoning. These examples are illustrative rather than exhaustive and do not imply that the corresponding pairing is limited to the listed tasks or is universally superior for them.

Auxiliary use. MLPs, attention blocks, and projection layers commonly map features into a shared space, align channel dimensions, weight branches, or perform fusion. These operations support cross-representation interaction but remain auxiliary unless they dominate feature extraction and aggregation.

3 Core Tasks

Core tasks refer to the fundamental problems through which point cloud models identify, localize, and organize objects or regions in 3D data. In this section, four core tasks are reviewed: classification (Section 3.1),

segmentation (Section 3.2), detection (Section 3.3), and tracking (Section 3.4). Classification assigns a global label to an object or scene, segmentation predicts point-level or region-level labels, detection localizes objects with 3D bounding boxes, and tracking extends detection over time by associating objects across consecutive frames. These tasks are grouped together because they form the basic pipeline for understanding static and dynamic 3D scenes.

3.1 Classification

Point cloud classification assigns one semantic label to an entire 3D object or scene by aggregating geometric information and, when available, additional point attributes into a global feature representation. Methods are first organized by their dominant input representation: projection-based and multi-view methods convert point clouds into one or more structured 2D views; spatially discretized methods organize them into regular or hierarchical 3D cells; hybrid methods combine multiple representations; and point-based methods retain individual points as the primary feature carriers. Point-based methods are then grouped by their dominant feature-processing architecture, including MLP-based, Convolution-based, GNN-based, and Transformer-based approaches. These categories are not strictly exclusive, since recent models may use auxiliary operators from other families. Learning settings such as self-supervised and few-shot learning, together with multimodal transfer, are treated as separate design dimensions. The following discussion compares these families in terms of assumptions, computational cost, limitations, and practical trade-offs.

3.1.1 Projection-Based and Multi-View Methods

Projection-based and multi-view methods organize point-cloud information into one or more structured 2D representations, such as depth, silhouette, or attribute maps, and process them using established image backbones. Multi-view methods additionally aggregate information across several rendered viewpoints. These approaches benefit from regular 2D computation and pretrained models, but remain sensitive to viewpoint selection, rendering quality, view coverage, and fusion strategy. Using too few views may miss discriminative structures, whereas increasing the number of views raises computational cost.

MVCNN (Su et al., 2015) established the standard multi-view pipeline by aggregating independently extracted view features through max pooling. Although efficient, this operation may discard complementary information and inter-view relations. GVCNN (Feng et al., 2018) addresses this issue by weighting and hierarchically grouping views, whereas MHBN (Yu et al., 2018b) captures second-order interactions among local features through bilinear pooling. MVACPN (Wang et al., 2022c) further employs attention-based pooling to retain informative and fine-grained cues. Compared with MVCNN, these methods provide richer fusion, but introduce additional computational and architectural complexity.

These approaches generally rely on fixed camera configurations, which may fail to capture the most informative regions of each object. MVTN (Hamdi et al., 2021) addresses this limitation by learning object-dependent viewpoints through differentiable rendering. Thus, while GVCNN, MHBN, and MVACPN improve how rendered views are fused, MVTN improves how the views themselves are selected.

Another limitation is that conventional projections may lose geometric information along the viewing direction. DTV-CNN (Xia, 2023) alleviates this problem by combining complementary depth and thickness views. In contrast to MVTN, which determines *where* to observe the object, DTV-CNN improves *what information* each projection represents.

Projection also enables knowledge transfer from pretrained 2D models. CLIP2Point (Huang et al., 2023a) uses rendered depth maps and image–depth contrastive learning to adapt CLIP representations to point-cloud classification. Similarly, Yan et al. (2023a) projects 3D features into multiple views and aligns them with pretrained 2D representations under self-supervised learning. While CLIP2Point adapts a vision–language model to depth inputs, Yan et al. (2023a) uses multi-view projections to supervise a native 3D encoder. Both reduce reliance on labeled 3D data, but remain dependent on projection quality and viewpoint coverage.

Overall, projection-based methods have evolved from simple view pooling toward structured fusion, adaptive viewpoint selection, richer projection types, and 2D-to-3D knowledge transfer. Nevertheless, the inherent

loss of 3D information generally makes them better suited to complete objects rendered under controlled conditions than for partial scans, severe occlusion, clutter, or uncontrolled acquisitions.

3.1.2 Spatially Discretized Methods

Spatially discretized methods organize 3D space into dense or sparse voxel grids, hierarchical cells, or multi-resolution structures and typically process them using 3D convolutions or other spatially structured operators. Although they provide structured neighborhoods, they suffer from a resolution trade-off: coarse grids lose geometric details, whereas fine grids require substantial memory and computation. Dense grids are particularly inefficient because most voxels are empty.

VoxNet (Maturana & Scherer, 2015) applies 3D CNNs to dense occupancy grids, offering a simple baseline but limiting spatial resolution. FPNN (Li et al., 2016) reduces exhaustive computation by learning probes over a continuous 3D field. OctNet (Riegler et al., 2017) and O-CNN (Wang et al., 2017) instead use octrees to avoid processing empty space; OctNet enables adaptive high-resolution convolution, whereas O-CNN concentrates computation near object surfaces. These methods improve spatial efficiency but require specialized indexing and irregular memory access.

MRCNN (Ghadai et al., 2019) adopts a multi-resolution strategy, combining coarse global grids with finer representations near object boundaries. Compared with octree-based methods, it provides a more explicit separation of global and local geometry, although its predefined resolutions are less adaptive. In recent years, purely voxel-based classification methods have become less common, with many studies shifting toward hybrid approaches.

Overall, dense voxel models are simple but scale poorly, while probing, octree-based, and multi-resolution methods improve efficiency or geometric detail at the cost of greater complexity. The remaining trade-off between regular convolution, quantization loss, and memory usage has motivated hybrid point-voxel architectures.

3.1.3 Point-based Methods

Point-based methods operate directly on unordered points, preserving precise coordinates and avoiding projection and quantization losses. Their main challenge is to achieve permutation invariance while modeling both local geometry and global shape. According to the dominant feature-processing operator, they are grouped into MLP-based, Convolution-based, Graph-based, and Transformer-based approaches. This categorization reflects the principal feature-transformation mechanism rather than every auxiliary component.

MLP-based Methods MLP-based methods, as discussed in Section 2, use shared point-wise mappings and symmetric aggregation. Global variants are efficient, whereas hierarchical variants rely on sampling and neighborhood grouping to capture local geometry, adding search cost and sensitivity to point density.

PointNet (Qi et al., 2017a) established this paradigm using shared MLPs and symmetric max pooling, while DeepSets (Zaheer et al., 2017) provided a general formulation for permutation-invariant set functions. Its T-Net improves alignment robustness, but global pooling does not explicitly model local geometry. PointNet++ (Qi et al., 2017b) addressed this limitation through hierarchical sampling, grouping, and local PointNet modules.

Momenet (Joseph-Rivlin et al., 2019), PointWeb (Zhao et al., 2019), and SRN (Duan et al., 2019b) enriched PointNet-style processing with geometric moments, adaptive neighborhood interactions, and structural dependencies. These mechanisms improve local sensitivity but remain dependent on the quality of the selected neighborhoods or structural groups.

Recent work emphasizes efficient MLP design. PointNeXt (Qian et al., 2022) modernizes PointNet++ through revised architectural blocks and improved training strategies, while PointMLP (Ma et al., 2022) shows that residual MLPs with lightweight geometric normalization can remain competitive with more complex graph or Transformer models. DualMLP (Paul et al., 2024) uses complementary MLP streams for local-global interaction.

PointGL (Li et al., 2024a) uses residual MLPs for point embedding and local graph pooling for neighborhood aggregation. It is categorized as MLP-based because its main learnable transformations are residual point-wise MLPs, whereas the graph component mainly provides lightweight, nearly parameter-free pooling rather than learned graph convolution. PointMIL (De Vries et al., 2025) is a cross-cutting multiple-instance learning framework rather than an independent architectural family. It can replace conventional global aggregation in different point-cloud backbones, while the dominant architecture remains determined by the underlying feature extractor. Accordingly, POINTMIL results should be categorized according to the selected backbone rather than MIL pooling itself.

Overall, MLP-based classifiers are computationally efficient and well suited to standard parallel hardware, but global variants may miss local structure, whereas hierarchical variants improve geometric modeling at the cost of additional sampling and neighborhood searches.

Convolution-based Methods Convolution-based methods apply localized, weight-shared operators to irregular neighborhoods. Unlike regular-grid convolution, their kernels must account for unordered coordinates and nonuniform sampling, making kernel design and neighborhood construction central to both accuracy and cost.

Continuous methods define kernels from relative point coordinates. PCNN (Atzmon et al., 2018) extends point functions to continuous fields, SPHNet (Poulenard et al., 2019) uses spherical-harmonic kernels for rotation invariance, and RS-CNN (Liu et al., 2019d) models predefined geometric relations. KPConv (Thomas et al., 2019) uses fixed or deformable kernel points, while ConvPoint (Boulch, 2020) learns continuous spatial weighting. PACConv (Xu et al., 2021b) instead dynamically assembles position-adaptive kernels from a shared weight bank according to relative point positions, avoiding the need to predict an independent kernel for every local neighborhood. DensePoint (Liu et al., 2019c) combines point convolution with dense feature connections. These methods avoid voxelization but remain sensitive to neighborhood scale, sampling density, and local interpolation or kernel construction.

Discrete methods impose local structure through ordering, rings, shells, bins, or grids. PointCNN (Li et al., 2018) learns an $\mathcal{X}$ -transformation, A-CNN (Komarichev et al., 2019) uses annular neighborhoods, RConv (Zhang et al., 2019a) combines rotation-invariant relations with spatial binning, and ShellNet (Zhang et al., 2019b) summarizes concentric shells. PointWiseCNN (Hua et al., 2018) applies pointwise convolution over local neighborhoods. These designs simplify local parameterization but depend on the imposed ordering or partitioning.

DeltaConv (Wiersma et al., 2022) introduces anisotropic differential operators, while CompositeNets (Floris et al., 2024) separately encode spatial coordinates and point features before combining them. Overall, continuous kernels adapt their weights directly to relative coordinates and therefore handle irregular geometry flexibly, but they require repeated kernel evaluation or interpolation. Discrete operators simplify local parameterization by imposing an ordering, binning, or geometric structure, but their performance may depend more strongly on how that structure is defined. Point convolutions provide strong local geometric priors, but repeated neighborhood queries and kernel evaluations are costly. Their performance may degrade under sparse or nonuniform sampling, inappropriate neighborhood scales, or noisy local geometry.

GNN-based Methods GNN-based methods construct or use graph-structured representations in which points form nodes and geometric or learned relationships define edges. GNN operators then propagate and aggregate features through this connectivity. Explicit graphs support relational modeling, but graph construction adds cost and makes performance dependent on neighborhood quality.

ECC (Simonovsky & Komodakis, 2017), SpecGCN (Wang et al., 2018a), RGCNN (Te et al., 2018), and PointGCN (Zhang & Rabbat, 2018) introduced edge-conditioned, spectral, or Laplacian-based graph processing. Their behavior, however, depends strongly on graph topology. DGCNN (Wang et al., 2019c) introduced EdgeConv and reconstructed graphs in feature space, while LDGCNN (Zhang et al., 2021a) improved feature reuse. Dynamic graphs adapt to learned semantics, but repeated neighbor searches increase computational cost. Moreover, unreliable intermediate features may produce unstable neighborhoods. These models mainly differ in whether graph connectivity is used to define local kernels, adapt aggregation weights,

preserve manifold structure, propagate information along extended paths, or reduce graph-processing cost. KCNet (Shen et al., 2018), PointNGCNN (Lu et al., 2020), and 3D-GCN (Lin et al., 2020) model local structure through graph kernels or neighborhood aggregation. PointManifold (Yang & Gao, 2020) emphasizes manifold continuity. CurveNet (Xiang et al., 2021a) treats the point cloud as an undirected graph and constructs curves as guided walks through its k -nearest-neighbor graph, enabling information to propagate along longer point sequences. PointViG (Zheng et al., 2024b) and Point-SkipNet (Saeid et al., 2025) emphasize lightweight graph processing.

GNN-based models offer strong relational priors, but static graphs are cheaper and less flexible, whereas dynamic graphs are more adaptive but incur greater runtime and memory costs and are more sensitive to noisy features.

Transformer-based Methods Transformer-based methods apply attention to points or point tokens to model contextual relationships without fixed convolutional kernels. They provide flexible local-global interaction, but their computational cost and geometric sensitivity depend strongly on positional encoding, tokenization, and attention scope.

Transformer classifiers primarily differ in how they restrict attention through local neighborhoods, sparse global representatives, serialized point sequences, or point patches. PAT (Yang et al., 2019b) combines MLP-based point embeddings with Group Shuffle Attention and Gumbel Subset Sampling. Point Transformer (Zhao et al., 2021a) uses local vector attention with relative geometric encoding, whereas PCT (Guo et al., 2021) introduces offset attention for unordered points. PTV2 (Wu et al., 2022) strengthens grouped local attention, LCPFormer (Huang et al., 2023b) improves communication across neighboring local regions, and SPoTR (Park et al., 2023a) introduces sparse global-context modeling. These strategies reduce the cost of dense global attention, but depend on neighborhood partitions or selected global representatives.

More recent designs emphasize scalability. PPT (Wang et al., 2025i) introduces point-patch tokenization, and SPT (Wu et al., 2025a) explores spiking Transformer classification. These approaches reduce attention cost but may lose fine interactions when their patching, or sparsification strategies fail to preserve local geometry.

Transformers provide rich contextual modeling and support large-scale pretraining, but full point-wise attention scales poorly. Local, sparse, serialized, or patch-based variants reduce this cost while introducing dependence on neighborhood and token-construction quality.

3.1.4 Hybrid Methods

Hybrid classifiers combine complementary representations so that one can compensate for information lost by another. PV-Ada (Zhu et al., 2022) combines point-voxel abstraction, local Transformer encoding, and adaptive pooling to improve robustness to noise, outliers, missing points, and occlusion. The point representation reduces quantization loss, while voxels provide structured neighborhoods. However, conversion and fusion add overhead.

PointView-GCN (Mohammadi et al., 2021) processes partial point clouds from multiple views and uses graph convolution to model within-view geometry and cross-view relationships. PointOfView (Ren et al., 2024a) similarly combines raw point and multi-view features for few-shot classification, in which point-view fusion defines the representation design and few-shot learning defines the training strategy. Hybrid models can recover complementary evidence but depend on accurate alignment and balanced fusion. Their multiple processing branches also increase computational and implementation complexity.

3.1.5 Conclusion, Challenges, and Future Directions of Classification

No classification family is universally superior. Projection-based and multi-view methods benefit from mature 2D backbones but depend on viewpoint coverage and fusion quality; spatially discretized methods provide structured spatial processing but suffer from quantization and memory costs; hybrid methods combine complementary representations but add alignment and fusion overhead; and point-based methods preserve coordinate-level geometry while requiring explicit local-global modeling. Among point-based approaches,

MLP-based models prioritize efficiency, convolution-based models provide strong local geometric priors, GNN-based models capture explicit relational structure, and Transformer-based models support broader contextual interaction.

Classification remains sensitive to noise, occlusion, clutter, partial observations, nonuniform sampling, and synthetic-to-real domain shifts. These factors affect the families differently: projection-based and multi-view models depend on visible coverage; spatially discretized models depend on resolution; point convolutions and GNNs depend on neighborhood stability; and Transformers depend on tokenization and positional encoding.

Future work should prioritize robust and efficient local-global modeling, domain generalization, limited-supervision learning, multimodal or self-supervised pretraining, and lightweight processing. More realistic benchmarks and protocol-aware comparisons are also needed because reported results vary with point count, input attributes, pretraining, augmentation, voting, and dataset splits.

3.2 Segmentation

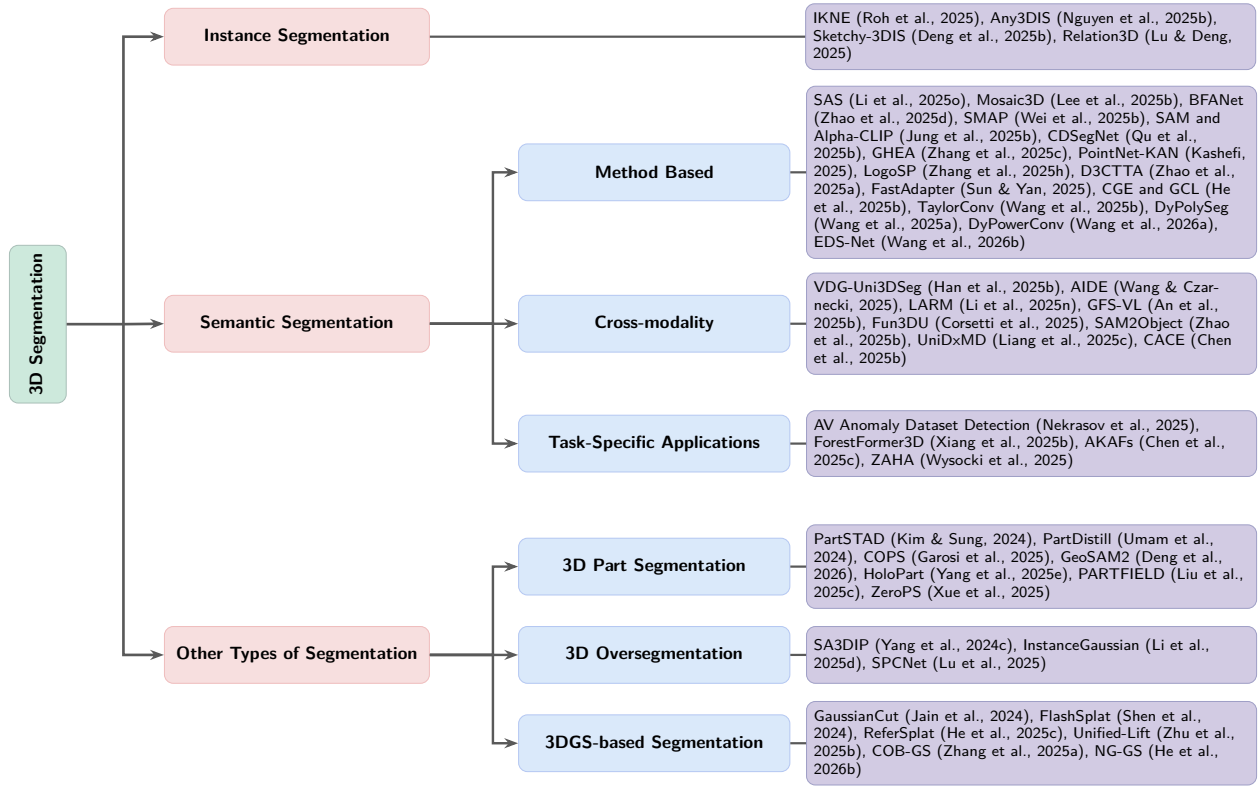


Figure 2: Taxonomy of recent 3D segmentation research directions and representative methods, datasets, and application-specific studies.

3D point cloud segmentation aims to partition a point cloud into meaningful subsets based on geometric properties or semantic content. By assigning semantic labels or grouping points according to their spatial relationships and contextual information, segmentation serves as a fundamental component of higher-level scene understanding and supports a broad range of downstream applications, including scene reconstruction and object recognition. While earlier surveys (Vinodkumar et al., 2023; Sarker et al., 2024) mainly focused on the historical development of 3D point cloud segmentation methods, this survey emphasizes recent advances in the field. In recent years, research has increasingly explored new directions, including open-vocabulary and few-shot learning, cross-modal knowledge transfer, domain adaptation, and the integration of vision-language and foundation models. As illustrated in Figure 2, we organize recent studies into instance segmentation,

semantic segmentation, and other segmentation tasks, including part segmentation, oversegmentation, and 3DGS-based segmentation.

3.2.1 Instance Segmentation

Instance segmentation extends 3D point cloud segmentation by partitioning points into distinct object instances while preserving geometric coherence. This requires reasoning about object-level structures in addition to local point features, which is particularly challenging in complex scenes due to occlusions, clutter, and partial observations.

Recent advances address these challenges by incorporating higher-level reasoning and enhanced relation modeling. Object-centric approaches explicitly encode instance-level features to capture holistic geometric and structural properties, improving instance grouping and boundary consistency (Roh et al., 2025). Multi-view strategies leverage 2D segmentation models and enforce cross-view instance consistency through mask tracking, thereby reducing fragmented 3D proposals (Nguyen et al., 2025b). Weakly supervised methods show that sparse annotations, such as sketch-based points and 3D bounding boxes can guide accurate instance segmentation while lowering labeling costs (Deng et al., 2025b). More recently, relation modeling frameworks have enhanced 3D instance segmentation by explicitly encoding geometric and spatial relationships between scene points and instance queries, using adaptive superpoint aggregation, contrastive refinement, and relation-aware self-attention to improve instance discrimination and consistency (Lu & Deng, 2025).

3.2.2 Semantic Segmentation

This section categorizes recent 3D point cloud semantic segmentation methods into three main directions: method-based approaches, cross-modality approaches, and task-specific applications. These categories are not strictly mutually exclusive, because some methods combine multiple learning settings, modalities, or architectural strategies.

Method-Based Method-based approaches improve semantic segmentation through different learning objectives, architectural designs, and adaptation strategies. We organize these methods into four subgroups: open-vocabulary and unsupervised segmentation, architecture and efficiency, generalization and adaptation, and few-shot segmentation.

Open-Vocabulary and Unsupervised Segmentation. Several works address the problem of undefined classes, particularly in open-vocabulary settings. Li et al. (2025o) aims to train and use 3D models by distilling knowledge from 2D models into 3D representations, while Yen et al. (2026) introduces a training-free approach that generates 3D proposals by merging 3D superpoints from multi-view 2D masks. By combining the outputs of 2D models, SAS (Li et al., 2025o) reduces the inherited limitations of the prior models. Similarly focused on refining the proposal-generation stage, Jung et al. (2025b) improves 3D proposal generation using robust 3D tracking and iterative removal of overlaps and utilizes Alpha-CLIP to obtain accurate, object-centric embeddings. These embeddings are then cross-referenced with the input text. However, the method is specifically designed for 3D instance segmentation.

Because relying heavily on these prior 2D networks can introduce significant overhead, Lee et al. (2025b) introduces Mosaic3D to reduce computational cost and latency, which is based on inherently 3D backbones. After applying SparseUNet, the CLIP-aligned point embeddings are compared with the input text embeddings from Recap-CLIP. Seeking to improve how these textual alignments are generated, Wei et al. (2025b) uses an off-the-shelf image captioner and introduces Sparse Masked Attention Pooling (SMAP) for point captioning to improve open-vocabulary segmentation. The generated captions are fed into a Caption2Tag module to create point-level tags, which are then used to infer semantic segmentation. Pushing beyond text alignments toward fully unsupervised learning, Zhang et al. (2025h) proposes LogoSP, a method for unsupervised 3D semantic segmentation that distills self-supervised 2D foundation models into 3D point features. The points are then progressively grouped into larger superpoints, which are organized into a graph from which global patterns are extracted to assign pseudo-labels.

Architecture and Efficiency. In contrast to open-vocabulary methods, several recent studies focus on improving core architectural efficiency and fully supervised segmentation performance through struc-

tural and computational modifications. Kashefi (2025) proposes PointNet-KAN to investigate the effect of Kolmogorov–Arnold Networks (KANs). PointNet-KAN replaces the shared MLPs of the original PointNet with shared KAN layers constructed using Jacobi polynomials. The study also includes PointNet-KAN-MLP, which combines a KAN encoder with an MLP decoder to mitigate overfitting in segmentation tasks, and PointNet-KAN-MLP++, which integrates KANs into the set abstraction modules of PointNet++ while retaining MLPs for the classification head.

Moving to geometric degradation within backbone architectures, Sun & Yan (2025) introduces FastAdapter, a general module designed to mitigate the geometric information degradation typically caused by fast down-sampling methods. To achieve this, the paper uses two components: Point-to-Anchor (P2A), which establishes anchor points to aggregate local features, and Anchor-to-Point (A2P), which injects these aggregated features back into the downsampled points. At the feature-extraction level, Zhao et al. (2025d) moves away from general performance metrics to focus on boundary feature analysis. By extracting multi-level features using OctFormer and an octree structure, the paper decouples semantic and boundary features to improve 3D semantic segmentation. Furthermore, to optimize the inference speed of segmentation architectures at the output level, Qu et al. (2025b) moves beyond the conventional multi-step process of diffusion models and proposes CDSegNet, a model based on the Conditional-Noise Framework (CNF). This model generates segmentation masks in a single step and improves robustness.

Generalization and Adaptation. Other recent research tackles challenges related to data robustness, scarcity, and cross-domain generalization across the machine learning lifecycle. To mitigate the initial challenge of data scarcity during training, Zhang et al. (2025c) proposes GHEA, a framework for generating new samples. They use generative models to produce new samples from existing data and then weight and mix the samples according to their hardness.

Even with augmented data, models must often handle unseen environments. To tackle domain generalization, He et al. (2025b) proposes Category-level Geometry Embedding (CGE) and Geometric Consistent Learning (GCL). Unlike global approaches, CGE maps features into a geometric space to capture specific structural properties unique to each semantic category. Furthermore, GCL attempts to inject physics-inspired awareness into the model. Lastly, for dynamic real-world deployments, Zhao et al. (2025a) proposes D3CTTA to address continual test-time adaptation for 3D LiDAR on the fly. D3CTTA is a backpropagation-free framework that manages density variations and feature correlations. Its novelty lies in combining distance-aware prototype learning with domain-specific feature decorrelation to mitigate category confusion across changing domains without gradient updates.

Few-Shot Segmentation. Few-shot point cloud semantic segmentation aims to segment novel categories using only a small number of annotated support samples. Several recent works build on the observation that pre-training a backbone such as DGCNN (Wang et al., 2019c) on seen categories introduces domain bias and computational overhead, while also failing to adequately capture local geometric structure under limited supervision. The methods differ in how they resolve two coupled sub-problems: (1) how to extract discriminative local structure without pre-training, and (2) how to close the feature gap between support and query sets once such features are extracted.

Treating local geometric structure as a function-approximation problem, Wang et al. (2025b) proposes Taylor-Conv to decompose features into low- and high-order terms, pairing its parametric variant with an Adaptive Push-Pull module to align prototypes via cross-attention. Extending this, DyPolySeg (Wang et al., 2025a) combines multiple high-order convolutions with Mamba blocks for long-range scene modeling, utilizing a Prototype Completion Module to bridge feature gaps through auto- and cross-correlation. To further improve gradient continuity over fine geometries, Wang et al. (2026a) introduces Dynamic Power Convolution (Dy-PowerConv) with learnable power functions, refining predictions through an iterative Visual Introspective Prototype module that explicitly models query-support differences. Finally, approaching feature alignment from a bio-inspired perspective, Wang et al. (2026b) treats support and query sets as co-adapting “species,” employing a Symbiotic Evolution Module to iteratively refine shared prototypes across multiple “generations.”

Cross-Modality Recent advances in 3D semantic segmentation increasingly adopt cross-modal learning (integrating language, images, and geometry) to overcome limited annotations, domain gaps, and closed-set semantics, thereby enabling more scalable and generalizable scene understanding. Vision-language priors drive much of this progress: VDG-Uni3DSeg (Han et al., 2025b) and AIDE (Wang & Czarnecki, 2025) align 2D VLM features with 3D geometry for open-vocabulary recognition, while LARM (Li et al., 2025n) and GFS-VL (An et al., 2025b) extend these ideas to generalized few-shot segmentation using language-driven relational cues and prototype-guided adaptive infilling of 3D VLM pseudo-labels, respectively. Transitioning to instance-level generalization, SAM2Object (Zhao et al., 2025b) consolidates multi-view SAM2 (Ravi et al., 2025) masks for zero-shot 3D instance segmentation, whereas Fun3DU (Corsetti et al., 2025) demonstrates that natural language task descriptions and Chain-of-Thought reasoning can guide functional 3D understanding. Beyond expanding recognition capabilities, cross-modal learning also enhances structural robustness: UniDxMD (Liang et al., 2025c) unifies 2D–3D representations for unsupervised domain adaptation, while CACE (Chen et al., 2025b) improves sim-to-real transfer through context-aware augmentation and 2D–3D consistency.

Task-Specific Applications Beyond general segmentation, several works tailor these techniques to specialized domains and real-world challenges, including anomaly detection, environmental monitoring, indoor reconstruction, and urban digital twins. To address the challenge of out-of-distribution objects, Nekrasov et al. (2025) introduces a high-resolution 3D LiDAR dataset for anomaly segmentation, benchmarking five baselines adapted from 2D segmentation. In environmental monitoring, Xiang et al. (2025b) proposes ForestFormer3D, an end-to-end framework for semantic instance segmentation of forest LiDAR data. By utilizing instance- and semantic-aware (ISA) query selection in a 5D embedding space and a score-based block-merging strategy for large-scale inference, the model effectively separates intertwined tree canopies in complex forest environments. For indoor reconstruction, Chen et al. (2025c) improves structural component identification, such as beams and columns, by introducing Architectural-Knowledge-Aware features (AKAFs); their two-stage framework fuses explicit global coordinates with relative semantic positions, consistently improving the accuracy of various backbones. Finally, in the context of urban digital twins, Wysocki et al. (2025) tackles the heterogeneity of building facades by introducing the ZAHA dataset and the Level of Facade Generalization (LoFG) taxonomy based on CityGML standards.

3.2.3 Other Types of Segmentation

Beyond semantic and instance segmentation, several specialized segmentation tasks have emerged to address different levels of scene and object understanding, including part-level decomposition, oversegmentation, and segmentation of 3D Gaussian Splatting (3DGS) representations.

3D part segmentation 3D part segmentation aims to decompose an object into semantically meaningful components, such as the legs of a chair or the wings of an airplane, requiring a finer understanding of object structure than semantic or instance segmentation. Existing approaches can be broadly divided into methods that operate directly in 3D space and methods that transfer knowledge from 2D foundation models. PARTFIELD (Liu et al., 2025c) learns a continuous 3D feature field for part grouping without relying on image projections, while COPS (Garosi et al., 2025), ZeroPS (Xue et al., 2025), and GeoSAM2 (Deng et al., 2026) leverage multi-view reasoning and foundation models such as SAM, SAM2, and GLIP to transfer part-level understanding from 2D to 3D representations. Building upon this direction, PartSTAD (Kim & Sung, 2024) and PartDistill (Umam et al., 2024) improve cross-modal knowledge transfer through task adaptation and distillation strategies, respectively. More recently, HoloPart (Yang et al., 2025e) addresses amodal part segmentation by employing a diffusion-based architecture to infer complete part geometry beyond visible surfaces. Overall, recent advances have been largely driven by foundation-model integration and multi-view reasoning, while generative methods are beginning to address occluded and incomplete object structures.

3D oversegmentation 3D oversegmentation partitions a scene into a large number of small, geometrically coherent regions, often referred to as superpoints, which serve as intermediate primitives for downstream segmentation tasks. Recent work has focused on improving the quality and utility of these representations. SPCNet (Lu et al., 2025) generates superpoints by serializing points along a space-filling curve to

preserve spatial locality and computational efficiency. Beyond their traditional role as a preprocessing step, oversegmented regions are increasingly being used as a fundamental representation for higher-level scene understanding. SA3DIP (Yang et al., 2024c) combines geometric and appearance cues to construct more reliable primitives for open-world instance discovery, reducing both under- and over-segmentation errors, while InstanceGaussian (Li et al., 2025d) performs oversegmentation directly on Gaussian representations before aggregating regions into category-agnostic object instances. These developments indicate a shift from viewing oversegmentation as a simple preprocessing operation toward treating it as a core representation for object-centric and open-vocabulary scene understanding.

3D Gaussian Splatting (3DGS) segmentation Although 3D Gaussian Splatting is not a conventional point-set representation, it represents scenes through explicit spatial Gaussian primitives and has become increasingly relevant to 3D segmentation and scene understanding. Accordingly, 3DGS segmentation aims to associate semantic, instance, or referring labels directly with Gaussian primitives, enabling object-level understanding in Gaussian-based scene representations. A major research direction focuses on lifting 2D segmentation results into 3D Gaussian space. FlashSplat (Shen et al., 2024) formulates this process as a globally optimal linear programming problem for efficient mask lifting, while Unified-Lift (Zhu et al., 2025b) introduces an object-aware end-to-end framework that establishes consistent associations between multi-view segmentations and Gaussian primitives. Another line of work addresses segmentation quality around object boundaries. COB-GS (Zhang et al., 2025a) adaptively splits ambiguous boundary Gaussians to better align object contours with scene geometry, whereas NG-GS (He et al., 2026b) incorporates a NeRF-guided continuous feature field to produce smoother and more consistent boundaries. Beyond automatic segmentation, GaussianCut (Jain et al., 2024) enables interactive object selection through graph-cut optimization, and ReferSplat (He et al., 2025c) extends the task to referring segmentation by localizing objects from natural-language descriptions. Collectively, these methods demonstrate a progression from efficient 2D-to-3D label lifting toward geometry-aware, interactive, and language-guided segmentation frameworks for Gaussian-based scene representations. For a broader discussion of segmentation as well as related editing and generation applications, we refer readers to the recent survey by He et al. (2026a).

3.2.4 Conclusion, Challenges, and Future Directions for Segmentation

Recent 3D segmentation research has expanded beyond conventional fully supervised semantic and instance segmentation toward open-vocabulary, few-shot, weakly supervised, domain-adaptive, multimodal, and Gaussian-based settings. Fully supervised methods generally retain strong performance under standard benchmark conditions, whereas limited-supervision and open-world approaches address more realistic challenges involving scarce annotations, unseen categories, and deployment-domain shifts. Because these methods differ in task definition, supervision, input modality, dataset, and evaluation metric, their reported results should primarily be compared within similar experimental settings rather than interpreted as a unified ranking.

Several challenges remain. Many open-vocabulary and cross-modal methods depend on 2D foundation models, multi-view rendering, or textual priors, introducing projection errors, cross-modal misalignment, and additional computational cost. Few-shot and weakly supervised methods must learn reliable local geometry from limited annotations, while domain-adaptive systems must handle changes in sensor characteristics, density, scene composition, and environmental conditions. Instance, part, and Gaussian-based segmentation further require consistent object boundaries and labels across incomplete observations, multiple views, or heterogeneous 3D primitives.

Future research should develop more efficient 3D-native models that retain the semantic generalization of foundation-model priors while reducing dependence on expensive 2D processing. Other important directions include unified open-world segmentation, stronger few-shot and weakly supervised learning, cross-domain adaptation without continual retraining, uncertainty-aware predictions, scalable processing of large scenes, and consistent segmentation across point, voxel, multi-view, and Gaussian representations.

Table 3: Taxonomy of 3D object detection methods organized along six complementary axes: representation and modality design, detection stage, anchor strategy, architectural paradigm, scene type, and supervision type.

Axis	Subcategory	Representative Methods
Representation /Modality	Point-based	PointRCNN (Shi et al., 2019), VoteNet (Qi et al., 2019), 3DSSD (Yang et al., 2020b), Point-GNN (Shi & Rajkumar, 2020), 3DETR (Misra et al., 2021), Group-free 3D (Liu et al., 2021b), 1A-SSD (Zhang et al., 2022c), Clusterformer (Pei et al., 2023), OV-3DETR (Lu et al., 2023), PTT (Huang et al., 2024a), FASTER (Dang et al., 2025), GeoFormer (Jin et al., 2025b), Part-A ² net (Shi et al., 2020b)
	Spatially discretized (voxel-based)	VoxelNet (Zhou & Tuzel, 2018), SECOND (Yan et al., 2018), Voxel R-CNN (Deng et al., 2021), VoTr (Mao et al., 2021b), FocalSparseConv (Chen et al., 2022e), SST (Fan et al., 2022b), DSVT (Wang et al., 2023a), VoxelNeXt (Chen et al., 2023e), SAFDNet (Zhang et al., 2024b), UniMamba (Jin et al., 2025a), ViKiENet (Yu et al., 2025b), FSHNet (Liu et al., 2025e)
	Spatially discretized (pillar/BEV-based)	PointPillars (Lang et al., 2019), PillarNet (Shi et al., 2022a), PillarNeXt (Li et al., 2023c), PillarHist (Zhou et al., 2025f), 3DPillars (Noh et al., 2025), PIXOR (Yang et al., 2018a), BEVFormer (Li et al., 2024h), BEVDepth (Li et al., 2023j), SparseBEV (Liu et al., 2023b), BEVNeXt (Li et al., 2024g)
	Projection-based and multi-view	MV3D (Chen et al., 2017), AVOD (Ku et al., 2018), LaserNet (Meyer et al., 2019), RangeDet (Fan et al., 2021), EVT (Lee et al., 2025d), OpenM3D (Hsu et al., 2025)
	Hybrid representations / multimodal designs	PV-RCNN (Shi et al., 2020a), PV-RCNN++ (Shi et al., 2023), PointAugmenting (Wang et al., 2021a), VISTA (Deng et al., 2022), CVFNet (Gu et al., 2022), TransFusion (Bai et al., 2022), Bridged Transformer (Wang et al., 2022d), FUTR3D (Chen et al., 2023b), GAFusion (Li et al., 2024d), RCBEVDet (Lin et al., 2024c), LiDAR-4D Radar Fusion (Chae et al., 2024), LiRaFusion (Song et al., 2024), Ev-3DOD (Cho et al., 2025), V2X-R (Huang et al., 2025g)
Detection Stage	One Stage	SA-SSD (He et al., 2020a), SST (Fan et al., 2022b), DSVT (Wang et al., 2023a), SparseBEV (Liu et al., 2023b), BEVNeXt (Li et al., 2024g)
	Two Stage	PV-RCNN (Shi et al., 2020a), PV-RCNN++ (Shi et al., 2023), Voxel R-CNN (Deng et al., 2021), LiDAR R-CNN (Li et al., 2021d), Pyramid R-CNN (Mao et al., 2021a), CT3D (Sheng et al., 2021a), DiffRefine (Shin et al., 2025)
Anchor Strategy	Anchor-based	VoxelNet (Zhou & Tuzel, 2018), PointPillars (Lang et al., 2019), SECOND (Yan et al., 2018), PV-RCNN (Shi et al., 2020a)
	Anchor-free	PointRCNN (Shi et al., 2019), CenterPoint (Yin et al., 2021)
Architectural Paradigm	Convolution-based	VoxelNeXt (Chen et al., 2023e), Rethinking Backbone Design for Lightweight 3D Object Detection in LiDAR (Chandorkar et al., 2025), Mixed Precision PointPillars for Efficient 3D Object Detection with TensorRT (Fuengfusin et al., 2026)
	Transformer-based	VoTr (Mao et al., 2021b), SST (Fan et al., 2022b), DSVT (Wang et al., 2023a), GeoFormer (Jin et al., 2025b)
	GNN-based	Point-GNN (Shi & Rajkumar, 2020),
Scene Type	Indoor	ImVoteNet (Qi et al., 2020a), H3DNet (Zhang et al., 2020b), 3DETR (Misra et al., 2021), Group-free 3D (Liu et al., 2021b), V-DETR (Shen et al., 2023), FCAF3D (Rukhovich et al., 2022)
	Outdoor	CIA-SSD (Zheng et al., 2021b), MPPNet (Chen et al., 2022d), FSD (Fan et al., 2022c), CenterFormer (Zhou et al., 2022c), PointPainting (Vora et al., 2020), 3D-CVF (Yoo et al., 2020), BEVFusion (Liang et al., 2022), SparseLIF (Zhang et al., 2024d), MambaFusion (Wang et al., 2025f)
Supervision	Weakly-supervised	WS3D (Meng et al., 2020), VS3D (Qin et al., 2020), Back to Reality (Xu et al., 2022), ViT-WSS3D (Zhang et al., 2023b), VG-W3D (Huang et al., 2024b), Prompt3D (Zhang et al., 2024i), (Han et al., 2024a)
	Semi-supervised	Sess (Zhao et al., 2020a), 3diourmatch (Wang et al., 2021b), ProficientTeachers (Yin et al., 2022a), NoiseDet (Chen et al., 2023f), Ssda3d (Wang et al., 2023f), PTPM (Wu et al., 2024c), DPL (Zhang et al., 2024e), Sp3d (Zhao et al., 2025c), PieAug (Lee et al., 2025a), GA-BEVFusion (Hazra et al., 2025), (Park et al., 2025)
	Self-/Unsupervised	Pointcontrast (Xie et al., 2020), GCC-3D (Liang et al., 2021), Randomrooms (Rao et al., 2021), (Ergelik et al., 2022), ProposalContrast (Yin et al., 2022b), Union (Lentsch et al., 2024), Lise (Zhang et al., 2024h), Liso (Baur et al., 2024), Patchcontrast (Shrout et al., 2025), Cmae-3d (Zhang et al., 2025e), DOTA (Xia et al., 2025a)

3.3 Detection

3D object detection estimates the 3D pose and extent of objects by predicting oriented 3D bounding boxes and class labels from point clouds such as LiDAR or RGB-D data, sometimes in combination with images (Mousavian et al., 2017; Shi et al., 2019; Yin et al., 2021). The task is challenging because point clouds are irregular and sparse, have range-dependent density, and suffer from occlusions and partial observations, making discriminative feature learning and accurate localization difficult (Qi et al., 2019; Shi et al., 2020a).

Because the number of proposed 3D detection methods is now very large, a single linear or purely chronological review is difficult to navigate. We therefore adopt a multi-axis taxonomy that reflects the most common design decisions and the trade-offs they introduce, helping readers compare methods within comparable settings and quickly identify approaches suited to their constraints. Specifically, we categorize detectors along six complementary criteria: **data representation** (point-based, spatially discretized, projection-based and

multi-view, and hybrid), **detection stage** (one-stage, two-stage), **anchor strategy** (anchor-based, anchor-free), **Architectural Paradigm** (MLP-based, convolution-based, GNN-based, and transformer-based), **scene type** (indoor, outdoor), and **supervision type** (fully, weakly, semi-, self-supervised).

Table 3 presents a comprehensive taxonomy of 3D detection methods organized according to these six complementary criteria. Because these axes overlap, many papers fit multiple categories. We therefore use the taxonomy as an overlapping tagging scheme rather than a strict partition, which better reflects the modular nature of 3D detectors and keeps the review easy to navigate while we discuss each criterion in its own subsection.

3.3.1 Data Representation

A core design choice in 3D object detection is the data representation used to convert sparse, irregular point clouds into learnable features, which directly affects geometric fidelity, receptive field, and runtime (Shi et al., 2019; Zhou & Tuzel, 2018; Lang et al., 2019). This section groups detectors into five common representations: **point-based** methods operate on raw (or RoI-sampled) points to preserve fine geometry, at the cost of irregular computation and expensive neighborhood aggregation (Shi et al., 2019; Qi et al., 2019; Shi et al., 2020b; Yang et al., 2020b; Shi & Rajkumar, 2020; Misra et al., 2021; Liu et al., 2021b; Zhang et al., 2022c; Pei et al., 2023; Lu et al., 2023; Huang et al., 2024a; Dang et al., 2025; Jin et al., 2025b); Within **spatially discretized representations**, **voxel-based** methods discretize space and learn from sparse voxel tensors, enabling scalable context modeling with sparse 3D convolutions or sparse transformers but introducing quantization effects (Zhou & Tuzel, 2018; Yan et al., 2018; Deng et al., 2021; Mao et al., 2021b; Chen et al., 2022e; Fan et al., 2022b; Wang et al., 2023a; Chen et al., 2023e; Zhang et al., 2024b; Jin et al., 2025a; Yu et al., 2025b; Liu et al., 2025e); and **Pillar/BEV-based spatially discretized** approaches organize features on metric ground-plane grids, enabling efficient planar processing while potentially reducing vertical detail (Lang et al., 2019; Shi et al., 2022a; Li et al., 2023c; Zhou et al., 2025f; Noh et al., 2025; Yang et al., 2018a; Li et al., 2024h; 2023j; Liu et al., 2023b; Li et al., 2024g). **Projection-based and multi-view** approaches map point clouds or sensor observations into structured 2D views, such as range images or multiple camera views, benefiting from efficient 2D processing while inheriting view-dependent information loss and occlusion artifacts (Chen et al., 2017; Ku et al., 2018; Meyer et al., 2019; Fan et al., 2021; Lee et al., 2025d; Hsu et al., 2025). Finally, **hybrid and multimodal designs** combine complementary information to balance accuracy and efficiency. Hybrid representations integrate multiple geometric encodings of the same 3D data, such as point-voxel features (Shi et al., 2020a; 2023), whereas multimodal methods fuse point clouds with images, radar, event data, or cooperative sensor inputs (Wang et al., 2021a; Deng et al., 2022; Gu et al., 2022; Bai et al., 2022; Wang et al., 2022d; Chen et al., 2023b; Li et al., 2024d; Lin et al., 2024c; Chae et al., 2024; Song et al., 2024; Cho et al., 2025; Huang et al., 2025g). Since these choices are modular, many detectors naturally combine multiple representations or modalities, so we treat this taxonomy as an overlapping tagging scheme rather than a strict partition.

3.3.2 Detection Stage

Most modern 3D detectors follow either a **one-stage** or **two-stage** design. **One-stage** methods predict 3D bounding boxes directly from dense scene features using a single detection head. This design usually provides a simpler pipeline and lower latency, which is suitable for real-time applications (He et al., 2020a; Fan et al., 2022b; Wang et al., 2023a; Liu et al., 2023b; Li et al., 2024g). These methods include convolution-based and query-based detectors, and some approaches also use lightweight refinement without a separate proposal refinement stage (Fan et al., 2022b; Liu et al., 2023b).

In contrast, **two-stage** detectors first generate object proposals and then refine them with RoI feature extraction and a second refinement module. This strategy can improve localization accuracy and recall, especially in sparse or long-range scenes, but it usually requires more computation (Shi et al., 2020a; Deng et al., 2021; Li et al., 2021d; Mao et al., 2021a; Sheng et al., 2021a; Shin et al., 2025). Recent two-stage methods further improve RoI features by using multi-scale features, point-voxel fusion, better proposal refinement, and iterative correction modules to reduce false positives and improve box alignment (Shi et al., 2023; Mao et al., 2021a; Sheng et al., 2021a; Shin et al., 2025).

3.3.3 Anchor Strategy

The design of object proposals in 3D detection heavily depends on the chosen anchor strategy, which determines how initial bounding box estimates are formulated. Historically, the field has been dominated by **anchor-based** methods (Zhou & Tuzel, 2018; Lang et al., 2019), where a dense grid of pre defined 3D bounding boxes is placed across the point cloud or its proxy representation such as voxels or pillars. The network then learns to predict class probabilities and regress residual offsets from these pre defined anchors to the ground-truth boxes. While this paradigm benefits from stable training and high recall, defining anchor sizes, ratios, and orientations requires extensive dataset-specific hyperparameter tuning. Furthermore, evaluating thousands of empty anchors in sparse 3D space introduces significant computational overhead.

To mitigate these drawbacks, a prominent shift has occurred toward **anchor-free** methods (Shi et al., 2019; Qi et al., 2019; Yin et al., 2021). These approaches eliminate the need for heuristic anchor generation by either modeling 3D objects as keypoints at their spatial centers and regressing bounding box properties directly from these features, or by utilizing bottom-up point-wise voting mechanisms. Accordingly, this section compares these two primary paradigms by evaluating their computational efficiency, flexibility in handling diverse object scales, and overall localization performance.

3.3.4 Architectural Paradigm

The architectural choice inherently governs the trade-off among inference speed, robustness to point sparsity, and spatial localization precision. Historically, **Convolution-Based Architectures** have been the most prevalent, typically applied after the data is rasterized into a regular grid. Standard 2D CNNs provide high efficiency for pillar-based representations, while sparse 3D convolutions dominate voxel-based methods by explicitly avoiding computations in empty space (Chandorkar et al., 2025; Shi et al., 2022b; Fuengfusin et al., 2026). The primary advantage of convolution-based architectures is their highly optimized inference speed, making them the standard for real-time applications. However, the requisite voxelization or projection processes inevitably discard fine-grained geometric details. This loss of resolution can degrade precise bounding box boundary estimation and limit the detection accuracy for small objects, such as pedestrians, especially at long ranges.

More recently, **Transformer-Based Architectures** have gained significant traction in the 3D domain (Mao et al., 2021b; Fan et al., 2022b). By leveraging self-attention mechanisms, architectures such as Voxel Transformer (VoTr) and Single Stride Transformer (SST) capture long-range dependencies and global contextual information across the sparse 3D scene, addressing the inherent receptive field limitations of standard local convolutions. The global receptive field enables transformers to excel under challenging conditions like severe occlusion and distant point sparsity. By utilizing scene-wide contextual cues, they can reliably infer object presence even when local evidence is fragmented. The main drawback is their high computational footprint and memory requirements, which can bottleneck the frames-per-second (FPS) rate in practical deployments.

Finally, **GNN-Based Architectures** offer a distinct paradigm by representing the point cloud directly as a graph (Shi & Rajkumar, 2020). In this formulation, points serve as nodes and local geometric neighborhoods define edges, allowing the network to natively preserve and exploit topological structures without prior voxelization. By directly processing raw points, GNN-based architectures achieve superior geometry preservation, which translates to highly precise 3D localization and accurate orientation (heading) estimation for complex shapes. Nonetheless, the substantial computational cost of dynamic neighborhood search across millions of points in large-scale outdoor scenes often restricts GNN-based architectures to localized bounding-box refinement stages or necessitates aggressive point downsampling.

3.3.5 Scene Type

Scene type plays an important role in 3D object detection, as it affects point cloud characteristics, occlusion levels, and computational requirements (Misra et al., 2021; Li et al., 2023c; Fan et al., 2022c). In this section, we focus only on methods that use point clouds as input, either independently or in combination with other sensors.

Indoor scenes usually contain denser point clouds because objects are observed at shorter distances. However, these environments are often cluttered, and objects may occlude one another (Qi et al., 2020a; Zhang et al., 2020b). Therefore, indoor detection methods commonly emphasize local geometric features and object-level reasoning (Zhang et al., 2020b; Liu et al., 2021b; Shen et al., 2023). Many approaches use point-based processing (Qi et al., 2020a; Zhang et al., 2020b) or transformer-based set prediction (Misra et al., 2021; Liu et al., 2021b; Shen et al., 2023), while several effective baselines are built on sparse voxel features with anchor-free detection heads (Rukhovich et al., 2022). When RGB-D data are available, some methods combine image features with point cloud information to improve semantic understanding in complex indoor environments (Qi et al., 2020a).

Outdoor driving scenes, in contrast, generate sparse and non-uniform point clouds over large areas, and point density decreases rapidly as distance increases (Yang et al., 2020b; Fan et al., 2021; 2022c). As a result, outdoor detection methods often rely on scalable representations such as bird’s-eye-view (BEV) maps, voxel grids, range-view projections, and fully sparse representations, which allow large receptive fields and efficient inference (Yin et al., 2021; Deng et al., 2021; Fan et al., 2021; Li et al., 2023c; Chen et al., 2023e; Fan et al., 2022c). Common architectures include fast single-stage or center-based detectors (Zheng et al., 2021b; Yang et al., 2020b; He et al., 2020a; Yin et al., 2021; Li et al., 2023c), as well as two-stage, temporal, or hybrid models that refine object localization, especially for distant objects (Deng et al., 2021; Li et al., 2021d; Chen et al., 2022d). Sparse attention mechanisms and transformer-based backbones are also used to capture broader context from sparse point clouds (Mao et al., 2021b; Fan et al., 2022b; Zhou et al., 2022c; Wang et al., 2023a). Moreover, LiDAR-camera fusion is widely applied in outdoor settings, since camera features provide useful semantic information and can improve detection in challenging cases such as distant, partially visible, or occluded objects (Vora et al., 2020; Yoo et al., 2020; Bai et al., 2022; Liang et al., 2022; Li et al., 2024d; Zhang et al., 2024d; Wang et al., 2025f).

In summary, indoor 3D object detection usually focuses on local geometry and object-centered modeling (Zhang et al., 2020b; Misra et al., 2021; Liu et al., 2021b; Shen et al., 2023), whereas outdoor detection places more emphasis on large-scale context, efficient sparse processing, and the frequent multi-sensor integration (Yin et al., 2021; Fan et al., 2022b; Wang et al., 2023a; Fan et al., 2022c; Liang et al., 2022; Zhang et al., 2024d).

3.3.6 Supervision

Most 3D point cloud object detectors are developed and evaluated in the fully-supervised setting (Mousavian et al., 2017; Shi et al., 2019; Yin et al., 2021; Pu et al., 2025), where large-scale datasets provide precise 3D bounding boxes for training. While this remains the dominant paradigm and the main source of state-of-the-art results, 3D box annotation is costly and time-consuming, motivating a growing body of work that reduces the need for dense labels. Accordingly, this section reviews limited-supervision 3D detection along three settings: **weakly-supervised** methods that replace full 3D boxes with weaker cues (e.g., 2D/image signals, clicks, tags, or partial labels) to enable 3D localization (Meng et al., 2020; Qin et al., 2020; Xu et al., 2022; Zhang et al., 2023b; Huang et al., 2024b; Zhang et al., 2024i; Han et al., 2024a); **semi-supervised** methods that combine a small labeled set with abundant unlabeled data using teacher–student learning, pseudo-labels, or consistency regularization (Zhao et al., 2020a; Wang et al., 2021b; Yin et al., 2022a; Chen et al., 2023f; Wang et al., 2023f; Wu et al., 2024c; Zhang et al., 2024e; Zhao et al., 2025c; Lee et al., 2025a; Hazra et al., 2025; Park et al., 2025); and **self-supervised or unsupervised** approaches that learn transferable 3D representations from unlabeled point clouds (e.g., reconstruction or masked modeling) for downstream detection, or attempt object discovery without explicit box annotations (Xie et al., 2020; Liang et al., 2021; Rao et al., 2021; Erçelik et al., 2022; Yin et al., 2022b; Lentsch et al., 2024; Zhang et al., 2024h; Baur et al., 2024; Shrout et al., 2025; Zhang et al., 2025e; Xia et al., 2025a).

3.3.7 Conclusion, Challenges, and Future Directions of Detection

In summary, each axis introduces fundamental trade-offs. Point-based representations preserve fine geometry but suffer from irregular memory access; voxel/pillar methods enable efficient convolution at the cost of quantization artifacts; hybrid designs balance both but increase system complexity. Two-stage detectors achieve higher accuracy through refined region proposal, while one-stage methods prioritize speed and simplicity.

Anchor-based strategies offer stable training but require heuristic design, whereas anchor-free approaches reduce hyperparameter tuning but may struggle with oriented objects. Multi-modal fusion (LiDAR–camera) currently sets state-of-the-art performance on benchmarks like nuScenes and Waymo, yet introduces sensor synchronization and calibration challenges. Transformer backbones are increasingly replacing CNNs for long-range context, but at higher computational cost. Outdoor detectors must handle extreme sparsity and scale variation, while indoor methods focus on occlusion and fine-grained layout. Finally, the supervision axis reveals a clear spectrum: fully supervised methods remain the accuracy leader, but weak, semi-, and self-supervised approaches are rapidly closing the gap, driven by the high cost of 3D annotation.

Remaining Challenges: (1) robustness to domain shift and sensor corruption — most methods degrade sharply on unseen weather or LiDAR configurations; (2) open-vocabulary and zero-shot 3D detection, where current models are tightly bound to fixed class vocabularies; (3) efficient long-range detection for high-resolution sensors (e.g., 128-beam LiDAR or 4D radar); and (4) truly unsupervised object discovery in point clouds without any human labels.

Promising Future Directions: unifying the six axes into learnable architectures that adapt representation and stage count per scene; leveraging large-scale pre-training (e.g., masked autoencoding) specifically for 3D geometry; integrating language and 3D for instruction-based detection; and developing benchmark suites that systematically evaluate trade-offs along each axis separately, enabling practitioners to select methods based on their specific constraints (e.g., speed vs. accuracy vs. label budget).

3.4 Tracking

3D object tracking is a fundamental computer vision task that continuously locates objects across consecutive frames in 3D space. Based on the number of targets, tracking methods can be categorized into Single-Object Tracking (SOT) and Multi-Object Tracking (MOT). Figure 3 illustrates this taxonomy.

3.4.1 Single-Object Tracking

SOT specifies a target in the initial frame and tracks it across subsequent frames. While most recent 3D SOT methods rely on deep neural networks (Wang et al., 2024c), a growing line of research explores multimodal tracking to combine geometric and visual information. Accordingly, this survey organizes SOT methods by input modality into LiDAR-only and RGB–LiDAR trackers. LiDAR-only methods are then grouped according to their dominant tracking paradigm into Siamese-based, Motion-based, and Transformer-based approaches. These categories are not strictly mutually exclusive.

LiDAR Trackers LiDAR-based tracking works with 3D point clouds captured by LiDAR sensors. These sensors are robust to lighting changes, but working with this data is challenging. LiDAR-based tracking methods are categorized into three frameworks: Siamese-based, motion-based, and transformer-based.

- **Siamese-based**

A shared Siamese backbone extracts features from both the template and the search area. The similarity between these features is then computed using a metric such as cosine similarity. SC3D (Giancola et al., 2019) combines a Siamese network with an autoencoder to encode sparse point clouds, after which a decoder reconstructs the 3D shape. However, this method requires a computationally expensive exhaustive search. 3D-SiamRPN (Fang et al., 2020) introduces two cross-correlation modules within a Region Proposal Network (RPN) to directly predict 3D bounding boxes, achieving real-time tracking. The Siamese paradigm can be combined with other architectural techniques such as VoteNet, voxelization, and transformers, particularly for handling 3D point clouds.

- **VoteNet-based:** These methods combine Siamese feature matching with VoteNet-style proposals, avoiding the computationally expensive exhaustive search. P2B (Qi et al., 2020b) improves SC3D’s speed and efficiency through an end-to-end learning approach that uses Hough voting (Qi et al., 2019) to guide proposal generation and verification. MLVSNet (Wang et al., 2021g) enhances P2B to handle the extreme sparsity of outdoor point clouds by utilizing multi-level

Hough voting, target-guided attention, and vote-cluster modeling to achieve better localization accuracy. BAT (Zheng et al., 2021a) addresses P2B’s inability to capture size and part information by leveraging 3D box information from the first frame, introducing a new BoxCloud method and Box-Aware Feature Fusion.

- **Voxel-based:** These methods handle unstructured and sparse LiDAR data through voxel-based processing combined with the Siamese paradigm. V2B (Hui et al., 2021) uses a Siamese network to learn shape-aware features and then projects the voxelized features into a dense bird’s-eye view representation.
- **Siamese-Transformer:** These methods enhance feature representation and model the relationship between the template and the search region through non-linear feature matching and attention mechanisms. PTT (Shan et al., 2021) introduces a transformer-based module leveraging self-attention, making P2B (Qi et al., 2020b) more robust to sparsity and disorder. PTTR (Zhou et al., 2022a) replaces PTT’s linear cosine similarity with cross-attention matching while proposing a coarse-to-fine approach, first performing broad localization to estimate the target region and then refining the target’s position and orientation.

VoteNet-based variants use voting and proposal generation to avoid exhaustive search, but their localization quality can degrade when target points are very sparse. Voxel- and BEV-based variants provide regular structures for efficient processing, although discretization may remove fine geometric details. Siamese-transformer variants improve long-range template-search interaction and global feature matching, but generally introduce additional computational and data requirements.

- **Motion-based**

Motion modeling trackers process consecutive frames to learn movement patterns that improve tracking efficiency. M2-Track (Zheng et al., 2022) is a two-stage approach that first localizes targets and then refines bounding boxes — an approach effective in cluttered scenes. DMT (Xia et al., 2023) improves BAT (Zheng et al., 2021a) by replacing its complex 3D region proposal network with two modules: a Motion Prediction Module, which provides an initial target estimate, and a Voting Module, which refines it. This approach achieves faster tracking, a lighter model, and higher accuracy. SiamCUT and MoCUT (Nie et al., 2024) introduce a unified network, AdaFormer, which encodes geometric variations across different object types. FocusTrack (Zhou et al., 2025e) proposes a one-stage tracker that jointly models motion and semantics. Its inter-frame motion-modeling module captures differences between frames and then guides a Focus-and-Suppress attention module. TrajTrack (Fan et al., 2025) is a two-frame tracker with trajectory-based motion modeling that uses TrajFormer (a transformer architecture) to predict future trajectories.

- **Transformer-based**

With the success of transformer architectures, researchers have increasingly adopted them for LiDAR tracking. PTT-Net (Shan et al., 2022) uses an attention mechanism with two stages: voting and proposal generation. CXTrack (Xu et al., 2023a), instead of cropping frames, processes full point clouds to keep context, which improves robustness to occlusions and appearance changes. MBPTrack (Xu et al., 2023b) addresses appearance changes with two innovations: a transformer-based memory module and a box-prior localization network. CorpNet (Wang et al., 2023c) simplifies tracking by using a single encoder that both extracts and correlates template-search features via multi-level self- and cross-attention, achieving better accuracy and real-time speed. HVTrack (Wu et al., 2024b) consists of three modules: Relative-pose-aware memory, base-expansion cross-attention, and contextual point self-attention. This method handles large frame-to-frame variations. PTTR++ (Luo et al., 2024b) extends PTTR with parallel point-wise and BEV branches, performing transformer-based template-search feature matching in both representations before selectively fusing their features.

Siamese-based trackers rely on appearance matching to generate proposals quickly, but degrade sharply as point clouds become sparser; motion-based trackers address this by tracking relative motion instead of shape, trading sparsity-robustness for a dependence on accurate foreground segmentation; transformer-based trackers instead target a different weakness — local correlation’s limited receptive field — using attention

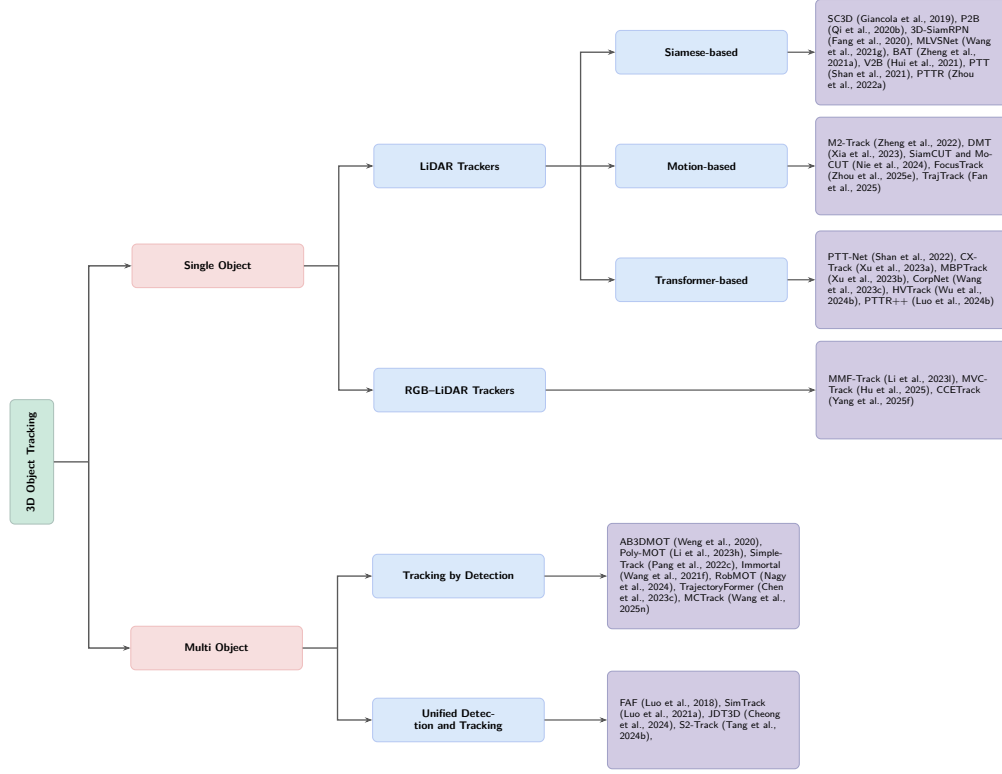


Figure 3: Taxonomy of 3D object tracking methods across single-object and multi-object tracking paradigms.

to capture global context; they handle distractors more effectively, at the cost of higher compute and data requirements.

RGB-LiDAR Trackers RGB-LiDAR trackers leverage the complementary advantages of both modalities. There has been limited research in this area. MMF-Track (Li et al., 2023f) provides more reliable tracking by creating textured pseudo-points and then using attention modules to connect modalities at different scales. MVCTrack (Hu et al., 2025) segments the target and uniformly samples virtual points, then assigns depth values and projects them into 3D space, which improves tracking of far-away or small objects. CCETTrack (Yang et al., 2025f) aligns the global context across modalities and performs fusion in the bird’s-eye view space, helping to better distinguish similar objects.

LiDAR models face challenges due to sparse, textureless 3D point clouds. For this reason, RGB-LiDAR fusion models have been developed to improve accuracy and robustness.

3.4.2 Multi Object Tracking

Multi-object tracking simultaneously localizes multiple objects and maintains consistent identities for them across consecutive frames. In addition to accurate object localization, MOT must address data association, object appearance and disappearance, occlusion, missed detections, and trajectory management. As illustrated in Figure 3, we organize 3D MOT methods into two principal paradigms: tracking by detection and unified detection and tracking.

Seminal works such as AB3DMOT (Weng et al., 2020) demonstrate that streamlined classical approaches can balance accuracy and speed. While probabilistic methods improved precision through uncertainty modeling, recent data-driven architectures achieve superior performance, particularly when handling sparse point cloud data. However, this substantial gain in accuracy often incurs higher computational costs, creating a trade-off between lightweight CPU-based algorithms and heavy, GPU-dependent deep learning models. A comparative summary of these paradigms is provided in Table 4.

Table 4: Comparison of the main 3D MOT paradigms in terms of their principal advantages and limitations.

Paradigm	Advantages	Limitations
Tracking by Detection	Detectors and trackers optimized independently Fast classical associations Traceable failures	Dependent on detection quality Features not jointly optimized temporally Requires heuristics
Unified Detection & Tracking	Joint training Attention handles occlusion Reduced reliance on hand-crafted association	Typically requires more training data Computationally expensive Lower modular interpretability

Tracking by Detection Tracking by Detection divides the pipeline into sequential detection, association, motion estimation, and track-management steps. While traditional baselines rely on Kalman filters, recent methods incorporate learned motion models or geometric constraints to better handle occlusion (Li et al., 2023h; Wang et al., 2021f). TrajectoryFormer (Chen et al., 2023c) constructs predictive trajectory hypotheses from temporally propagated and current-frame detection boxes to improve trajectory–detection association and recover short-term missed detections. MCTrack (Wang et al., 2025n) provides a detector-agnostic tracking framework that combines motion prediction and multi-stage association and is designed to generalize across different autonomous-driving datasets. However, performance remains strongly dependent on detection quality, and detector features are not necessarily optimized for temporal association. Confidence-aware matching and robust track-management strategies are therefore used to reduce false associations and error propagation (Pang et al., 2022c; Nagy et al., 2024).

Unified Detection and Tracking Unified frameworks integrate detection and association to reduce the sequential error propagation commonly observed in TBD pipelines. Early approaches, such as FAF (Luo et al., 2018), used shared backbones to improve inference speed but often retained separate post-processing steps. SimTrack (Luo et al., 2021a) further integrated detection and tracking by directly predicting new and tracked objects from consecutive point clouds, reducing reliance on explicit heuristic association. More recently, JDT3D (Cheong et al., 2024) adopted a LiDAR-based joint detection-and-tracking framework in which detection and tracking are trained end-to-end through persistent object queries. S2-Track (Tang et al., 2024b) further simplified the end-to-end query-based tracking pipeline by improving query initialization, propagation, and matching.

3.4.3 Conclusion, Challenges, and Future Directions of Tracking

Recent 3D tracking research has progressed from appearance-based matching and modular tracking-by-detection pipelines toward motion-aware, transformer-based, multimodal, and jointly optimized frameworks. These developments have improved target localization, temporal association, and robustness across both SOT and MOT settings.

Despite substantial progress, reliable 3D tracking remains difficult under sparse observations, prolonged occlusion, rapid motion, large appearance changes, and missed detections. SOT methods must preserve the target identity while adapting to geometric variation, whereas MOT systems must jointly manage association, track initialization and termination, identity switches, and long-term trajectory consistency. Current methods also face a trade-off between the accuracy of large temporal or transformer-based models and the latency and memory constraints of real-time deployment.

Future research should extend tracking beyond closed-set assumptions toward open-world tracking of novel objects without task-specific retraining (Ishaq et al., 2025). Cooperative perception may reduce single-agent occlusion by sharing observations across platforms (Chiu et al., 2024), but requires robust synchronization, communication-efficient feature exchange, and mechanisms for handling inconsistent or missing information. Additional priorities include uncertainty-aware association, long-term memory, robustness to sensor and domain shifts, multimodal tracking under partial sensor failure, and lightweight models suitable for embedded deployment.

4 Geometric Modeling, Alignment, and Pose Estimation

Geometric modeling, alignment, and pose estimation tasks are grouped together because they all focus on the geometric structure and spatial relationships of 3D point clouds rather than only semantic labeling. These tasks involve preserving, generating, recovering, aligning, or interpreting point cloud geometry. This section reviews compression (Section 4.1), generation (Section 4.2), reconstruction (Section 4.3), registration (Section 4.4), and 6DoF pose estimation (Section 4.5) under this shared geometric perspective.

4.1 Compression

3D Point Cloud Compression (3DPCC) is a critical task in modern 3D data processing pipelines, particularly as 3D sensors such as LiDAR, depth cameras, and SLAM systems generate massive volumes of spatial data. A point cloud typically comprises a set of 3D coordinates, optionally enriched with attributes such as color or reflectance. The compression process aims to reduce storage and transmission costs while preserving essential geometric and attribute information for reconstruction or downstream tasks. To systematically analyze existing methods, we categorize 3DPCC models along four dimensions: the type of compression (lossy or lossless), the data component being compressed (geometry, attributes, or joint), the temporal nature of the data (static, dynamic, or dynamically acquired), and the representation strategy used (e.g., projection-based, point-based, octree-based, voxel-based, or sparse tensor-based).

Compression type distinguishes methods that reconstruct point clouds with some allowable distortion (lossy) from those that reproduce them exactly (lossless). Most current deep learning-based methods fall within the lossy category because learned quantization generally provides higher compression gains. The compressed data component indicates whether a method targets geometry (spatial coordinates), attributes, such as color or reflectance, or both components jointly. Joint compression remains comparatively underexplored but is becoming increasingly important in applications such as immersive media and robotics. Data type distinguishes static point clouds from dynamic point cloud sequences and dynamically acquired point clouds. Dynamic point clouds represent objects or scenes whose geometry or attributes evolve over time, whereas dynamically acquired point clouds are captured continuously or sequentially by a sensing platform, often under sensor or platform motion. These categories are not strictly disjoint, since a continuously acquired point cloud stream may also contain moving objects or time-varying scene geometry and may therefore be regarded as both dynamic and dynamically acquired. Finally, representation strategy describes how point cloud data are structured before or during compression, including projection-based, point-based, octree-based, voxel-based, and sparse tensor-based representations.

Projection-based methods convert 3D point clouds into 2D representations, such as planar images, range images, or depth maps, enabling the use of conventional image or video compression codecs to exploit spatial and temporal redundancies (Van Der Hooft et al., 2019; Liu et al., 2020b; 2021a; Gao et al., 2022; Wang & Liu, 2022; Wang et al., 2024d). In contrast, **point-based** methods operate directly on raw point clouds, allowing the processing of irregular and unordered point sets without requiring intermediate structured representations (Huang & Liu, 2019; Wiesmann et al., 2021; You et al., 2024; Xie et al., 2024a). **Octree-based** methods organize point clouds into hierarchical octree structures by recursively subdividing 3D space into occupied cubes, where each node is encoded using compact occupancy codes that can be efficiently compressed with entropy models (Schnabel & Klein, 2006; Wang et al., 2021c; Que et al., 2021; Fu et al., 2022; Pang et al., 2022b; Fang et al., 2022; Song et al., 2023a; Nguyen & Kaup, 2023; Fan et al., 2023; Akhtar et al., 2024; Jin et al., 2024b; Cui et al., 2023; Wang et al., 2025o; Zheng et al., 2026a; Jiang et al., 2026). **Voxel-based** methods transform point clouds into regular volumetric grids, representing spatial information through binary voxel occupancy, which facilitates neighbor indexing and the application of 3D convolutions while leveraging previously decoded voxels for improved compression efficiency (Zhang et al., 2014; Thanou et al., 2015; Quach et al., 2019; Wang et al., 2021d; Que et al., 2021; Pang et al., 2022b; Nguyen & Kaup, 2023; Ruan et al., 2024; Rudolph et al., 2024). To address the high computational cost associated with dense voxel representations, **sparse tensor-based** methods employ sparse voxelization and sparse convolutions to process only occupied voxels (Pang et al., 2022b; Nguyen & Kaup, 2023; Wang et al., 2023b; Zhang et al., 2023e; Xie et al., 2024b; Mao et al., 2025; Huo et al., 2025; Fu et al., 2026). These methods significantly reduce computational complexity and have demonstrated strong performance in point

cloud compression, typically requiring quantized integer point clouds to enable efficient sparse convolution operations. To complement the preceding representation-oriented discussion, Table 5 provides a consolidated overview of the reviewed studies, categorizing methods by compression type, compressed data component, and data type. This organization enables a more systematic comparison of the 3DPCC literature across complementary methodological perspectives.

Table 5: Representative 3D point cloud compression studies categorized by compression type, compressed data component, and data type.

Compression Type	Compressed Data Component	Data Type	Representative Methods
Lossy	Geometry	Static	(Huang & Liu, 2019), (Quach et al., 2019), (Wang et al., 2021c), VoxelContext-Net (Que et al., 2021), (Wang et al., 2021d), OctAttention (Fu et al., 2022), GRASP-Net (Pang et al., 2022b), (Song et al., 2023a), OctFormer (Cui et al., 2023), ECM-OPCC (Jin et al., 2024b), (Xie et al., 2024b), (Zheng et al., 2026a)
		Dynamic	(Akhtar et al., 2024), OctMamba (Jiang et al., 2026)
		Dynamically acquired	(Wang & Liu, 2022), (Fan et al., 2023), msLPCC (Wang et al., 2024d)
	Attribute	Static	(Zhang et al., 2014), PCAC-GAN (Mao et al., 2025), (Huo et al., 2025), DeepRAHT (Fu et al., 2026)
	Joint	Static	YOGA (Zhang et al., 2023e), (Ruan et al., 2024), (Rudolph et al., 2024)
		Dynamic	(Thanou et al., 2015), (Gao et al., 2022)
		Dynamic + dynamically acquired	(Van Der Hooft et al., 2019), (Liu et al., 2020b), (Liu et al., 2021a)
Lossless	Geometry	Dynamic	Octree-STCM (Wang et al., 2025o)
	Attribute	Dynamic	(Wang et al., 2023b), 4DAC (Fang et al., 2022)
	Joint	Static	(Schnabel & Klein, 2006), (Nguyen & Kaup, 2023)

4.1.1 MPEG Point Cloud Coding Standards

MPEG has developed standardized point cloud coding frameworks to support efficient compression and interoperable representation of diverse 3D content. Alongside these standards, a range of benchmarking frameworks, software tools, and learned coding methods have been introduced to address challenges encountered in their evaluation, extension, and practical deployment. Accordingly, this subsection reviews MPEG point cloud coding standards and their related developments from a challenge-driven perspective, with particular attention to evaluation fairness, reproducibility, coding efficiency, cross-content generalization, temporal modeling, and interoperability.

The MPEG point cloud coding ecosystem has primarily developed along two complementary branches. Geometry-based Point Cloud Compression (G-PCC) directly encodes point cloud geometry and attributes using native 3D coding tools. In contrast, Video-based Point Cloud Compression (V-PCC) projects volumetric content onto 2D video representations to leverage established video coding technologies. These standardized frameworks provide the basis against which many subsequent coding, evaluation, and extension efforts have been developed.

Although G-PCC and V-PCC provide standardized coding frameworks, differences between sparse LiDAR scans, dense surface-based point clouds, and dynamic volumetric content create challenges for unified evaluation, cross-content generalization, and efficient adaptation. To address this, researchers proposed benchmarking frameworks assessing both compression efficiency and computational cost. Zaghetto et al. (2022) introduced a general evaluation interface, while Fan et al. (2022d) developed D-DPCC, a deep PCC framework for LiDAR using multiscale motion fusion and adaptive interpolation to better model temporal dynamics. These advancements enabled deep models to outperform traditional handcrafted motion compensation techniques.

Fragmentation across implementation frameworks such as TensorFlow and PyTorch has made reproducibility and fair comparison difficult. To overcome this issue, Pang et al. (2022a) introduced pccAI, a PyTorch-based suite for deep 3DPCC model development and evaluation. Improving compression quality also became a major focus. Pang et al. (2022b) proposed GRASP-Net, which separates point cloud compression into base and

enhancement layers, combining CNN and MLP modules for richer feature extraction. Zaghetto & Graziosi (2021) and Lodhi et al. (2021) contributed adaptive entropy models that use neural networks for context-based probability estimation, optimizing rate-distortion trade-offs dynamically. Together, these methods target different bottlenecks in learned compression: pccAI Pang et al. (2022a) improves reproducibility, GRASP-Net Pang et al. (2022b) enriches the learned representation, and adaptive entropy models Zaghetto & Graziosi (2021); Lodhi et al. (2021) reduce coding redundancy through more accurate probability estimation.

Generalization between sparse LiDAR scans and dense object point clouds remains difficult because these data types differ substantially in density, spatial distribution, and acquisition characteristics. To address this limitation, Wang & Ma (2022) presented SparsePCGC, a unified framework for lossy and lossless compression using sparse tensors. This framework was extended by SparsePCGCv2 (Wang et al., 2022b), which added KNN-based neighborhood attention for better context modeling. For dynamic PCC, Akhtar et al. (2024) introduced DPCGC-SC, which uses sparse CNNs to propagate inter-frame geometry and predict latent codes, enhancing temporal coherence. Attribute compression has also progressed: Deep-PCAC (Sheng et al., 2021b) employed variational autoencoders and sparse convolution for color compression, while Pinheiro et al. (2023) proposed NF-PCAC, a normalizing flow model for high- and low-bitrate settings. Geometry compression was improved by Zhou et al. (2023d), who jointly optimized down-sampling and up-sampling for better reconstruction. Qi & Gao (2024) introduced a variable-rate architecture with scale-aware transformations, and Yu & Gao (2024) developed a plug-in entropy estimator. Zuo et al. (2024a) further enhanced dynamic PCC with a temporal contextual pyramid for cross-frame information fusion at lower computational costs. Overall, these developments reveal three complementary design directions: sparse representations improve scalability and cross-content generalization, temporal models exploit inter-frame redundancy, and component- or rate-specific methods increase coding flexibility, albeit with different computational and implementation costs.

MPEG point cloud compression standardization has continued to evolve through improvements to both geometry-based and video-based coding frameworks. Enhanced G-PCC (E-G-PCC) (Group, 2024), under development as MPEG-I Part 38, extends the original geometry-based point cloud compression framework with tools better suited for dynamic point cloud sequences and improved temporal coding (Group, 2024; JTC1/SC29, 2026). In parallel, the video-based branch was updated through (JTC1/SC29, 2025), which specifies Visual Volumetric Video-based Coding (V3C) and Video-based Point Cloud Compression (V-PCC) for immersive volumetric media representation. Together, these standards reflect MPEG’s continued focus on improving compression efficiency, dynamic scene support, and interoperability for point cloud and volumetric video applications.

4.1.2 Conclusion, Challenges, and Future Directions of Compression

In conclusion, point cloud compression methods involve different trade-offs depending on their representation and target component. Projection-based approaches benefit from mature 2D codecs but may introduce distortion during projection, while point-based methods preserve irregular geometry at the cost of more difficult entropy modeling. Octree-based models provide efficient hierarchical coding, particularly for geometry and lossless compression, whereas voxel-based methods enable regular 3D processing but incur high memory and computational costs. Sparse tensor representations reduce this overhead by operating only on occupied regions, although their performance remains sensitive to quantization and voxel resolution. Therefore, the most suitable representation depends on point density, coding mode, and available computational resources.

Several challenges still restrict the practical use of learned compression. Research remains largely focused on lossy geometry coding, while lossless, joint geometry-attribute, and dynamic compression are less mature. Generalization across dense objects, sparse LiDAR scans, and unseen acquisition conditions also remains difficult. In dynamic settings, temporal prediction can improve compression efficiency but may increase latency and propagate errors between frames. Evaluation is another concern because geometry and attribute quality require different metrics and should therefore be reported separately together with bitrate, complexity, and memory consumption.

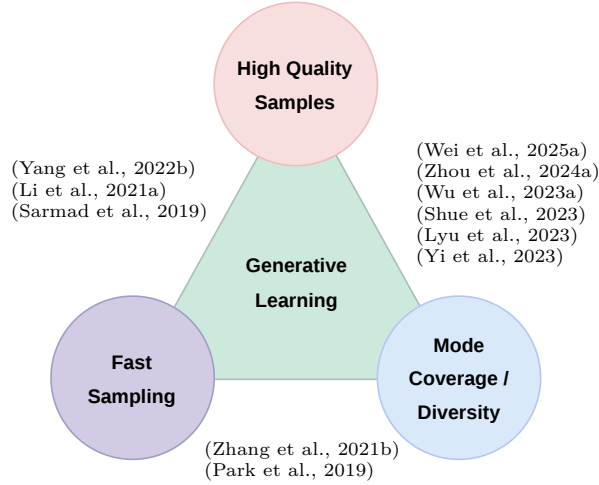


Figure 4: Representative generative families mapped across the speed–fidelity–coverage trilemma. GAN-based methods emphasize fast inference, diffusion models favor fidelity and coverage, and latent models balance efficiency with coverage. Placements illustrate qualitative tendencies rather than quantitative comparisons across tasks, datasets, output formats, or hardware.

Future work should develop lightweight and adaptive frameworks that support different compression modes, data components, and point cloud types within a unified pipeline. Joint methods should model dependencies between geometry and attributes more explicitly, while dynamic codecs require efficient temporal mechanisms with limited latency and error accumulation. Further advances are also needed in scalable entropy models, variable-rate coding, sparse computation, and standardized evaluation protocols. Reporting rate–distortion performance together with computational and perceptual criteria, as discussed in Section 7.2, would improve reproducibility and facilitate real-world deployment.

4.2 Generation

Point cloud generation synthesizes plausible 3D geometry either directly as unordered point sets or through structured and continuous representations from which point clouds or surfaces can be obtained, optionally with attributes such as color or normals. We categorize methods by their conditioning and input assumptions into three settings: **unconditional generation**, which samples shapes from noise or a learned latent space; **point cloud completion**, which infers missing geometry from partial, sparse, or noisy observations; and **weakly conditional generation**, which uses high-level or indirect cues such as category labels, text, images, sketches, or sensor-specific information. These task categories may use overlapping architectures, generative objectives, modalities, and output representations. These models provide geometric priors for 3D asset synthesis and editing, as well as ill-posed tasks such as denoising, upsampling, and completion, helping bridge the gap between incomplete or noisy observations and plausible high-fidelity geometry. Figure 4 illustrates the qualitative speed–fidelity–coverage trade-offs among major generative families.

Generation differs from 3D reconstruction in its objective. Generation produces plausible geometry from a learned distribution or weak conditioning signal, whereas reconstruction aims to recover the specific scene or object supported by one or more observations, often together with camera calibration. Reconstruction is discussed separately in Section 4.3.

4.2.1 Unconditional Generation

Unconditional generation learns a distribution of complete shapes and synthesizes new instances without observing a target object. Early point-based methods commonly used adversarial learning because it supports fast, single-pass sampling. SP-GAN (Li et al., 2021a) imposed a sphere-guided structural prior to encourage dense correspondence and part-aware manipulation. Such structural constraints help organize unordered outputs, but adversarial models remain sensitive to unstable training, mode collapse, and uneven point density, and restrictive priors may limit their ability to represent diverse shape topologies.

Recent methods increasingly adopt diffusion because progressive denoising offers stable optimization and broad shape coverage, although with higher sampling cost. FrePolad (Zhou et al., 2024a) used frequency-rectified latent diffusion to preserve global structure and fine details, while flow distillation enabled one-step generation with performance comparable to standard multi-step diffusion models (Wu et al., 2023a). Diffusion has also shifted from raw coordinates toward structured latent spaces: triplane models generate neural fields from compact 2D features (Shue et al., 2023), whereas surface-aware methods diffuse sparse semantic points and decode them into meshes (Lyu et al., 2023). These structured formats simplify modeling and support surface extraction, but latent or decoding errors may propagate to the final geometry.

Compared with adversarial generators, diffusion models generally provide more stable training, broader shape coverage, and stronger geometric refinement, but require substantially more computation during sampling. Adversarial methods remain preferable for fast generation, whereas diffusion is better suited to applications where fidelity and diversity are more important than inference speed.

4.2.2 Point Cloud Completion

Point cloud completion aims to recover a full 3D shape from an incomplete or noisy point cloud. Classical completion methods commonly rely on geometric regularities, symmetry, templates, or retrieved shapes, and their performance can degrade when these assumptions do not match the observed object. Learning-based methods mitigate some of these limitations by estimating data-driven shape priors, although their predictions remain dependent on the training distribution and may be unreliable for ambiguous or out-of-domain observations.

Early approaches mainly used direct encoder-decoder architectures. PCN (Yuan et al., 2018) introduced coarse-to-fine point-set prediction, while decoder mechanisms from FoldingNet, AtlasNet, and PU-Net (Yang et al., 2018b; Groueix et al., 2018; Yu et al., 2018a) influenced completion through folding, parametric surface generation, and point expansion. However, these methods originally addressed reconstruction, surface generation, or upsampling rather than completion itself. Such global-code decoders are comparatively simple, but compressing the observation into a single latent representation can discard local details and produce over-smoothed or category-average completions.

Later methods adopted hierarchical generation to better preserve structural organization. TopNet (Tchapmi et al., 2019) modeled point groupings through tree-structured decoding, RL-GAN-Net (Sarmad et al., 2019) encouraged realistic outputs through adversarial and reinforcement learning, and PF-Net (Huang et al., 2020) progressively reconstructed large missing regions using multi-scale fractal generation. These methods improve coarse-to-fine structural organization, but inaccuracies introduced at early generation stages can propagate through subsequent refinement.

Transformer-based methods further reduced reliance on a single global representation. PoinTr (Yu et al., 2021b) formulated completion as geometry-aware set-to-set translation, while SeedFormer and AnchorFormer (Zhou et al., 2022b; Chen et al., 2023h) used patch seeds and discriminative anchors to retain regional structure during progressive reconstruction. Compared with direct decoders, these methods provide stronger global reasoning and retain more regional information, but their token interactions and refinement stages generally introduce greater computational and memory costs.

A parallel multimodal branch combines partial point clouds with RGB evidence to reduce uncertainty in unseen regions. ViPC, XMFNet, and EGIInet (Zhang et al., 2021b; Aiello et al., 2022; Xu et al., 2024a) progressively improved image guidance, localized interaction, and cross-modal alignment, while MAENet and DuInNet (Liu et al., 2025d;h) further strengthened feature fusion and output uniformity. Such visual cues

are useful when missing geometry can be inferred from the corresponding image, but inaccurate viewpoints, occlusion, or domain mismatch may provide misleading information. Their additional image encoders and fusion modules also increase data and computational requirements.

PCDreamer (Wei et al., 2025a) exploited pretrained multi-view diffusion priors to generate visual constraints for the unobserved geometry and used a consolidation stage to reduce inconsistencies among generated views. Diffusion priors can enlarge the range of plausible completions and provide strong external shape knowledge, but they require multiple generation and consolidation operations, increasing inference cost and potentially introducing inconsistencies or hallucinated structures.

The main challenges remain the recovery of large missing regions, preservation of observed geometry, robustness to noise and uneven sampling, and ambiguity among multiple plausible completions. Practical models must also generalize from synthetic benchmarks to real sensor data, accommodate diverse categories and scales, and generate dense and uniform outputs suitable for downstream tasks.

4.2.3 Weakly Conditional Generation

Weakly conditional generation controls synthesis through high-level or indirect cues such as language, sketches, images, category labels, or sensor-specific information without fully specifying the target geometry. At the category level, PC-cGAN (Yang et al., 2022b) used a two-stage BranchGCN-based generator to synthesize shapes from class labels and improve control over the generated object class. Category labels provide stronger semantic control than unconditional sampling but do not specify instance-level geometry.

In this setting, 3D Gaussian splatting represents an asset using explicit point-like Gaussian primitives whose geometry, opacity, and appearance can be optimized through differentiable rendering. Its efficient rasterization and compatibility with point-cloud priors make it attractive for controllable asset generation, although rendering fidelity does not necessarily imply accurate surface geometry. GaussianDreamer (Yi et al., 2023) combined 2D and 3D diffusion priors with point cloud guidance to generate text-conditioned 3D Gaussian assets, while GaussianAnything (Lan et al., 2025b) used a point-cloud-structured latent space to synthesize editable surfel Gaussians from text or single images. Related diffusion frameworks support artistic stylization (Yang et al., 2026c;b), whereas sketch-and-text-guided diffusion directly generates colored point sets (Wu et al., 2023d). These methods improve controllability, but text remains geometrically ambiguous, sketches lack depth, and image guidance depends on viewpoint and visibility. Gaussian-based outputs also favor rendering and editing over explicit metric geometry.

Sensor-conditioned methods must additionally reproduce acquisition-specific patterns. RangeLDM (Hu et al., 2024) combined range-view latent diffusion with geometry-aware adversarial learning for LiDAR generation, upsampling, and inpainting, while Text2LiDAR (Wu et al., 2024e) used an equirectangular transformer for text-guided panoramic generation. Range-view processing reduces computational cost but may introduce projection distortions and dependence on sensor-specific configurations.

Overall, semantic cues provide flexible but weak geometric control, images and sketches add spatial evidence but remain viewpoint-dependent, and sensor-conditioned methods better preserve acquisition structure at the cost of specialized representations and additional computation.

4.2.4 Conclusion, Challenges, and Future Directions of Generation

Point cloud generation methods involve trade-offs among quality, efficiency, conditioning, and representation. GAN-based models provide fast inference but may suffer from unstable training, mode collapse, and irregular geometry (Li et al., 2021a; Yang et al., 2022b). Diffusion models often achieve strong geometric refinement and broad sample coverage but require heavier training and slower sampling unless distillation is used (Zhou et al., 2024a; Hu et al., 2024; Wei et al., 2025a). Transformer-based completion and multimodal fusion can improve global reasoning and reduce ambiguity, although they introduce additional computational cost and remain sensitive to sparsity, noise, and domain shift (Yu et al., 2021b; Aiello et al., 2022; Li et al., 2025k). The chosen 3D format also influences usability: implicit fields support continuous surfaces, while Gaussian and surfel primitives favor rendering and editing (Park et al., 2019; Chen & Zhang, 2019; Yi et al., 2023; Lan et al., 2025b).

Major challenges include the unordered and grid-free structure of point clouds, non-uniform density, and range-dependent LiDAR sparsity. Models must balance local surface fidelity with global plausibility, particularly under severe incompleteness, where strong priors may hallucinate geometry. Evaluation is also fragmented: CD and EMD dominate point-set completion, while other settings use heterogeneous perceptual, semantic, rendering, and sensor-specific metrics. Existing benchmarks do not consistently or directly measure topology, physical plausibility, sensor realism, controllability, and uncertainty. Generalization across CAD models, real scans, RGB-D data, LiDAR, and different noise patterns remains limited (Li et al., 2025k; Hu et al., 2024).

Future work should develop unified and efficient models that share geometric priors across generation, completion, denoising, inpainting, and conditional synthesis. Distilled samplers, sparse computation, lightweight decoders, shared latent spaces, and modular conditioning may reduce deployment cost (Wu et al., 2023a; Du et al., 2025; Wei et al., 2025a). Sensor-aware objectives, domain adaptation, uncertainty-aware generation, and broader evaluation protocols are also needed to improve robustness, controllability, and real-world applicability.

4.3 Reconstruction

3D reconstruction recovers a digital representation of an object or environment from images, depth measurements, point clouds, or combinations of these observations. The output may jointly model geometry and view-dependent appearance, or provide a surface suitable for geometric analysis. Accordingly, we distinguish between *scene-level reconstruction*, commonly evaluated through novel-view synthesis and mapping, and *surface reconstruction*, which seeks explicit or implicit geometry that can be processed independently of the input views (Farshian et al., 2023; Liu et al., 2025f; Xu et al., 2025b). This distinction is important because photorealistic rendering does not necessarily imply accurate depth, connectivity, or topology.

4.3.1 Scene-Level Reconstruction

Scene-level neural reconstruction is mainly based on *implicit neural fields* or *explicit neural primitives*. Both are relevant to point-cloud processing: neural fields can be sampled into point sets or surfaces, whereas 3D Gaussian Splatting (3DGS) represents a scene as an unordered set of point-like primitives with position, anisotropic extent, opacity, and appearance attributes.

Implicit Neural Fields Neural Radiance Fields (NeRF) model a scene as a continuous function that maps spatial coordinates to volume density and combines spatial coordinates with viewing directions to predict view-dependent color, with images generated through differentiable volume rendering (Mildenhall et al., 2021). This formulation supports high-quality view synthesis, but usually requires scene-specific optimization and repeated network evaluations along each camera ray.

Later methods address different limitations of the original model. Mip-NeRF and NeRF++ improve scale-aware sampling and unbounded-scene representation, while NeRF-W and BAD-NeRF handle appearance variation, blur, or camera-pose errors (Barron et al., 2021; Zhang et al., 2020a; Martin-Brualla et al., 2021; Wang et al., 2023d). These methods improve robustness under particular capture conditions, but generally add optimization variables or computational overhead.

Efficiency-oriented methods reduce repeated MLP evaluations through compact encodings, tensor factorization, or accelerated sampling. InstantNGP, TensoRF, and NeRFacc are representative examples (Müller et al., 2022; Chen et al., 2022a; Li et al., 2023d). Their benefits are not identical: faster training does not necessarily imply lower storage or faster final rendering.

Sparse-view methods such as FreeNeRF regularize the optimization of an individual scene, whereas PixelNeRF and IBRNet learn image-conditioned representations across multiple scenes (Yang et al., 2023; Yu et al., 2021a; Wang et al., 2021e). Generalizable methods reduce per-scene optimization but are more sensitive to distribution shifts. Dynamic approaches such as Nerfies and D-NeRF introduce deformation models to represent temporal changes (Park et al., 2021; Pumarola et al., 2021), although they require additional assumptions about motion smoothness or canonical structure.

Table 6: Comparison of representative neural scene-reconstruction approaches and their principal trade-offs in fidelity, efficiency, generalization, and dynamic modeling.

Scene Formulation	Category	Representative Approaches	Main Trade-off
Neural Field	Base / Fidelity	NeRF (Mildenhall et al., 2021), Mip-NeRF (Barron et al., 2021), NeRF-W (Martin-Brualla et al., 2021)	Continuous and high-fidelity modeling, but usually with scene-specific and computationally expensive optimization.
Neural Field	Efficiency / Acceleration	Instant-NGP (Müller et al., 2022), TensorRF (Chen et al., 2022a), NerfAcc (Li et al., 2023d)	Faster training or rendering, with representation-specific memory, approximation, and discretization costs.
Neural Field	Sparse-View / Generalizable	FreeNeRF (Yang et al., 2023), PixelNeRF (Yu et al., 2021a), IBRNet (Wang et al., 2021e)	Fewer input views or less per-scene optimization, but greater dependence on regularization and training distributions.
Neural Field	Dynamic / 4D	Nerfies (Park et al., 2021), D-NeRF (Pumarola et al., 2021)	Temporal modeling of non-rigid scenes, at the cost of additional motion assumptions, memory, and optimization complexity.
Gaussian Primitives	Geometry / Rendering Quality	SuGaR (Guédon & Lepetit, 2024), GSDF (Yu et al., 2024a), Mip-Splatting (Yu et al., 2024b)	Better surface alignment or rendering quality, but with additional constraints, representations, or optimization stages.
Gaussian Primitives	Compression / Efficiency	LightGaussian (Fan et al., 2024b), CompGS (Navaneet et al., 2024)	Reduced storage and faster rendering, with a possible loss of fine geometry or view-dependent appearance.
Gaussian Primitives	Sparse-View / Generalizable	FSGS (Zhu et al., 2024d), PixelSplat (Charatan et al., 2024)	Reconstruction from fewer views, but stronger reliance on geometric regularization or learned cross-scene priors.
Gaussian Primitives	Dynamic / Online	4DGS (Wu et al., 2024a), Deformable 3DGS (Yang et al., 2024d), GS-SLAM (Yan et al., 2024)	Efficient dynamic rendering or online mapping, but sensitivity to motion estimation, pose drift, and primitive growth.

Overall, scene-specific NeRF variants retain stronger adaptation to individual environments but require optimization for each scene. Generalizable models reduce or eliminate per-scene optimization at the cost of greater dependence on the training distribution, while sparse-view and dynamic models introduce additional priors without fully resolving ambiguities in unseen geometry.

Explicit Neural Primitives In contrast to continuous neural fields, 3DGS represents a scene using explicit anisotropic Gaussian primitives optimized through differentiable rasterization (Kerbl et al., 2023). It generally provides faster rendering and more direct local manipulation than NeRF, but the primitives are optimized mainly through image-space losses and do not inherently form a connected surface.

Geometry- and rendering-oriented methods introduce additional constraints on primitive placement or filtering. SuGaR encourages Gaussians to align with an underlying surface, GSDF combines Gaussian primitives with a signed-distance field, and Mip-Splatting reduces scale-dependent rendering artifacts (Guédon & Lepetit, 2024; Yu et al., 2024a;b). These methods address different failure modes: surface regularization improves geometric consistency, whereas anti-aliasing primarily improves projected image quality.

Compression methods such as LightGaussian and CompGS reduce the storage and rendering cost of large Gaussian sets through pruning or quantization (Fan et al., 2024b; Navaneet et al., 2024). Aggressive compression, however, may remove primitives needed to represent fine geometry or view-dependent appearance.

Sparse-view methods introduce geometric or learned priors, dynamic models attach deformation fields to Gaussian primitives, and Gaussian-based SLAM systems jointly optimize camera poses and scene structure (Zhu et al., 2024d; Charatan et al., 2024; Wu et al., 2024a; Yang et al., 2024d; Yan et al., 2024). Large-scale approaches use scene partitioning or additional LiDAR constraints (Lin et al., 2024a; Jiang et al., 2024a). Their main failure modes include pose drift, uncontrolled primitive growth, temporal inconsistency, and sensor-calibration errors.

Compared with NeRF, 3DGS generally provides lower rendering latency and easier local editing, but can require substantial memory and lacks inherent topology. NeRF is often more suitable for continuous offline scene modeling, whereas 3DGS is attractive for interactive rendering and online mapping. Neither family guarantees accurate geometry in sparse, reflective, textureless, or incorrectly calibrated regions.

The categories in Table 6 may overlap; each method is assigned according to its principal contribution.

4.3.2 Surface Reconstruction

Surface reconstruction recovers geometry that can be processed independently of the input viewpoints. We organize the reviewed methods according to how the surface is recovered: directly from point samples, through explicit surface prediction, using implicit surface functions, or with Gaussian primitives coupled to additional geometric constraints. These categories describe reconstruction strategies and output formulations rather than the general representation taxonomy introduced earlier. They differ in their assumptions about input quality, connectivity, topology, smoothness, and learned geometric priors.

Surface Reconstruction from Point Samples Classical methods such as Poisson reconstruction and Ball-Pivoting construct surfaces directly from sampled points (Kazhdan et al., 2006; Bernardini et al., 2002). Poisson reconstruction often produces smooth and watertight outputs, but may oversmooth details or fill unobserved regions. Ball-Pivoting and related local triangulation methods preserve the observed geometry more directly, but remain sensitive to noise, irregular sampling, incomplete coverage, and missing or inaccurate normals. Thus, these methods depend strongly on the quality and density of the input samples and generally cannot infer reliable geometry in substantially unobserved regions.

Explicit Surface Reconstruction Explicit-surface approaches produce surface geometry through parametric patches or by deforming an initial mesh. AtlasNet represents a shape as a collection of parametric surface patches, while Pixel2Mesh deforms a template mesh using image features (Groueix et al., 2018; Wang et al., 2018b). These formulations can provide surfaces that are more directly usable in graphics or simulation, but template-based methods may restrict topology and patch-based models may produce seams, overlaps, or self-intersections.

Implicit Surface Reconstruction DeepSDF and Occupancy Networks describe surfaces through continuous signed-distance or occupancy functions (Park et al., 2019; Mescheder et al., 2019). They support varying topology and are not limited by a fixed output resolution. However, training and mesh extraction require many spatial queries, and thin structures may be smoothed when observations or sampling density are insufficient.

Gaussian-Assisted Surface Reconstruction Standard 3DGS does not provide explicit topology. Surface-oriented approaches therefore align Gaussian primitives with geometry or couple them with implicit fields. SuGaR constrains Gaussians toward a surface, while GSDF and 3DGSR combine Gaussian rendering with signed-distance representations (Guédon & Lepetit, 2024; Yu et al., 2024a; Lyu et al., 2024b). These approaches retain efficient rendering and improve geometric coherence, but require additional memory and optimization. Errors in camera poses or Gaussian placement may also propagate to the extracted surface.

In summary, surface reconstruction from point samples directly uses the observed geometry but remains strongly dependent on sampling density, coverage, noise, and normal quality. Explicit-surface approaches produce surface patches or meshes that are more directly usable, but may impose topology constraints or introduce seams, overlaps, and self-intersections. Implicit surfaces represent continuous geometry without requiring a fixed topology, although they incur substantial spatial-query and surface-extraction costs. Gaussian-assisted approaches retain efficient rendering but require additional geometric constraints to support reliable surface extraction.

4.3.3 Conclusion, Challenges, and Future Directions of Reconstruction

Scene-level and surface reconstruction optimize related but different objectives. Neural fields emphasize continuous appearance modeling, while Gaussian primitives prioritize efficient explicit rendering. Surface reconstruction strategies instead differ in whether geometry is recovered directly from point samples, represented through explicit surface patches or meshes, modeled through implicit functions, or extracted using Gaussian primitives with additional geometric constraints. The appropriate formulation therefore depends on whether the main objective is photometric fidelity, surface accuracy, completeness, topology, latency, or memory efficiency.

Sparse observations, inaccurate camera poses, occlusions, reflective materials, and dynamic content remain common sources of error. Point-sample outputs lack inherent connectivity, explicit-surface methods may impose restrictive topology, and implicit fields require costly sampling and surface extraction. Gaussian-assisted approaches additionally require geometric regularization before reliable surfaces can be extracted. Large-scale and temporal settings further introduce memory growth, pose drift, and consistency problems.

Hybrid explicit-implicit representations may provide a practical balance between fast Gaussian rendering and continuous geometric regularization (Yu et al., 2024a; Lyu et al., 2024b). Adaptive primitive allocation, pruning, and quantization may reduce memory costs, while depth or LiDAR measurements can provide stronger geometric constraints when calibration is reliable. Dynamic reconstruction may further benefit from jointly modeling motion and temporally consistent surfaces.

4.4 Registration

3D point cloud registration refers to the problem of estimating a spatial transformation, typically a rigid transformation composed of rotation and translation, that aligns a source point cloud with a target point cloud in a common coordinate system while preserving geometric consistency. Over the past decades, substantial progress has been made toward this goal, and many solutions, including traditional and deep learning-based approaches, have been proposed (Zhang et al., 2024j; Yang et al., 2024b). Registration approaches can be broadly categorized into traditional geometry-based methods and modern learning-based approaches, as summarized in Table 7. In the following, we discuss both traditional and learning-based registration strategies, highlighting their main principles, strengths, and limitations.

- **Classical Registration:** This category encompasses geometry-based methods for rigid registration of 3D point clouds. Local refinement methods are primarily centered on the Iterative Closest Point (ICP) algorithm (Besl & McKay, 1992) and its improved variants (Rusinkiewicz & Levoy, 2001; Yuan et al., 2025). These methods rely on explicit point correspondences or distance-minimization principles to estimate rigid transformations and are well suited to scenarios with moderate overlap, limited noise, and sufficiently accurate initialization. Their strengths include simplicity, interpretability, and efficiency in well-conditioned cases.

In contrast, global and robust geometry-based methods aim to reduce dependence on accurate initialization and improve robustness to incorrect correspondences, outliers, and partial overlap (Zhou et al., 2016; Bustos & Chin, 2017; Stechschulte et al., 2019; Wang et al., 2024b; Di Lauro et al., 2024). Although these methods generally provide a wider convergence range than local ICP-based refinement, feature matching, correspondence verification, or transformation-hypothesis evaluation may introduce additional computational cost. Their performance also remains sensitive to low overlap, repetitive geometry, and the quality of the extracted geometric features (Denayer et al., 2024).

- **Learning-based Registration:** Learning-based approaches leverage data-driven models to extract robust feature representations and estimate rigid transformations from point cloud data. Supervised methods learn point correspondences, overlap information, or global geometric representations from labeled data, enabling accurate alignment under noise and partial overlap when the training and test distributions are sufficiently related (Ao et al., 2023; Wang et al., 2024g; Wu et al., 2023c; Yao et al., 2024).

In contrast, unsupervised and self-supervised approaches optimize alignment objectives without explicit ground-truth transformations, often using geometric consistency or probabilistic modeling to address challenging scenarios such as low overlap, partial observations, and large-scale scenes (Mei et al., 2023; Liu et al., 2024c; Zheng et al., 2024a; Jiang et al., 2024c; Yuan et al., 2023). Their main principle is to learn representations that capture local and global geometric structures while reducing dependence on labeled transformations. Learning-based methods can improve robustness under conditions represented during training, but important limitations remain, including dependence on the training data distribution, higher computational and memory costs, and sometimes limited interpretability compared with classical geometry-based methods.

Table 7: Comparison of classical geometry-based and learning-based approaches for rigid point cloud registration. The methods are organized by registration family, subcategory, and compared in terms of their main mechanisms, representative methods, advantages, limitations, and computational trade-offs.

Family	Subcategory	Main Mechanism	Representative Methods	Advantages	Limitations and Computational Trade-offs
Classical geometry-based	Local refinement	Iterative correspondence assignment and local distance minimization	ICP (Besl & McKay, 1992) and its improved variants (Rusinkiewicz & Levoy, 2001; Yuan et al., 2025)	Simple, interpretable, efficient, and effective after sufficiently accurate initialization	Sensitive to poor initialization, low overlap, outliers, and repetitive geometry; may converge to incorrect local minima
Classical geometry-based	Global and robust alignment	Geometric feature matching, robust correspondence verification, or transformation-hypothesis evaluation	Fast global and robust registration methods (Zhou et al., 2016; Bustos & Chin, 2017; Stechschulte et al., 2019; Wang et al., 2024b; Di Lauro et al., 2024)	Reduces dependence on accurate initialization and improves robustness to incorrect correspondences	Feature extraction, correspondence verification, or hypothesis search may be computationally expensive; sensitive to weak geometric features and very low overlap
Learning-based	Supervised	Learning point correspondences, overlap confidence, descriptors, or global transformation representations from labeled data	Buffer (Ao et al., 2023), end-to-end registration (Wang et al., 2024g), RORNet (Wu et al., 2023c), and PARE-Net (Yao et al., 2024)	Can learn robust local and global geometric features and improve alignment under noise and partial overlap	Requires labeled training data and may be sensitive to domain, sensor, density, or scene-distribution shifts; generally introduces additional training and inference cost
Learning-based	Unsupervised and self-supervised	Geometric consistency, probabilistic alignment, or reconstruction-based objectives without explicit ground-truth transformations	UDPReg (Mei et al., 2023), progressive distance extension (Liu et al., 2024c), RegFormer (Zheng et al., 2024a), GTINet (Jiang et al., 2024c), and PointMBF (Yuan et al., 2023)	Reduces dependence on labeled transformations and can exploit large collections of unlabeled point clouds	Training objectives may admit inaccurate or degenerate alignments; higher optimization complexity and limited interpretability may remain

The comparison in Table 7 shows that no single registration family is universally preferable. Local classical methods are efficient and interpretable but require sufficiently accurate initialization, whereas global geometry-based methods improve robustness at the cost of more expensive feature matching or hypothesis evaluation. Supervised learning-based methods can learn stronger geometric representations but depend on labeled data and the similarity between training and test distributions. Unsupervised and self-supervised methods reduce annotation requirements, although their geometric objectives may produce ambiguous or degenerate solutions under low overlap or repetitive structures. In practice, hybrid pipelines that combine learned feature extraction or global alignment with geometrically constrained local refinement can provide a useful balance between robustness, interpretability, and computational efficiency.

4.4.1 Conclusion, Challenges, and Future Directions of Registration

Point cloud registration remains essential to 3D vision, with classical methods favoring efficiency under well-conditioned settings and learning-based approaches improving alignment through learned geometric features. However, registration remains difficult under low overlap, noise, outliers, partial observations, repetitive geometry, and poor initial poses. Classical methods may fail under these conditions, whereas learning-based methods can require large training datasets, substantial computational resources, and representative training distributions. Future research should therefore focus on developing registration methods that are more accurate, efficient, robust to real-world noise and low-overlap scenarios, less dependent on labeled data, and easier to interpret and deploy in large-scale 3D scenes.

4.5 Pose Estimation

In this section, we discuss pose estimation in 3D space. For rigid objects, pose estimation usually aims to recover the object’s spatial transformation with respect to the camera or reference coordinate frame, whereas human pose estimation is commonly formulated in terms of articulated joints, body structure, or mesh recovery (Liu et al., 2024a). We divide this topic into two main parts: object pose estimation and human pose estimation, which are discussed separately in the following subsections.

4.5.1 Object Pose Estimation

Object pose estimation involves recovering an object’s 3D translation and rotation with respect to the camera or reference coordinate frame, which is commonly formulated as a rigid 6-DoF pose estimation problem in $SE(3)$ (Xiang et al., 2017). In many applications, once an object has been identified, its precise pose must also be estimated. Depending on the formulation, the task may include estimating object size or scale, particularly in category-level pose-and-size estimation settings. However, point clouds generated from depth images or LiDAR are often noisy, and real-world settings typically provide only partial observations, which

constitute a major challenge for accurate pose estimation (Wang et al., 2019a). Point density may vary across different parts of the object, while occlusion can result in missing surface regions (He et al., 2020b). In addition, geometric or rotational symmetries may make multiple poses observationally equivalent, leading to ambiguity in pose estimation (Hodan et al., 2020).

In this survey, we organize object pose estimation methods according to their generalization scope into three groups: instance-level, category-level, and novel-object-level methods. These categories differ in their assumptions regarding the target objects, the priors available during training and inference, and their ability to generalize beyond the training set. Their main characteristics and trade-offs are summarized in Table 8.

- **Instance-level:** In instance-level settings, the goal is to estimate the pose of a known object instance that is represented during training. Many works in this area learn instance-specific RGB-D or point-cloud features for accurate pose estimation (He et al., 2021b). Some methods also reduce their dependence on real annotated data by synthesizing point-cloud segments from available 3D object models (Gao et al., 2021). The strong object-specific prior generally supports accurate pose estimation, but it limits direct generalization to previously unseen objects.
- **Category-level:** In category-level settings, the goal is to estimate the pose of unseen object instances belonging to a known semantic category. The primary challenge is intra-category variation, namely differences in shape, scale, and appearance among objects belonging to the same class (Zou et al., 2024). A common strategy is to map observed objects into a shared canonical space or normalized object coordinate representation, where 9-DoF pose-and-size estimation can be performed by estimating rotation, translation, and scale (Wang et al., 2019b). Other category-level methods further improve robustness by learning stronger shape, geometric, or contextual priors across object instances (Chen et al., 2020b; Li et al., 2025h). Although these methods provide broader generalization than instance-level approaches, their performance may degrade when test objects differ substantially from the learned category-level priors.
- **Novel-object-level:** In novel-object-level settings, the target instance has not been observed during task-specific training, and in some formulations the test objects may also belong to categories that are absent from the training set. A CAD or object model is often assumed to be available at test time, although recent methods relax this requirement by using reference observations or model-free formulations (Wen et al., 2024; Lee et al., 2025c). The main emphasis is on generalization and robustness to novel objects, clutter, occlusion, and limited supervision (Cai et al., 2022b; Chen et al., 2024a; Caraffa et al., 2024). Because these methods reduce the strong object-specific assumptions of instance-level approaches, they commonly introduce a trade-off between broad generalization, instance-specific accuracy, and inference cost.

Table 8: Comparison of instance-level, category-level, and novel-object settings for rigid object pose estimation. The settings are compared in terms of representative strategies, advantages, limitations, and computational trade-offs.

Setting	Inference Target	Typical Prior	Representative Strategies	Advantages	Limitations and Computational Trade-offs
Instance-level	Known object instances	Object-specific training data and, in many cases, a 3D object model	Instance-specific RGB-D or point-cloud feature learning and synthetic point-cloud generation (He et al., 2021b; Gao et al., 2021)	High accuracy for known objects and effective use of detailed object geometry	Requires object-specific data or retraining, provides limited transfer to novel objects, and may require costly annotation or synthetic-data generation
Category-level	Unseen instances from known semantic categories	Category-level shape, canonical-coordinate, or contextual priors	Canonical-space mapping, normalized object coordinates, and shared geometric or contextual priors (Wang et al., 2019b; Chen et al., 2020b; Zou et al., 2024; Li et al., 2025h)	Generalizes across different instances of the same category and can jointly estimate object pose and size	Sensitive to large intra-category variation, symmetry, scale ambiguity, and mismatch with the learned category prior; commonly requires category-level supervision
Novel-object-level	Previously unseen objects and, in some settings, objects from unseen categories	CAD models, reference observations, or pretrained geometric and visual features	Viewpoint encoding, model or reference matching, zero-shot estimation, and foundation-model features (Cai et al., 2022b; Wen et al., 2024; Chen et al., 2024a; Caraffa et al., 2024; Lee et al., 2025c)	Reduces or removes the need for object-specific retraining and supports more open-world deployment	May require CAD models or reference acquisition; matching and iterative refinement can increase inference time, while symmetry and partial observations introduce greater ambiguity

The progression from instance-level to category-level and novel-object-level estimation reflects a gradual reduction in object-specific assumptions. Instance-level methods encode detailed prior knowledge through object-specific training and therefore generally provide the highest accuracy for a fixed set of objects. Category-level methods replace this prior with shared canonical or geometric representations, improving generalization across instances while introducing sensitivity to intra-category variation. Novel-object-level methods further reduce task-specific training requirements by relying on CAD models, reference observations, or pretrained visual and geometric representations. This broader generalization is commonly accompanied by increased matching or refinement costs and by greater ambiguity under symmetry, occlusion, and limited observations. Consequently, no setting is universally preferable: the appropriate formulation depends on the expected object inventory, the availability of test-time priors, computational constraints, and the required level of generalization.

4.5.2 Human Pose Estimation

Human pose estimation aims to localize and model the articulated configuration of the human body by estimating the spatial positions of key joints or body parts, typically in 2D or 3D (Kappan et al., 2026; Galaaoui et al., 2025). Unlike object pose estimation, which generally focuses on recovering a rigid transformation, human pose estimation addresses a highly non-rigid and deformable structure with complex kinematic constraints, self-occlusion, and large variations in body motion. This distinction introduces different challenges and requires specialized representations and learning strategies. The task is central to numerous applications, including human–computer interaction, motion capture, action recognition, animation, sports analytics, and healthcare. Comprehensive surveys have reviewed the evolution of human pose estimation from classical pictorial structure models to modern deep learning approaches (Liu et al., 2024d; Kappan et al., 2026).

In this subsection, we focus specifically on methods that directly process point clouds or LiDAR-derived point sets. These methods can be broadly categorized according to how they model the articulated body and exploit temporal or structural priors: frame-based joint estimation, temporal point-cloud sequence modeling, and joint pose-and-body-shape estimation, although some methods may span multiple categories. Sensor type, such as dense body scans, event point clouds, or sparse LiDAR observations, is treated as an input characteristic rather than as a separate methodological category. The main paradigms and their trade-offs are summarized in Table 9.

- **Frame-based joint estimation** methods estimate human joint locations from an individual point cloud or a limited observation without explicitly modeling long-term temporal dependencies. Early methods directly learn representations from raw 3D points and regress joint coordinates (Zhou et al., 2020b; Chen et al., 2022b). More recent approaches adapt point-cloud architectures to sparse and irregular LiDAR observations, including LidPose (Weng et al., 2023; Kovács et al., 2024), LPFormer (Ye et al., 2024), and density-aware pretraining methods such as DAPT (An et al., 2025a). These methods are suitable for real-time inference, but their accuracy may degrade when important body regions are missing or severely occluded.
- **Temporal point-cloud sequence modeling** methods aggregate information across multiple frames to capture motion continuity and recover joints that may be ambiguous or invisible in a single observation. SPiKE explicitly models point-cloud sequences for human pose estimation (Ballester et al., 2025), while LiDARCapV2 incorporates temporal and human–object interaction information into LiDAR-based pose recovery (Zhang et al., 2024f). Temporal context can improve consistency and robustness under sparse observations, although it introduces additional memory and computational cost and may be affected by inaccurate temporal association or abrupt motion.
- **Joint pose and body shape estimation** methods recover both skeletal joint locations and a more complete representation of body geometry, such as a human mesh or parametric body model. PointHPS adopts a cascaded framework for jointly estimating human pose and shape from point clouds (Cai et al., 2023), while LiveHPS addresses human pose and shape recovery from scene-level LiDAR observations in unconstrained environments (Ren et al., 2024b). Explicit body priors help reconstruct occluded regions and produce more complete outputs, but they also increase model

complexity and may introduce errors when clothing, body proportions, or observed poses differ substantially from the assumed body model.

Table 9: Comparison of point-cloud-based human pose estimation paradigms. The methods are organized by their use of frame-level, temporal, and body-model information and compared according to their inputs, prediction targets, structural priors, advantages, limitations, and computational trade-offs.

Paradigm	Input and Temporal Information	Prediction Target / Prior	Representative Methods	Advantages	Limitations and Computational Trade-offs
Frame-based joint estimation	Individual body scans, event point clouds, or sparse LiDAR frames; no explicit long-term temporal modeling	3D joint coordinates; point-based, sensor-aware, or density-aware geometric features	Direct point-cloud regression (Zhou et al., 2020b; Chen et al., 2022b), LidPose (Weng et al., 2023; Kovics et al., 2024), LPPFormer (Ye et al., 2024), and DAPT (An et al., 2025a)	Low latency, simpler inference, and direct use of point-level geometric information	Sensitive to missing body regions, self-occlusion, point density, and sensor scan patterns; limited ability to recover joints that are not visible in a single frame
Temporal point-cloud sequence modeling	Consecutive point-cloud or LiDAR frames with explicit temporal context	3D joints with motion, temporal-consistency, or interaction priors	SPiKE (Ballester et al., 2025) and LiDARCapV2 (Zhang et al., 2024f)	Improves motion consistency and can recover information missing from individual sparse frames	Requires additional memory and computation; assumes reliable temporal association and may propagate tracking errors across frames
Joint pose and body shape estimation	Point clouds or scene-level LiDAR observations; temporal information may also be used	3D joints together with body mesh or parametric body shape; explicit body-model prior	PointHPS (Cai et al., 2023) and LiveHPS (Ren et al., 2024b)	Produces a more complete articulated body representation and regularizes occluded or unobserved regions	Higher architectural or optimization complexity; sensitive to mismatch between the assumed body model and clothing, body proportions, or severe occlusion

These paradigms differ mainly in how they balance inference efficiency, temporal robustness, and output completeness. Frame-based methods are computationally attractive for real-time applications but must infer the articulated pose from a single incomplete observation. Temporal methods improve consistency by aggregating evidence across frames, although this benefit comes with additional memory, latency, and dependence on reliable temporal correspondence. Joint pose-and-shape approaches introduce stronger structural priors and provide a complete body representation, but they are generally more computationally demanding and may be affected by body-model mismatch. From a representation–architecture perspective, sparse LiDAR inputs benefit from density-aware point encoders, temporal observations favor sequence or attention-based aggregation, and full-body reconstruction requires architectures that combine local point evidence with global kinematic or parametric body constraints.

4.5.3 Conclusion, Challenges, and Future Directions of Pose Estimation

Pose estimation remains important in 3D vision and encompasses both the rigid rotation and translation of objects and the articulated configurations of the human body. Despite recent progress, pose estimation remains difficult under noise, occlusion, partial observations, and missing data. Object pose estimation is further complicated by shape variation and object symmetries, whereas human pose estimation faces challenges such as body motion, self-occlusion, and sparse LiDAR measurements. Future research should focus on developing methods that are more accurate, robust in real-world scenes, generalizable to unseen objects and complex human poses, less dependent on large labeled datasets, and efficient enough for real-time deployment.

5 Foundation Models and Scene Understanding

Foundation models and scene understanding are discussed together because both move point cloud research beyond conventional geometry-focused perception toward semantic and multimodal 3D understanding. This section reviews foundation models (Section 5.1) and scene understanding (Section 5.2), emphasizing how point clouds are increasingly integrated with images, language, and large pretrained models for open-vocabulary recognition, grounding, captioning, question answering, and reasoning. Within the foundation models discussion, we first discuss parameter-efficient fine-tuning techniques, which have become essential for adapting large pre-trained 3D models to downstream tasks with minimal computational overhead, and then present a comprehensive taxonomy of 3D foundation models organized according to their functional role in the 3D understanding pipeline.

5.1 Foundation Models

Foundation Models (FMs) are large-scale pretrained models that learn general-purpose representations from diverse data and can be reused across tasks with minimal fine-tuning. Rather than optimizing for a single objective, FMs capture broad structural, semantic, or reasoning priors that transfer effectively to downstream problems (Deng et al., 2025c).

In 3D vision, FMs extend beyond models trained on 3D data to include those pretrained on 2D images and language, whose representations can be leveraged for 3D perception, representation learning, and reasoning. Three classes of FMs, distinguished by their input modalities and training objectives, have become particularly influential.

Vision Language Models (VLMs) jointly model visual and textual information, enabling open-vocabulary perception, semantic grounding, and cross-modal alignment (Xu et al., 2023c; Luan et al., 2025). Large Language Models (LLMs) provide reasoning, planning, and task decomposition capabilities, increasingly used to reason over perceptual outputs and enable flexible interaction with 3D environments (Xu et al., 2025a). Vision Foundation Models offer strong visual and geometric priors without language, providing transferable cues for objectness, boundaries, and structure (Jung et al., 2025a).

While prior surveys have cataloged applications of 2D or language FMs to 3D tasks (Ma et al., 2024b; Thengane et al., 2025; Shao et al., 2025; Liu et al., 2025g), the relationship between point clouds and FMs has remained conceptually fragmented. To bring clarity, we explicitly distinguish between the source modality of the FM, the adaptation paradigm, and its functional role in the 3D pipeline.

An orthogonal consideration to source modality is the adaptation strategy. To efficiently adapt these large pretrained models to downstream tasks, parameter-efficient fine-tuning (PEFT) techniques have emerged as a practical solution (Ding et al., 2023). Unlike full fine-tuning, which updates all parameters and is computationally expensive, PEFT methods freeze most pretrained parameters and adapt the model by updating a small parameter subset or introducing lightweight trainable modules (Han et al., 2024b; Xu et al., 2026c). After covering PEFT as a training paradigm, we organize 3D FMs by their four functional roles in relation to point clouds (Table 10). These functional roles are not mutually exclusive, and each method is assigned according to its dominant role in the 3D pipeline.

5.1.1 Parameter-Efficient Fine-Tuning for Point Clouds

Full fine-tuning of pretrained models is resource-intensive. Parameter-Efficient Fine-Tuning (PEFT) reduces adaptation cost by freezing most pretrained parameters and updating only a small parameter subset or lightweight task-specific modules (Tang et al., 2024c). Recent PEFT methods for point clouds can be organized into three main categories.

Prompt Tuning methods insert learnable prompt tokens into each transformer layer. IDPT introduces instance-aware dynamic prompts (Zha et al., 2023), while DAPT constructs internal prompts capturing instance-specific features (Zhou et al., 2024c).

Adapter-Based Tuning methods couple a frozen backbone with lightweight trainable adaptation components. In the broad taxonomy adopted here, these components may operate on spatial features, graph structures, intermediate feature sequences, low-rank weight updates, or spectral representations. **Spatial adapters** such as Point-PEFT combines a Point-prior Prompt with a Geometry-aware Adapter that aggregates features within spatial neighborhoods, achieving better performance than full fine-tuning with only 5 % of trainable parameters (Tang et al., 2024c). Graph Feature Tuning (GFT) learns a dynamic graph from tokenized inputs using a lightweight graph convolution network (Dhakal et al., 2026). Side Token Adaptation on a neighborhood Graph (STAG) (Furuya, 2026) employs a graph convolutional side network operating in parallel with a frozen backbone, reducing temporal and spatial costs. **Mamba based adapters** have recently emerged as an efficient alternative: Point Mamba Adapter (PMA) (Zha et al., 2025b) constructs an ordered feature sequence from all layers of the pretrained model and leverages the Mamba architecture to fuse complementary semantics across layers, promoting comprehensive point cloud understanding with minimal parameters. **Low-rank adapters** inject low-rank matrices into pretrained weights. PointLoRA combines

Table 10: A comprehensive taxonomy of 3D foundation models, categorized according to their functional role in the 3D understanding pipeline.

Functional Role	Approach	Methods	Contribution	Open Problems
Representation-Defining FMs	Geometry-centric 3D FMs (GFMs)	DUST3R (Wang et al., 2024e), MUST3R (Cabon et al., 2025), Fast3R (Yang et al., 2025b), VGGT (Wang et al., 2025g), Distill3R (Leblanc & Poullis, 2026)	End-to-end geometry & camera pose learning from multi-view images	Generalization across camera configs; large-scene scalability
	Point Cloud Adapted Vision-based FMs	Seal (Liu et al., 2023e), Uni-Adapter (Tamjidi et al., 2025)	Adapts 2D vision FMs to 3D, reducing annotation/training needs	Limited 3D geometric reasoning; fine-grained structure loss
	3D Vision-Language FMs	AiDe (Wang & Czarnecki, 2025), 3D-LLaVA (Deng et al., 2025a), GreenPLM (Tang et al., 2025b), FoundationMorph (Pan et al., 2025a), TotalFM (Yamamoto & Kikuchi, 2026), ProxyTransformation (Peng et al., 2025), LF-3DVG (Wu et al., 2025b), FAC++ (Liu et al., 2025b)	Aligns 3D features with language for open-vocabulary understanding	Depends on pretrained 2D VLMs; struggles with OOD/noisy data
	Action- or Embodiment-Centric 3D FMs	Pointvla (Li et al., 2026a), 3D CAVLA (Bhat et al., 2025), Geovla (Sun et al., 2025a), Spatialvla (Qu et al., 2025a), Evo-0 (Lin et al., 2025)	Injects 3D representations into VLA models for embodied reasoning	High computational cost; large sim-to-real gap
Representation-Guiding FMs	-	GFS-VL (An et al., 2025b), EAIL (Zhang et al., 2025b), PCN-TI (Song et al., 2023b), Ges3ViG (Mane et al., 2025), Text2Loc++ (Xia et al., 2025b), GSP (Chen et al., 2025a)	FMs as external guidance; core 3D encoder remains unimodal	Guidance-only; no integrated 3D-language representation
	2D VLMs As Semantic Priors	SemAbs (Ha & Song, 2022), Fine-tuning with UAPs (Rudner et al., 2024)	Inference-time semantic guidance without 3D-language training	External only; cannot learn joint 3D-text features
	Vision FMs As Semantic Priors	BALViT (Hindel et al., 2025), Partslip (Liu et al., 2023d), Vipoccc (Feng et al., 2025a), HFIT (Guo et al., 2026b)	Distills 2D knowledge via pseudo-labels to reduce annotation cost	Projection information loss; significant 2D-to-3D domain gap
Reasoning-Centric FMs	-	LARM (Li et al., 2025n), ZeroKey (Gong et al., 2025), Reason3D (Huang et al., 2025c), 3D-AffordanceLLM (Chu et al., 2025), SeqAfford (Yu et al., 2025a), LiDAR-LLM (Yang et al., 2025d), RadarLLM (Lai et al., 2025), PULLM (Zhang et al., 2025g), 3D-R1 (Huang et al., 2025f)	LLMs as high-level reasoners for instruction-following & QA tasks	High LLM computational cost; no point-level feature learning
Point-Cloud-Scaffolded FMs	-	3D-LOTUS++ (Garcia et al., 2025), Gpt4scene (Qi et al., 2025), VisionCube (Wang et al., 2025d)	Uses reconstructed 3D structure as scaffolding for spatial reasoning	Necessity of point clouds debated; vision-only alternatives may suffice

low-rank adaptation with multi-scale token selection. It embeds LoRA layers within the most parameter-intensive components of point cloud transformers. This approach achieves competitive performance with only 3.43 % of trainable parameters (Wang et al., 2025l). **Spectral-domain adapters** transfer tuning from the spatial to spectral domain. Point cloud Graph Spectral Tuning (PointGST) introduces a lightweight Point Cloud Spectral Adapter (PCSA) to fine-tune parameters in the spectral domain, de-correlating confusion among tokens (Liang et al., 2025a).

Reparameterization-Based Tuning methods replace dense update weight matrices with sparse approximations. Monarch Sparse Tuning (MoST) (Han et al., 2025a) introduces Point Monarch matrices that capture local geometric features of irregular points while introducing no additional inference overhead.

The choice among these methods involves clear trade-offs. MoST and PointGST prioritize accuracy with minimal parameters but require specific architectural assumptions. DAPT, STAG, and PMA focus on

computational efficiency. Point-PEFT and PointLoRA offer balanced general-purpose adaptation, while IDPT’s instance-aware prompting adds training-time complexity. Despite these advances, open challenges remain: preserving geometric fidelity under sparse or noisy data, enabling cross-task generalization, and integrating PEFT with emerging 3D vision-language FMs.

5.1.2 Representation-Defining Foundation Models

In these works, the FM serves as the primary representation learner, generating dense 3D representations directly from raw data in a unified feed-forward pass.

Geometry-centric 3D Foundation Models (GFMs) generate dense 3D representations such as depth or point maps from multi-view images. Their key innovation reduces reliance on known camera parameters and explicit multi-stage geometric pipelines, replacing complex multi-step reconstruction with a fast unified process. DUS3R (Wang et al., 2024e), MUS3R (Caban et al., 2025), Fast3R (Yang et al., 2025b), and VGGT (Wang et al., 2025g) use transformer-based feed-forward architectures for parallel multi-view processing, inferring a common scene representation that encodes appearance and perspective. This unified representation supports camera parameters, depth maps, point maps, novel view synthesis, and point tracks. Challenges remain in accuracy, generalization, and real-time efficiency (Cong et al., 2025; Leblanc & Poullis, 2026). DUS3R (Wang et al., 2024e) established a point-map-based formulation without known camera parameters, while Fast3R (Yang et al., 2025b) prioritizes speed at the cost of some accuracy.

Point Cloud Adapted Vision-based Foundation Models leverage large-scale pretrained models to overcome limitations in depth estimation and scene understanding, including expensive hardware and poor generalization (Xu et al., 2026d). Seal (Liu et al., 2023e) and imaging-based projection methods enable segmentation and point cloud understanding with high scalability and generalizability (Dong et al., 2025). Adaptive strategies like Uni-Adapter (Tamjidi et al., 2025) improve robustness to noisy or out-of-distribution data. However, evaluations such as UniQA-3D (Zuo et al., 2025) show these models still fall short of robust human-level 3D reasoning.

3D Vision-Language Foundation Models incorporate language directly into representation learning. Point cloud features are explicitly aligned with language through contrastive, distillation (Hinton et al., 2015; Mansourian et al., 2025), or alignment objectives (Wang & Czarnecki, 2025; Deng et al., 2025a; Tang et al., 2025b), producing a language-aware embedding space. This enables downstream tasks including medical image analysis (Pan et al., 2025a; Yamamoto & Kikuchi, 2026), visual grounding (Peng et al., 2025; Wu et al., 2025b; Liu et al., 2025a), and robotic interaction (Liu et al., 2025b), where language serves as the direct interface for querying, controlling, or reasoning about the 3D world.

Action or Embodiment-Centric 3D Foundation Models address limitations of Vision-Language-Action (VLA) models in spatial understanding by injecting 3D representations into existing architectures via low-perturbation point cloud injection (Li et al., 2026a), depth awareness and chain reasoning (Bhat et al., 2025), independent geometry encoder (Sun et al., 2025a), Ego3D position encoding and Adaptive Action Grids (Qu et al., 2025a), or implicit depth perception without additional sensors (Lin et al., 2025).

5.1.3 Representation-Guiding Foundation Models

Here, FMs provide external priors while the point cloud encoder remains the core representation learner. Various approaches leverage pretrained VLMs to guide 3D representations. The GFS-VL framework (An et al., 2025b) improves point cloud segmentation by combining 3D VLM knowledge with multi-shot samples. Some studies emphasize the use of 3D representation as a spatial anchor for other data modalities (Zhang et al., 2025b). In point cloud completion, Song et al. (2023b) and Zhou et al. (2025a) both use the CLIP (Radford et al., 2021) model to guide the geometric reconstruction of incomplete shapes. Ges3ViG incorporates human pointing gestures alongside language to guide recognition Mane et al. (2025). Text2Loc++ uses hierarchical natural language descriptions for 3D localization (Xia et al., 2025b). Finally, the training-free approach proposed by Chen et al. (2025a) improves the efficiency of vision-language models for detecting

out-of-distribution data in 3D spaces. In all these works, FMs guide perception without defining the point cloud representation.

2D VLMS As Semantic Priors use language as an inference-time semantic prior. The FM is frozen, the point cloud encoder is task-trained, and no alignment between point cloud and language representations is attempted (Ha & Song, 2022; Rudner et al., 2024).

Vision Foundation Models as Semantic Priors transfer 2D knowledge to reduce annotation costs. For point cloud segmentation, predictions are mapped to 3D via pooling and pseudo-label generation (Dong et al., 2025). BALViT combines frozen image models with LiDAR via 2D-3D adapters Hindel et al. (2025). For part segmentation, GLIP knowledge transfers to point clouds through rendering and label transfer (Liu et al., 2023d). For 3D structure prediction, relative depth from vision models is converted to metric depth (Feng et al., 2025a). HFIT extracts and merges heterogeneous RGB-depth features without retraining Vision Transformers (Guo et al., 2026b).

5.1.4 Reasoning-Centric Foundation Models

The FM here functions exclusively as a high-level reasoning engine, decoupled from geometric representation learning. It operates on top of the point cloud system without modifying or learning point-level features, interpreting semantic concepts and user intentions while specialized encoders handle geometric details. LLMs augment point cloud understanding by enriching category prototypes and detecting keypoints (LARM (Li et al., 2025n) and ZeroKey (Gong et al., 2025)). Reason3D (Huang et al., 2025c), 3D-Affordance LLM (Chu et al., 2025), and SeqAfford (Yu et al., 2025a) reformulate segmentation and affordance detection as instruction-reasoning problems. This extends to real-world data via LiDAR-LLM (Yang et al., 2025d) and RadarLLM (Lai et al., 2025). PULLM (Zhang et al., 2025g) leverages LLMs for point cloud upsampling, while 3D-R1 (Huang et al., 2025f) employs reinforcement learning and dynamic view selection to enhance reasoning and generalization. Methods like LARM and ZeroKey are lightweight and suitable for real-time applications, while Reason3D and 3D-R1 achieve higher reasoning accuracy at significantly greater computational cost. These works point toward models that combine precise 3D perception with language reasoning capabilities.

5.1.5 Point-Cloud-Scaffolded Foundation Models

Point clouds here serve as temporary structural scaffolding—reconstructed or employed for geometric context, but as external input rather than learned representation. Zhang et al. (2025d) questions whether point clouds truly enhance spatial reasoning in LLMs. 3D-LOTUS++ (Garcia et al., 2025) integrates point cloud-based motion planning with language and vision for robotic manipulation. Conversely, GPT4Scene (Qi et al., 2025) reconstructs a point cloud from video and renders a BEV image to support VLM-based spatial reasoning. VisionCube (Wang et al., 2025d) demonstrates that structured 3D representations from images can effectively scaffold complex multi-step spatial reasoning.

5.1.6 Conclusion, Challenges, and Future Directions of Foundation Models

Table 10 summarizes the four functional roles. Representation-Defining FMs excel in dense geometry, Reasoning-Centric FMs in language interaction, while Guiding and Scaffolded FMs serve as external priors or structural context. None currently handles all tasks robustly. For instance, GFMs struggle under unseen camera configurations; 3D VLMs degrade significantly on out-of-distribution data; and Reasoning-Centric approaches face prohibitive inference costs that limit real-time deployment. These limitations highlight the need for task-specific selection rather than a one-size-fits-all solution. PEFT provides a practical enabler for deploying these large models with minimal overhead.

Despite transformative impact, challenges remain in scalability, generalization, robustness across sensor modalities, and computational cost. Future progress requires unified models that jointly learn geometry and semantics from multimodal data, supported by rigorous benchmarks for spatial reasoning and out-of-distribution robustness. Advancing PEFT to preserve geometric fidelity, enable cross-task generalization, and integrate with vision-language models will be essential for robotics and augmented reality deployment.

5.2 Scene Understanding

3D scene understanding aims to extract semantic, relational, and contextual knowledge from three-dimensional environments by jointly reasoning over geometry and high-level concepts. In this survey, we organize 3D scene understanding tasks into four complementary categories: 3D captioning (Section 5.2.1), 3D grounding (Section 5.2.2), 3D question answering (3DQA) (Section 5.2.3), and 3D reasoning (Section 5.2.4). These tasks differ in their output structure and reasoning requirements, ranging from holistic scene-level description (3D captioning) to interactive query-based understanding (3DQA), precise language-conditioned localization (3D grounding), and multi-step inference over spatial, semantic, or physical relationships (3D reasoning). This categorization provides a structured view of the field by grouping tasks according to the form of language supervision and the depth of reasoning involved, enabling clearer comparison of methodologies and facilitating analysis of how different models trade off expressiveness, interpretability, and computational complexity across 3D scene understanding problems. An overview of this taxonomy of 3D scene understanding tasks is illustrated in Figure 5.

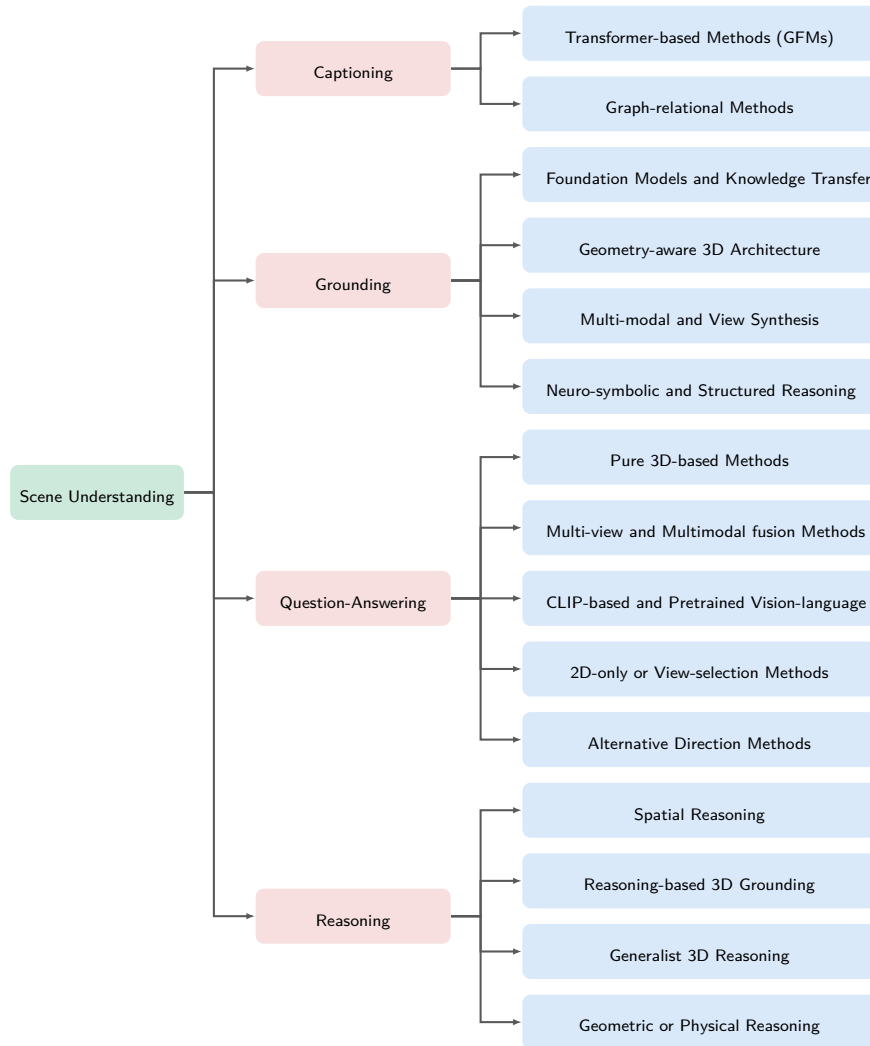


Figure 5: Taxonomy of 3D scene understanding tasks. Scene understanding methods are organized into four functional categories based on their output structure and reasoning depth: 3D Captioning (holistic description), 3D Grounding (language-conditioned localization), 3D Question Answering (interactive querying), and 3D Reasoning (compositional inference).

5.2.1 Captioning

3D captioning maps 3D object or scene representations, typically point clouds or RGB-D scans, to natural-language descriptions. Depending on the task level, a model may describe an entire scene or an individual object, whereas 3D dense captioning additionally localizes multiple object instances and generates a caption for each localized region. Surveys of 2D captioning are generally organized according to their underlying deep learning methodologies (Hossain et al., 2019; Stefanini et al., 2022; Ghandi et al., 2023). Building on the broader categorization of Yu et al. (2023b), we provide a more granular analysis of recent 3D captioning methods, including emerging outdoor settings and unified grounding-captioning frameworks. The reviewed methods are organized into two broad groups: transformer-based and graph-relational methods. These categories are not mutually exclusive, because recent frameworks often combine transformer decoding, relational reasoning, and pretrained vision-language priors. Accordingly, each method is discussed according to its principal architectural emphasis.

Transformer-based methods in 3D captioning have developed in three main directions. First, integrated detection-and-description methods such as Vote2Cap-DETR (Chen et al., 2023a) and ToD3Cap (Jin et al., 2024a) jointly perform object localization and caption generation within unified query-based frameworks. These end-to-end transformer models reduce reliance on the hand-crafted components of traditional detect-then-describe pipelines, although their query interactions and multimodal processing may increase computational requirements. Second, context- and geometry-aware transformers (Wang et al., 2022a; Kim et al., 2024b) use specialized attention mechanisms to model object relationships and scene structure. Third, multimodal and pretrained transformers such as CG-MLLM (Huang & Xu, 2026), X-Trans2Cap (Yuan et al., 2022), 3D-VisTA (Zhu et al., 2023b), 3D CoCa (Huang et al., 2025f), and 3D CoCa v2 (Tang et al., 2026a) adopt different alignment mechanisms, including cross-modal knowledge transfer, 3D vision-language pre-training, frozen vision-language priors, and joint captioning and representation-learning objectives. Overall, integrated methods emphasize the joint optimization of localization and caption generation, context- and geometry-aware transformers strengthen spatial and relational reasoning, and multimodal pretrained approaches primarily improve semantic alignment between 3D representations and language.

Graph-relational methods such as Scan2Cap (Chen et al., 2021b) and MORE (Jiao et al., 2022) use graph neural networks and structured scene relations to model interactions among objects, an idea also extended to related tasks such as 3D scene-change captioning (Liao et al., 2021). Subsequent attention-based relational models (Kim et al., 2024b) capture object relationships without requiring explicitly constructed graphs. However, graph construction can become expensive in cluttered scenes, particularly when dense all-pairs relationships are modeled, for which the number of candidate edges grows quadratically with the number of instances. A newer direction uses unified relational-attention frameworks such as 3DJCG (Cai et al., 2022a), which combines relation-aware feature enhancement with lightweight transformer heads for joint dense captioning and visual grounding. Related multi-view methods further incorporate complementary 2D cues to improve dense captioning (Ma et al., 2025c). This evolution has enabled more flexible relational modeling and the integration of complementary visual information (Jia et al., 2023). Alongside these architectural developments, diffusion-based techniques have been explored through semantic conditioning, data augmentation, progressive generation, and view selection to improve caption quality and visual-language alignment (Luo et al., 2023; Cioni et al., 2023; Luo et al., 2024a; Shu et al., 2025). Nevertheless, graph-relational methods may still struggle to generalize to unseen objects and scene layouts beyond the training distribution.

Conclusion, Challenges, and Future Directions of Captioning

3D captioning has evolved from staged detect-then-describe pipelines toward transformer and graph-relational architectures that increasingly incorporate multimodal pretraining and generative techniques. Integrated models jointly perform localization and captioning, relational methods capture scene structure, and pretrained vision-language models improve semantic alignment.

However, performance remains sensitive to localization errors, cluttered scenes, uncommon objects, and complex spatial relations. Large multimodal and language backbones also increase computational cost.

Existing benchmarks provide limited scene and annotation diversity, while conventional n-gram metrics inadequately capture factual consistency, spatial grounding, and semantic correctness.

Future work should focus on lightweight geometry-aware models, open-vocabulary and few-shot learning, unified grounding-captioning frameworks, and broader real-world benchmarks. Evaluation should also measure localization and relational accuracy, hallucination, uncertainty, and out-of-distribution robustness.

5.2.2 Grounding

Visual grounding (Rohrbach et al., 2016; Mao et al., 2016; Li et al., 2022b; 2023k; Wang et al., 2025k; Li et al., 2025m) is a foundational task in the field of integrating computer vision and natural language processing. 3D visual grounding aims to identify and localize the object referred to by a natural-language expression within a 3D scene, typically by selecting an object proposal or predicting its bounding box, instance label, or segmentation mask. Because of these capabilities, it provides a language-conditioned interface for downstream 3D scene understanding, interactive environments, and robotic perception. Broad applications such as comprehensive understanding of 3D scenes (Man et al., 2024b; Zhi et al., 2025; Ost et al., 2026), creating interactive virtual, or augmented reality systems (Gopakumar et al., 2024; Jang et al., 2024; Zhai et al., 2025) and controlling robots with natural language commands (Huang et al., 2025a; Tang et al., 2025a) are among the areas that directly benefit from advances in this field.

This review organizes recent 3D grounding methods according to the principal mechanism used to bridge natural-language queries and 3D scene evidence. Foundation-model and knowledge-transfer approaches introduce pretrained semantic knowledge, geometry-aware architectures explicitly encode spatial structure, multimodal and multi-view methods enrich the available visual evidence, and neuro-symbolic or structured-reasoning methods model compositional and relational constraints. These directions are not mutually exclusive, as recent systems often combine several of these mechanisms; therefore, each method is discussed according to its dominant methodological emphasis.

Foundation models and knowledge transfer leverage pretrained Vision-Language Models (VLMs) or Large Language Models (LLMs) to enhance 3D visual grounding. By reusing previously learned semantic and visual knowledge, these approaches reduce dependence on large task-specific 3D annotations, although many still require object proposals, geometric preprocessing, or task-specific adaptation. Existing methods transfer pretrained representations (Bakr et al., 2022; Wang et al., 2023h), employ LLMs as semantic reasoners (Yang et al., 2024a), or use foundation models as robust feature extractors (Chen et al., 2024c; Liu et al., 2025i). Their broad semantic knowledge supports open-vocabulary grounding, but performance may depend on proposal quality and cross-modal alignment, while large models introduce additional latency and computational cost.

Geometry-aware 3D architectures tackle the geometric, structural, and sparsity challenges of point cloud data, including transformer adaptations (Zhao et al., 2021b; He et al., 2021a; Chen et al., 2023g), novel attention modules (Li et al., 2025e; Zhang et al., 2025f; Dey et al., 2025), and enhanced relational encoders (Chen et al., 2020a; Yuan et al., 2021; Xiao et al., 2024; Wang et al., 2025m) that produce robust 3D feature representations well suited to grounding tasks. The strengths of geometry-aware architectures on established benchmarks are offset by their dependence on costly 3D annotations, susceptibility to distribution shifts. Moreover, dense global attention can scale quadratically with the number of point tokens or object proposals, although local or sparse attention can partially reduce this cost.

Multimodal and view synthesis methods combine and enrich the 3D point cloud representation with information from other sensory modalities to achieve stronger groundings. This additional information can include 2D images (Lyu et al., 2024a; Xiao et al., 2025), multi-view projections (Huang et al., 2022b; Guo et al., 2023; Geng et al., 2025), segmentation masks (Chen et al., 2024b), and CLIP embeddings (Hegde et al., 2023; Zhang et al., 2024k; Li et al., 2025c). This fusion enriches the scene representation, leading to improved object detection accuracy and greater consistency across different viewpoints.

Neuro-symbolic and structured reasoning integrates neural networks with symbolic logic or structured knowledge bases, such as spatial commonsense, which is an emerging direction for complex reasoning. This hybrid method facilitates explicit reasoning, allowing models to compositionally understand the elements

within a 3D scene. This capability is essential for tasks like interpreting long-horizon, logically constrained linguistic queries and ensuring verifiable predictions (Zellers et al., 2021; Chen et al., 2022c; Hsu et al., 2023; Zhu et al., 2024a; Jahangard et al., 2025; Chen et al., 2025d; Huang et al., 2026a). It addresses key limitations of purely data-driven models, such as their poor generalization to new scenarios and lack of interpretability. Overall, foundation models favor semantic generalization, geometry-aware architectures emphasize spatial accuracy, multimodal methods reduce ambiguity through complementary cues, and structured-reasoning approaches improve compositionality and interpretability. Their main trade-offs involve 3D supervision, computational cost, cross-modal alignment, and scalability to diverse scenes.

Conclusion, Challenges, and Future Directions of Grounding

3D visual grounding has progressed through foundation-model transfer, geometry-aware architectures, multimodal fusion, and structured reasoning. These directions respectively strengthen semantic generalization, spatial modeling, target disambiguation, and compositional interpretation, enabling more accurate language-conditioned object localization.

However, performance remains sensitive to noisy or sparse 3D observations, inaccurate object proposals, complex spatial expressions, and distribution shifts. Large pretrained and multimodal models also increase computational cost, while current benchmarks and metrics provide limited coverage of open-world scenes, reasoning difficulty, and localization robustness.

Future work should prioritize open-vocabulary and zero-shot grounding, lightweight multimodal models, robust evaluation protocols, and adaptation to unseen objects and environments. Monocular 3D grounding remains particularly challenging because a single image provides incomplete depth and scale information (Zhan et al., 2024). Although recent VLM-based methods have improved this setting, reliable metric localization and generalization across cameras and scene distributions remain open challenges (Qu et al., 2026).

5.2.3 Question Answering

3D question answering (3DQA) has recently emerged as a significant research topic, requiring models to answer natural language questions about 3D scenes. Unlike 2D question answering (Antol et al., 2015; Kim et al., 2025), which relies on flat images, 3DQA captures richer spatial relationships between objects, making it essential for applications in robotics, augmented reality, and embodied AI.

With the growth of annotated benchmarks, several methods have been proposed for 3DQA (Ye et al., 2022; Luo et al., 2025; Zhou et al., 2025d; Zhang et al., 2026b; Xu et al., 2026a). 3DQA-TR (Ye et al., 2022) introduced an early transformer-based framework that fuses appearance, geometry, and linguistic features. DSPNet (Luo et al., 2025) integrates multi-view and point-cloud features, while HCNQA (Zhou et al., 2025d) incorporates intermediate reasoning checkpoints as supervision signals. Additional works have explored alternative directions, including state-space models (Qorsham et al., 2025), Gaussian splatting (Thai et al., 2025), active learning (Zhou et al., 2025c), and self-supervised learning (Li et al., 2023g).

Building on advances in 2D question answering, several methods adapt existing 2D models to 3DQA. Some approaches bridge the 2D–3D gap by selecting diverse views from 3D scenes and fusing multiple modalities (Mo & Liu, 2024; Guo et al., 2025). Others adapt CLIP representations to 3DQA (Parelli et al., 2023; Delitzas et al., 2023), while additional studies evaluate the zero-shot capabilities of existing 2D vision–language models on 3D scenes (Singh et al., 2024; Huang et al., 2025b; Wang et al., 2025e). Table 11 categorizes existing methods according to their dominant representational and reasoning strategies. These categories are not mutually exclusive, because recent approaches may combine direct 3D encoding, multi-view fusion, pretrained vision–language models, and alternative learning strategies.

Conclusion, Challenges, and Future Directions for Question Answering

In summary, 3DQA approaches can be broadly distinguished by how they represent and reason over 3D scenes. Pure 3D-based methods explicitly model geometry and spatial structure, making them well suited for questions requiring precise localization or spatial relationships, but they often suffer from limited scalability

Table 11: Categorization of 3D question answering approaches according to their dominant representational and reasoning strategies.

Dominant Strategy	Core Idea	Representative Methods	Key Trade-off
Pure 3D-based	Explicitly model geometry and spatial structure using point cloud encoders.	3DQA-TR (Ye et al., 2022), HCNQA (Zhou et al., 2025d)	Precise spatial reasoning vs. high annotation cost and limited scalability.
Multi-view / Multimodal Fusion	Combine 2D appearance with 3D geometry for richer scene representation.	DSPNet (Luo et al., 2025), (Mo & Liu, 2024), (Guo et al., 2025)	Better contextual understanding vs. increased system complexity.
CLIP-based / VLM Transfer	Adapt or distill pretrained 2D vision-language models to 3D scenes.	3D-CLIP (Parelli et al., 2023), Multi-CLIP (Delitzas et al., 2023)	Strong semantic generalization vs. potential loss of fine-grained 3D spatial accuracy.
2D-only / View Selection	Project 3D scenes to 2D views and apply 2D VLMs in zero-shot settings.	(Singh et al., 2024), (Huang et al., 2025b), (Wang et al., 2025e)	Efficient, zero-shot capability vs. view coverage dependence and occlusion struggles.
Emerging Directions	Explore novel architectures or learning paradigms.	State-space models (Qorsham et al., 2025), Gaussian Splatting (Thai et al., 2025), Active Learning (Zhou et al., 2025c)	Potential efficiency gains vs. maturing, limited evaluation.

and high annotation cost. Multi-view and multimodal fusion methods improve robustness by combining 2D appearance and 3D geometry, achieving better contextual understanding at the expense of increased system complexity and data requirements. CLIP-based and pretrained vision-language approaches offer strong semantic generalization and reduced training cost by leveraging large-scale 2D pretraining, though they may lose fine-grained 3D spatial accuracy. 2D-only or view-selection methods provide efficient and often zero-shot solutions, but their performance depends heavily on view coverage and may struggle with occluded or geometry-intensive queries. Overall, method selection depends on the balance among spatial precision, semantic generalization, and computational efficiency required by the target application.

Key challenges remain, including limited dataset scale and diversity, difficulty in multi-step spatial reasoning, poor cross-domain generalization, and fragmented evaluation protocols. Promising future directions include developing large-scale 3D foundation models, compositional reasoning frameworks, integration of physical reasoning, and unified evaluation benchmarks that systematically test spatial, semantic, and multi-step reasoning across diverse scene types.

5.2.4 Reasoning

3D reasoning refers to the capability of a model to infer non-trivial properties, relations, or outcomes in three-dimensional environments that are not directly observable from raw geometric input, such as point clouds or multi-view observations (Zhu et al., 2024a). Unlike standard 3D perception tasks that focus on recognition or localization, 3D reasoning requires compositional, multi-step inference over spatial relations, semantic concepts, geometry, or physical constraints. Depending on the nature of the inferred knowledge and the role reasoning plays in the task, existing works can be broadly categorized into four main subcategories: spatial reasoning, reasoning-based grounding, generalist 3D reasoning, and geometric or physical reasoning. These categories are not mutually exclusive, because recent methods may combine spatial inference, language-guided grounding, general-purpose vision–language modeling, and explicit geometric constraints. Accordingly, each work is discussed according to its primary reasoning objective. Table 12 summarizes representative works in each category.

Spatial reasoning focuses on inferring relative spatial relationships among objects—such as distance, orientation, containment, or visibility often requiring relational or viewpoint-dependent inference beyond direct perception (Chen et al., 2022c; Hong et al., 2023a; Wang et al., 2023e; He et al., 2024; Ma et al., 2024a; Zhu et al., 2024e; Man et al., 2024a; Ma et al., 2025b;a; Zha et al., 2025a; Chen et al., 2026). **Reasoning-based 3D grounding** uses compositional reasoning to decompose complex linguistic descriptions into semantic attributes and spatial constraints in order to produce structured 3D outputs, such as object instances or part-level masks, particularly in open-vocabulary settings (Zhu et al., 2024a; Kareem et al., 2024; Chen et al., 2024b; Sun et al., 2024a; Huang et al., 2025c; Feng et al., 2026). Unlike direct grounding, these methods address implicit or multi-step queries that require inferring the target or its relations before localization. **Generalist 3D reasoning** aims to build unified 3D vision–language models capable of handling diverse

tasks, such as question answering, grounding, and instruction following within a single framework, where reasoning is often integrated across tasks and may be either implicit or explicitly supervised (Hong et al., 2023b; Jiang et al., 2024b; Fu et al., 2025; Huang et al., 2025f; Kang et al., 2025; Deng et al., 2025a; Zhu et al., 2025a; Xue et al., 2026; Yang et al., 2026a). In contrast, **geometric or physical reasoning** covers methods that infer latent 3D structure or enforce geometric and physical consistency, including shape, pose, layout, support, stability, occlusion, and feasibility. These approaches often rely on explicit geometric representations, differentiable projection, or physics-inspired constraints (Chaudhuri et al., 2011; Jia et al., 2013; Schwing et al., 2013; Suwajanakorn et al., 2018; Qin et al., 2019b; Liu et al., 2019b).

Table 12: Categorization of 3D reasoning methods by their primary reasoning paradigm. Methods are grouped into spatial reasoning, reasoning-based grounding, generalist reasoning, and geometric/physical reasoning, each distinguished by the nature of inference required and the role of language and geometry.

Category	Core Idea	Representative Methods	Key Trade-off
Spatial reasoning	Infer relative spatial relationships among objects (distance, orientation, containment) requiring relational or viewpoint-dependent inference.	ViL3DRel (Chen et al., 2022c), 3D-CLR (Hong et al., 2023a), Super-CLEVR-3D (Wang et al., 2023e), LLM-TPC (He et al., 2024), SpatialPIN (Ma et al., 2024a), PQ3D (Zhu et al., 2024e), SIG3D (Man et al., 2024a), SpatialReasoner (Ma et al., 2025b), CVP Chen et al. (2026)	Explicit modeling of spatial relationships vs. additional complexity in representing and integrating relational information.
Reasoning-based grounding	Decompose complex linguistic descriptions into semantic attributes and spatial constraints to produce structured 3D outputs (object instances, part-level masks).	Scanreason (Zhu et al., 2024a), PARIS3D (Kareem et al., 2024), Reasoning3D (Chen et al., 2024b), Reason3D (Huang et al., 2025c), CoRe (Feng et al., 2026)	Explicit alignment of language with 3D structure vs. high computational cost and reliance on complex pipelines.
Generalist reasoning	Build unified 3D vision-language models capable of handling diverse tasks (QA, grounding, instruction following) within a single framework; reasoning is largely implicit.	3D-LLM (Hong et al., 2023b), MORE3D (Jiang et al., 2024b), Scene-llm (Fu et al., 2025), 3D-R1 (Huang et al., 2025f), Robin3D (Kang et al., 2025), 3D-LLaVA (Deng et al., 2025a), LLaVA-3D (Zhu et al., 2025a), UniBVR (Yang et al., 2026a), Descrip3D (Xue et al., 2026)	Broad task coverage and strong generalization vs. limited reliability in fine-grained geometric reasoning.
Geometric/Physical reasoning	Inference grounded in 3D geometry or physical principles (shape, pose, stability, support, occlusion, physical feasibility) using explicit geometric representations or physics-inspired constraints.	(Chaudhuri et al., 2011), (Jia et al., 2013), Box-in-Box (Schwing et al., 2013), KeypointNet (Suwajanakorn et al., 2018), MonoGRNet (Qin et al., 2019b), SoftRas (Liu et al., 2019b)	Strong geometric consistency and task precision vs. task-specific assumptions and limited language-level generality.

Conclusion, Challenges, and Future Directions of Reasoning

In summary, 3D reasoning has advanced from specialized spatial and physical inference toward unified, language-driven generalist models capable of handling diverse tasks such as question answering, grounding, and instruction following within a single framework. However, key trade-offs persist between geometric precision and semantic flexibility, while generalist models often struggle with fine-grained spatial reasoning and physical plausibility. High computational costs, reliance on large task-specific datasets, and fragmented evaluation protocols further hinder scalability and real-world deployment. Future research should focus on developing truly holistic foundational models that jointly learn geometry, semantics, and physics from multimodal data, alongside neuro-symbolic approaches that combine neural networks with explicit reasoning for interpretability and systematic generalization. Data-efficient learning through self-supervision and synthetic data, as well as standardized benchmarks that comprehensively test spatial, physical, and compositional reasoning, will be essential for advancing robust and reliable 3D reasoning systems.

6 Robustness, Generalization, and Reliability

Robustness is a critical requirement for reliable 3D point cloud understanding systems operating in real-world environments. Due to sensor noise, partial observations, distribution shifts, and potential adversarial manipulation, 3D machine learning models must maintain stable and consistent performance under challenging conditions. This section examines robustness from multiple perspectives. It first discusses security threats in 3D ML pipelines (Section 6.1), highlighting potential vulnerabilities and attack surfaces. It then analyzes robustness to noise and occlusion (Section 6.2), which commonly arise in practical sensing scenarios. Next, rotation invariance and equivariance in point cloud processing are reviewed (Section 6.3), as they are essential for handling arbitrary object orientations. Domain adaptation strategies (Section 6.4) are also explored to address distribution shifts across datasets and environments. Finally, anomaly detection (Section 6.5)

is covered as a mechanism for identifying unexpected or out-of-distribution samples and improving system reliability.

6.1 3D ML Pipeline Security Threats

A 3D perception pipeline consists of multiple stages, including physical sensing, preprocessing and transmission, training, inference and downstream decision-making. Each stage is susceptible to various types of security risks; therefore, security analysis should consider any possible attack threats together with the defenses available at each stage, in order to maintain a secure ML pipeline. To frame adversarial machine learning in 3D vision, it is useful to start from a pipeline-level view and then zoom into finer subcategories. Inspired by (Kim & Kaur, 2024), and as illustrated in Figure 6, threats can be grouped by where they occur in the pipeline: (i) attacks on sensors (e.g., cyber/physical LiDAR manipulation), and (ii) attacks on perception/decision models.

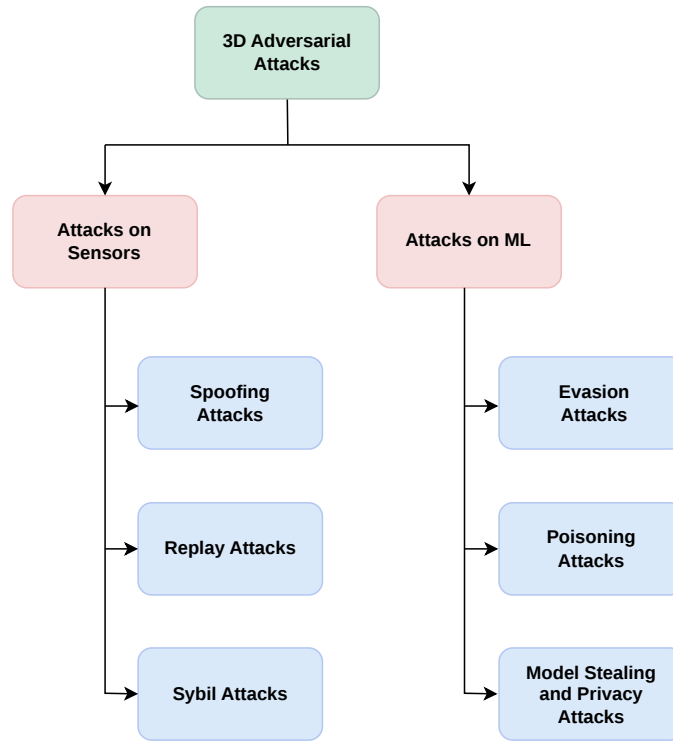


Figure 6: High-level overview of security threats across the 3D machine learning pipeline. Sensor-level threats include spoofing, replay, and Sybil attacks, whereas model-level threats include evasion, poisoning and backdoor attacks, and model extraction.

Existing point cloud security research has devoted substantial attention to model-layer evasion and poisoning attacks, whereas sensor-layer studies have more often focused on physically realizable LiDAR manipulation. Each branch imposes different threat models and calls for defenses with specific assumptions. In the following sections, we first focus on the model-layer branch and summarize Evasion attacks followed by their corresponding defenses in sections 6.1.1 and 6.1.2 respectively, then we discuss Poisoning attacks and defenses in section 6.1.3. Afterwards we revisit the same pipeline-level view in the LiDAR setting in section 6.1.4. It is worth noting that model-extraction and model-theft attacks explored widely in the 2D literature, have received substantially less attention in point cloud models, unlike Evasion and Poisoning attacks, and existing point cloud security literature rarely treats it as a first-class threat. The available studies primarily address

ownership verification and watermarking, which be discussed in section 6.1.3, rather than attacks that reconstruct model behavior through repeated queries. This limited coverage leaves open questions concerning query-based extraction, surrogate-model fidelity, and protection of proprietary 3D models.

6.1.1 Evasion Attacks, Attributes and Methodologies

Evasion attacks modify test-time inputs to produce adversarial examples that remain perceptually plausible but induce incorrect, often high-confidence predictions. Adversarial examples were first demonstrated in early work on 2D convolutional networks (Szegedy et al., 2013) and were subsequently extended to 3D point cloud models by (Liu et al., 2019a) and (Yang et al., 2019a). Building upon the taxonomies of (Naderi & Bajić, 2023) and (Li et al., 2024f), we can categorize adversarial attacks by attack attributes and attack methodologies, as discussed below:

- **Evasion Attack Attributes:** Each adversarial example attack algorithm can have its own characteristics according to the threat model it is practiced in. These attack attributes include:
 - **Representation Type:** Attacks may be applied to different 3D data representations. In this survey, we focus primarily on attacks targeting point-cloud representations.
 - **Attacker Knowledge:** Depending on the amount of knowledge the attacker holds about the target model, attacks can be black-box, white-box, or gray-box.
 - **Attacker Objective:** Depending on whether the attacker wants the model to predict the adversarial example in a specific class or any class other than the original one, the attacks can be categorized as targeted or untargeted, respectively.
 - **Type of Perturbation:** We can have point addition, deletion, displacement, transformation or object/scene attribute modification, which can be discussed from the attack methodology perspective as well.
 - **Attack Location:** The attack perturbation can occur in an on-surface or off-surface manner, depending on the adversarial perturbation’s closeness to the 3D object’s surface.
 - **Attack Physicality:** Attacks can be digital - more related to the ML model - or physical - relevant to physical systems - which will be checked in the next subsection.
- **Evasion Attack Methodologies:** Attack methodologies can differ along several independent dimensions, and these dimensions should be reported separately because an attack can belong to more than one of the categories mentioned here. Optimization-based - or gradient-based - attacks (Lou et al., 2024; Tang et al., 2024a; Xiang et al., 2019; Naderi et al., 2026; Bu & Liu, 2026) and generative-based attacks (Tang et al., 2023; Zhou et al., 2020a; Hamdi et al., 2020; Wang et al., 2026d) describe the process by which adversarial samples are produced, whereas transform-based attacks (Liu et al., 2023a; Zhao et al., 2020b; Zhang et al., 2024g; Tang et al., 2026b), and perturbation-based attacks which include point-shift/displacement, point-add and point-drop/deletion algorithms (Liu & Hu, 2023; Huang et al., 2022a; Liu et al., 2020a), describe the type of input modification. We also have heuristic attacks which can use some type of search-based algorithm or leverage some gradients as well (Cao et al., 2019b; Wicker & Kwiatkowska, 2019), and lastly, digital simulation-based attacks (Tu et al., 2020; Cao et al., 2019b), and physical system-based attacks (Cao et al., 2019a; Sato et al., 2025) describe the execution setting of the attack algorithm.

6.1.2 Evasion Defense Methodologies

According to their intervention stage, evasion defenses in general can be organized into data preprocessing methods and deep model modification methods; Data preprocessing methods can take place in either the training (Miao et al., 2024; Zhang et al., 2022a; Sun et al., 2021a) or testing phase (Sun et al., 2023; Lin et al., 2024b; Zhou et al., 2019; Li et al., 2026b; Fan et al., 2026; Luo et al., 2026) and include input purification, and runtime attack detection. Deep model modification methods consider tampering with the model architecture or weights with different approaches and include robust model training/adversarial training and certified prediction (Zhang et al., 2023d; Chu et al., 2022; Li et al., 2022a; Huang et al., 2026c;

Gui et al., 2026); Input preprocessing is generally model-agnostic but may remove legitimate geometric details, whereas adversarial training can improve model-level robustness at the cost of additional training and possible clean-data degradation. Cross-modal consistency checking and system-level fallback mechanisms provide additional lines of defense.

6.1.3 Poisoning Attacks and Defenses

Poisoning attacks manipulate the training process by introducing malicious data, labels, or updates. Backdoor attacks are a specific poisoning subtype in which under the common setting, triggered samples are injected into the training set by the adversary, so that the compromised model behaves normally on benign inputs but produces attacker-chosen behavior in the presence of the aforementioned specific trigger pattern. In the context of ML pipelines, the two terms are sometimes used interchangeably, even though there remains a parent-child relationship between the two concepts. To facilitate comparison across existing studies, we summarize 3D backdoor attacks based on a number of complementary attributes in the following taxonomy and then discuss existing 3D backdoor defenses to mitigate the mentioned attacks.

- **3D Backdoor Attacks and their Attributes**

3D backdoor attacks can be characterized based on four attributes:

- **Attack Goal:** Whether the attacker wants to change the label of the injected triggered data in the training set (Xiang et al., 2021b; Gao et al., 2025; Fan et al., 2024a; Guo et al., 2026a) or keep the label clean (Li et al., 2021c; Tian et al., 2021; Wen et al., 2021), or even having the goal of poisoning the utility/availability of the model, so that the model fails even on clean test inputs (Wang et al., 2024f). Some studies also use backdoors as a watermarking mechanism rather than as an attack. (Wei et al., 2024).

Label-flipping attacks are easier to implement but also, more easily exposed through label or sample auditing at the same time, whereas clean-label attacks are generally harder to detect, because poisoned samples retain apparently correct labels, although in this case, more carefully optimized triggers may be required as well. Availability-oriented poisoning differs from both, because its objective is broad performance degradation rather than trigger-specific misclassification.

- **Trigger Carrier:** Triggers can be embedded in geometric channels of the data (Xiang et al., 2021b; Bian et al., 2024; Gao et al., 2025; Guo et al., 2026a), other non-geometric channels such as intensity or color (Ning et al., 2024), some latent space representation of the data (Feng et al., 2025b; Jiang et al., 2025; Fan et al., 2024a) or operate at sensor-level (Chaturvedi et al., 2026; Zhang et al., 2022b).

Geometric triggers directly modify coordinates/shape of the 3D data, and may be detectable through outlier or surface-consistency analysis. Non-geometric/Attribute-based triggers can be less geometrically noticeable, however, they require access to the other types of channels mentioned before. Triggers operating at latent space depend on model internals - and so need more access - and are generally less transferable; Likewise, sensor-level triggers necessitate being involved in the acquisition process so they require more access as well, but they offer stronger practical real-world feasibility.

- **Realizability:** We can have purely digital triggers, defined in data space (Xiang et al., 2021b; Bian et al., 2024; Gao et al., 2025; Guo et al., 2026a) or physical triggers, implemented in real world (Zhang et al., 2022b; Li et al., 2023e).
- **Trigger Behavior:** We can have a single trigger mapped to a single type of misbehavior (Xiang et al., 2021b; Bian et al., 2024; Gao et al., 2025; Guo et al., 2026a), or a trigger causing multiple misbehaviors (Shan et al., 2025; Wang et al., 2024f).

- **3D Backdoor Defenses**

As defenders, similar to section 6.1.2, we can detect backdoors or defend against them by acting at different pipeline stages; For example, try to fix the input test data through Reconstruction techniques if it is corrupted (Lian et al., 2025; Xiang, 2022), detecting backdoors at inference time

by different monitoring approaches (Hu et al., 2023; Xiang, 2022) or moving the pipeline state towards safer states by tampering with different sections of it (Karim et al., 2024a;b; Xiang, 2022). We could even apply backdoor-aware fine-tuning or model repair, if trusted training data is available and access to the training process exists, or utilize outlier suppression methods (Lan et al., 2025a; Xiang, 2022), though these methods can fail against clean-label or semantically plausible triggers. In general, backdoor defenses can be categorized into training-data auditing, sanitization through reconstruction, model inspection, input purification, runtime detection, model repair or fine-tuning, and system-level fallback. Choosing the right defense method depends on the amount of access the defender has to the original training data, model weights, trusted clean samples, or suspected trigger examples.

6.1.4 Overview of LiDAR Attacks and Defenses

LiDAR is an active sensor. It emits laser pulses and measures their time of flight to construct a 3D point cloud, often augmented with intensity/reflectivity and timestamp information. In the context of LiDAR security, threats can be systematically categorized according to the component targeted by the attacker: the physical measurements acquired by the LiDAR sensor through interference with the optical signal; the scene interpretation produced by the perception model through manipulation of point-cloud inputs or model behavior; and the knowledge acquired by the system over time through poisoning or backdoor attacks during training or model updates. Guesmi & Shafique (2024) provide a broad view of physically realizable attacks on LiDAR perception and the survey by (Kim & Kaur, 2024) also provides a good starting point. Similar to previous sections, after introducing different attack methods on LiDAR sensors, we will address the existing defenses and remedies for these attacks as well.

- **LiDAR Attacks** Most LiDAR attacks ultimately try to cause either Phantom so-called ghost objects by injecting returns, so the system detects something that is not there, or Object hiding/removal, by suppressing or distorting returns, so a real object is missed. These attacks can be studied from different axes, based on which aspect we are viewing them in:
 - **Attack surface/Attack phase:** Based on where the attack takes place, whether it occurs in the acquisition phase, training phase, or inference phase.
 - **Attack Objective:** Based on what the attack is aiming to do, whether it’s phantom object creation, object hiding, localization displacement, availability degradation and model extraction.
 - **Realizability:** Based on whether the attack is on a digital level, hardware-in-the-loop simulation or physical demonstration.

If the attack is taking place in the acquisition phase, it can be done as sensor-level spoofing, tampering with the optical signal and LiDAR measurements by injecting extra light or disrupting real returns, impacting detection and even upstream tasks (Sun et al., 2020a; Sato et al., 2024; Nagata et al., 2025; Capraru et al., 2026); Or it can be done by placing or modifying real objects to induce a harmful point cloud (Tu et al., 2020). If it is a training-time poisoning/backdoor attack, it can be done by implanting behaviors in the model that only activate under specific triggers as discussed in section 6.1.3, in case of having access to the training or fine-tuning process (Li et al., 2023e; Chaturvedi et al., 2026). And finally, it can be aiming to compromise and degrade the system’s overall availability beyond targeted misdetections (Zhu et al., 2024c; Wang et al., 2026d).

- **LiDAR Defenses** Similar to previous sections, by mirroring the processing pipeline, LiDAR defenses can be organized by different stages they operate in, namely acquisition-stage prevention, sensor signal validation, point cloud consistency checking, runtime anomaly detection, robust model training, poisoning protection, and system-level response. Each defense should specify its targeted attack surface, required sensors or model access, latency, and residual failure modes. Below, the LiDAR defense taxonomy is addressed in a more general form:
 - **Sensor- and signal-level hardening:** “*Reducing what an attacker can inject*”, using physics-based checks, like whether occlusions/shadows make sense, because fake points usually can’t

match these patterns perfectly (Sun et al., 2020a), or adding hardware-level protections. However, these protection mechanisms do not provide a complete or guaranteed defense against such attacks (Sato et al., 2024). Finally, applying simple safety rules in software, like rejecting impossible distances/time-of-flight, and requiring consistency across multiple returns or across time.

- **Runtime detection and monitoring:** “*Catching spoofed frames/objects*”, these defenses assume some attacks will still get through and focus on *detecting* inconsistencies fast enough to trigger safe fallback behavior, such as catching spoofing by checking what should stay consistent in the real world, by flagging fake objects (Sun et al., 2020a), looking for point-level stability (Cho et al., 2023; Wang et al., 2026c), and detecting injected “ghost” objects by monitoring motion trajectories (You et al., 2021a).
- **Robust learning for 3D perception:** “*Making attacks harder to succeed*”, by improving adversarial robustness of 3D perception via smoothing and robustification techniques (Srinivasan et al., 2021), or using adversarial training and augmentation.
- **Defending against poisoning/backdoors:** “*Securing the learning pipeline*”, Process controls provide an important preventive layer against poisoning, either by auditing data using dataset/version control, signed updates and separation of training vs. evaluation data. However, they do not replace model inspection/trigger hunting or runtime backdoor detection tests for rare “shortcut” features, when the provenance of training data or pretrained weights cannot be fully trusted (Li et al., 2023e).

It is worth mentioning that most runtime attack detection procedures and defenses, are useful only when they are coupled with an explicit risk-mitigation policy, and ultimately need a system response, such as reducing vehicle speed, requesting additional confirmations from independent sensors or switching to a more conservative planning mode, as a defense without a safe fallback plan is only a partial defense.

6.1.5 Conclusion, Challenges, and Future Directions for 3D ML Pipeline Security Threats

The security of 3D machine learning pipelines extends beyond the robustness of individual neural networks. Vulnerabilities may arise during sensing, data preprocessing and transmission, training, inference, or downstream decision-making. The studies reviewed in this section show that sensor-level manipulation, adversarial evasion, poisoning and backdoor attacks, and model extraction require different threat models and defense assumptions. Effective protection therefore requires coordinated mechanisms across the full pipeline rather than isolated model-level defenses.

Several challenges remain unresolved. Existing security evaluations are often fragmented across datasets, architectures, sensing configurations, and attack assumptions, making direct comparisons difficult. Many attacks are evaluated only in digital settings and may not capture sensor characteristics, environmental conditions, temporal behavior, or interactions with downstream planning and control. Conversely, physically realizable attacks are frequently tested under restricted experimental conditions and may not generalize across sensors or deployment environments. Adaptive adversaries, multi-stage attacks, compound failures, and attacks against multimodal systems also remain underexplored. In particular, combining LiDAR, cameras, radar, and other sensing modalities may improve redundancy while simultaneously creating cross-modal attack surfaces and opportunities for corrupted information to propagate between data streams. Model extraction, intellectual-property protection, ownership verification, and watermarking also require further investigation for point cloud and foundation-model settings.

Future research should develop unified security frameworks that connect sensor-level threats, training-time attacks, inference-time manipulation, and system-level responses. Evaluation protocols should include adaptive adversaries, realistic physical constraints, temporal consistency, compound attacks, and cross-modal corruption, while clearly reporting attacker and defender capabilities. Promising directions include certified robustness, secure data and model provenance, backdoor-aware adaptation, cross-modal consistency checking, runtime anomaly detection, and defenses designed jointly with explicit fallback policies. Security mechanisms should also be evaluated with respect to clean-data performance, computational cost, latency,

required access, and residual failure modes. Progress along these directions can support more standardized evaluation and more resilient and trustworthy 3D perception pipelines.

6.2 Robustness to Noise and Occlusion

Real-world 3D point clouds are inevitably affected by measurement noise, quantization artifacts, spurious returns, and outliers, while also exhibiting structured missing regions due to limited viewpoints, clutter, and self-occlusion. Robustness to such degradations is critical because downstream tasks, including classification, detection, segmentation, registration, and completion, rely on geometric cues that become unreliable under partial visibility and corruption. Unlike images, point clouds are irregular and often sparse; consequently, perturbations or missing subsets of points may remove entire semantic parts or distort local geometry, leading to performance degradation. Robustness should therefore be treated as an explicit design consideration in 3D perception systems rather than assumed to emerge automatically from architectural scaling or training data size.

Before deep learning, robustness was addressed mainly through classical geometric estimation for registration and surface modeling. The Iterative Closest Point (ICP) framework (Besl & McKay, 1992) and its robust variants addressed noise and partial overlap through correspondence pruning and outlier rejection, while RANSAC-style consensus estimation (Fischler & Bolles, 1981) became a standard mechanism for handling heavily contaminated matches. In parallel, Moving Least Squares (MLS) surface fitting (Alexa et al., 2001; Fleishman et al., 2005) provided principled local smoothing under noisy measurements, and handcrafted local descriptors such as Spin Images (Johnson & Hebert, 2002), FPFH (Rusu et al., 2009), and SHOT (Salti et al., 2014) enabled matching and recognition under clutter and occlusion by relying on partial neighborhood statistics. These approaches established robustness primarily through explicit geometric modeling and filtering (Alexa et al., 2001; Johnson & Hebert, 2002; Rusu et al., 2009; Fleishman et al., 2005). Their main advantages are interpretability and independence from task-specific training, as they rely on fundamental differential geometry and local statistics (Besl & McKay, 1992; Johnson & Hebert, 2002; Rusu et al., 2009). However, their performance remains sensitive to initialization (Besl & McKay, 1992; Rusu et al., 2009), correspondence quality (Besl & McKay, 1992), neighborhood scale (Alexa et al., 2001; Johnson & Hebert, 2002; Rusu et al., 2009), and rejection thresholds (Fleishman et al., 2005; Rusu et al., 2009). This sensitivity is particularly pronounced under low overlap (Johnson & Hebert, 2002; Rusu et al., 2009), structured outliers such as sharp feature discontinuities (Fleishman et al., 2005), or substantial density variation (Rusu et al., 2009).

With the advent of learning-based methods, robustness increasingly shifted toward data-driven modeling of corruptions. Early works such as robust normal estimation (Boulch & Marlet, 2016) and PCPNet (Guerrero et al., 2018) learned noise-aware local geometry directly from data, while PointCleanNet (Rakotosaona et al., 2020) demonstrated learned denoising and outlier removal for dense point clouds. The modern deep learning era for raw point sets began with PointNet (Qi et al., 2017a) and its hierarchical extension PointNet++ (Qi et al., 2017b). Although not designed explicitly as robustness mechanisms, their permutation-invariant set processing and multi-scale aggregation biases have been shown to mitigate sensitivity to non-uniform sampling and partial visibility. Similarly, convolutional operators such as KPConv (Thomas et al., 2019) stabilize feature extraction under irregular sampling by enforcing structured local support, while RandLA-Net (Hu et al., 2020) pairs aggressive random downsampling with a local feature aggregation module to preserve geometric detail on large-scale, noisy outdoor scans; both indirectly contribute to robustness in downstream tasks. Compared with classical solvers that rely on manually-designed geometric priors or explicit statistical noise models (Boulch & Marlet, 2016; Rakotosaona et al., 2020), data-driven models can capture complex corruption patterns that are difficult to specify analytically, better accommodating non-linear real-world noise distributions (Boulch & Marlet, 2016). However, their robustness is inherently tied to the diversity and realism of the corruptions represented in the training data (Guerrero et al., 2018; Boulch & Marlet, 2016). Consequently, these models may exhibit degraded performance when transferred across different sensors, acquisition domains, or varying sampling densities not seen during training (Qi et al., 2017b; Boulch & Marlet, 2016).

Transformer-based architectures subsequently introduced broader contextual aggregation for incomplete observations. Point Transformer (Zhao et al., 2021a) can use contextual relationships beyond individual local

neighborhoods, reducing its dependence on local evidence that may be missing or corrupted. Masked point modeling approaches such as Point-BERT (Yu et al., 2022) train encoders to infer masked structure from the surrounding context and are frequently used for pretraining in partial-input tasks. In parallel, completion-oriented pipelines treat occlusion as a reconstruction problem, as exemplified by PCN (Yuan et al., 2018). At the evaluation level, benchmark-driven analyses have exposed substantial robustness gaps: ModelNet40-C (Sun et al., 2022a), Robo3D (Kong et al., 2023), and ModelNet-O (Fang et al., 2024) show that performance on clean datasets may substantially overestimate reliability under common corruptions and controlled occlusion patterns. For 3D object detection specifically, the KITTI-C, nuScenes-C, and Waymo-C benchmarks (Dong et al., 2023) reveal comparable degradation under sensor-level corruptions, motivating lightweight point-level preprocessing such as Statistical Outlier Removal (SOR) (Rusu et al., 2008) as a first line of defense prior to detection.

Building on these findings, recent methods introduce robustness at different stages of the processing pipeline. Ensemble and augmentation strategies such as EPiC (Levi & Gilboa, 2023) and PointCutMix-R (Zhang et al., 2022a), which synthesizes training samples by exchanging point subsets between shapes, improve prediction stability through diverse partial observations, whereas Set-Mixer (Zhang et al., 2024a) modifies feature aggregation to reduce the influence of individual noisy points. At inference time, PointCVaR (Li et al., 2024e) performs risk-optimized outlier removal, while PointSP (Li et al., 2025b) strengthens point sampling without retraining or modifying the underlying architecture. UPP (Ai et al., 2025) instead integrates denoising and completion with downstream analysis through parameter-efficient point-level prompting.

Registration robustness is pursued through both learned priors and refined classical solvers. PointDiffomer (She et al., 2024) uses graph neural diffusion and transformer-based correspondence modeling to improve registration under noise and geometric perturbations, whereas ComPC (Huang et al., 2025e) leverages pre-trained 2D diffusion priors for test-time completion of severely partial inputs. In contrast, certifiable solvers such as TEASER (Yang et al., 2020a) retain explicit guarantees for registration under extreme outlier ratios, while classical correspondence-based alignment continues to be refined for efficiency and robustness, as in SAC-IA variants that improve feature-point selection and parallelize the underlying kd-tree search (Wei, 2023). These mechanisms are therefore complementary rather than interchangeable: training and architectural strategies introduce limited inference overhead but depend on corruption coverage, inference-time procedures offer greater adaptability at additional computational cost, and generative completion can infer missing structure but may produce geometrically plausible yet incorrect regions.

Table 13 provides a task-oriented view of robustness mechanisms against noise and occlusion, organizing representative methods by both application domain and intervention strategy. Each row corresponds to a core point cloud task such as classification, detection, segmentation, normal estimation, denoising, registration, and completion. The columns indicate where robustness is introduced in the processing pipeline: (1) training-time strategies such as data augmentation, ensemble learning, and self-supervised pretraining; (2) model and architecture designs with built-in robustness properties; (3) inference-time preprocessing and adaptation, including sampling protocols, outlier filtering, and test-time adaptation; (4) classical geometric solvers; and (5) generative priors that leverage learned shape distributions for reconstruction.

The distribution of entries reveals several patterns. Classification spans the widest range of robustness mechanisms in Table 13, although this may partly reflect the greater availability of standardized corruption benchmarks. Registration combines classical solvers with explicit geometric and outlier assumptions and learned approaches such as PointDiffomer (She et al., 2024). Completion similarly ranges from direct reconstruction with PCN (Yuan et al., 2018) to generative completion with ComPC (Huang et al., 2025e). Detection and segmentation contain fewer representative methods evaluated under consistent corruption protocols.

6.2.1 Conclusion, Challenges, and Future Directions of Robustness to Noise and Occlusion

Robustness to noise and occlusion cannot be achieved through a single technique. Classical methods offer interpretable geometric assumptions, while learned and generative approaches better model complex corruptions and missing geometry but depend on representative data, computation, and domain transfer.

Table 13: Pipeline-oriented taxonomy of robustness mechanisms against noise and occlusion in 3D point cloud processing. Representative methods are organized by task and stage at which robustness is introduced, including training, architectural design, inference-time adaptation, geometric estimation, and generative reconstruction.

Task	Training Strategy	Model / Arch.	Preprocess. / Adapt.	Geom. Solvers	Generative
Classification	EPiC (ens) (Levi & Gilboa, 2023) PointCutMix-R (aug) (Zhang et al., 2022a) Point-BERT (pretrain) (Yu et al., 2022)	Set-Mixer (arch) (Zhang et al., 2024a)	PointSP (samp) (Li et al., 2025b) PointCVaR (filt) (Li et al., 2024e) UPP (adapt) (Ai et al., 2025)	–	–
Detection	Corruption aug. (aug) (Dong et al., 2023)	–	SOR (filt) (Rusu et al., 2008)	–	–
Segmentation	–	KPConv (arch) (Thomas et al., 2019) RandLA-Net (arch) (Hu et al., 2020) Point Transformer (arch) (Zhao et al., 2021a)	–	–	–
Normal/Denoise	–	PCPNet (model) (Guerrero et al., 2018)	PointCleanNet (filt) (Rakotosaona et al., 2020) MLS (filt) (Alexa et al., 2001)	–	–
Registration	–	PointDifformer (model) (She et al., 2024)	FPFH (feat) (Rusu et al., 2009)	ICP (solver) (Besl & McKay, 1992) RANSAC (solver) (Fischler & Bolles, 1981) TEASER (solver) (Yang et al., 2020a) SAC-IA (solver) (Wei, 2023)	–
Completion	–	PCN (model) (Yuan et al., 2018)	MLS (samp) (Alexa et al., 2001)	–	ComPC (gen) (Huang et al., 2025e)

Abbreviations: aug = augmentation, ens = ensemble, pretrain = self-supervised pre-training, arch = architecture, model = learned model, samp = sampling, filt = filtering/outlier removal, adapt = test-time adaptation, feat = feature descriptor, solver = geometric solver, gen = generative model.

Despite rapid progress, several challenges remain unresolved. Benchmarks such as ModelNet40-C (Sun et al., 2022a) and Robo3D (Kong et al., 2023) reveal that strong clean-set accuracy may substantially overestimate reliability under realistic corruptions, yet constructing corruption models that are physically plausible and semantics-preserving across sensors and tasks remains challenging. Occlusion realism constitutes a second bottleneck: although ModelNet-O (Fang et al., 2024) formalizes controlled self-occlusion, discrepancies between synthetic partiality and deployment-time visibility patterns limit transferability when viewpoints or acquisition conditions shift.

Algorithmically, robustness mechanisms remain difficult to integrate into efficient end-to-end pipelines. Multi-stage enhancement may introduce domain mismatch or additional failure modes, while parameter-efficient adaptation may not adequately suppress severe input defects, degrading representations from low-quality scans (Ai et al., 2025). Multimodal and generative pretraining can reduce data discrepancies (Nguyen et al., 2025a; Huang et al., 2025e), but their scalability, computational cost, and uncertainty remain concerns, particularly when reconstructed geometry is plausible but incorrect. Compound degradations are also insufficiently studied: detection performance can decline sharply under combined corruptions (Li et al., 2023f), and completion models trained on synthetic data may fail when noise and occlusion occur together (Tesema et al., 2025). Future work should therefore develop unified and efficient robustness pipelines, uncertainty-aware reconstruction, and evaluation protocols that reproduce realistic combinations of sensor failures.

6.3 Rotation Invariance and Equivariance

Rotation invariance in 3D point cloud analysis refers to the property that, for tasks whose target output is independent of orientation, a model produces identical outputs for an input point cloud and any of its rotated versions. This property is important due to the arbitrary poses induced by sensor viewpoints, object placement, or scene acquisition processes (Fei & Deng, 2024). Early studies on rotation-robust point cloud analysis primarily focused on achieving rotation invariance through geometric descriptors, such as 3D Zernike descriptors, which encode the overall shape in a rotation-invariant manner (Novotni & Klein, 2004). Recent networks for learning rotation-invariant geometric features in 3D point cloud analysis can be grouped into several methodological paradigms.

- **Invariant point-feature-based** methods extract invariant features by transforming raw coordinates into rotation-invariant representations. Notable examples include PPFNet (Deng et al., 2018b) and its unsupervised variant PPF-FoldNet (Deng et al., 2018a), which learn robust local descriptors for tasks such as matching and registration. Other representative works, such as PaRot (Zhang et al., 2023a), PRIN (You et al., 2021b), SGMNet (Xu et al., 2021a), and the general framework proposed in (Luo & Gao, 2024), further improve rotation invariance by disentangling shape and pose, constructing invariant geometric relations, or normalizing local point neighborhoods.
- **Rotation-invariant convolutional and graph-based** methods design local operators that aggregate geometric information in a manner insensitive to global orientation. Representative works include RConv (Zhang et al., 2019a), RConv++ (Zhang et al., 2022d), RI-GCN (Kim et al., 2020), RISurConv (Zhang et al., 2024l), and SectorConv (Lin et al., 2026).
- **Rotation-invariant transformer** methods leverage self-attention mechanisms and feature decoupling strategies to maintain consistent representations under arbitrary rotations. Representative models include RITNet (Yang et al., 2022a), RoITr (Yu et al., 2023a), MaskLRF (Furuya, 2024), self-supervised invariant transformers (Furuya et al., 2024), and RotInv-PCT (He et al., 2025a).

These paradigms differ in how they balance geometric information, robustness, and computational cost. Invariant feature-construction methods can be efficient and interpretable because they encode distances, angles, point-pair relations, or normalized local geometry before feature aggregation. However, they may discard directional information that is useful for pose-sensitive tasks. Invariant local operators preserve more neighborhood structure and can scale to large point clouds, but their performance can depend on sampling density, neighborhood construction, and the stability of local geometric relations. Attention-based methods can exploit broader context to disambiguate locally similar structures, although they typically require more memory and computation than local operators. Therefore, no single paradigm is universally preferable: the appropriate design depends on whether the task requires compact invariant descriptors, scalable local processing, or global contextual reasoning.

Rotation equivariance requires that a model’s output transforms predictably under rotations of the input, such that the extracted features rotate consistently with the point cloud. This property provides a strong inductive bias for learning 3D geometry (Fei & Deng, 2024). Depending on whether only rotations or both rotations and translations are considered, point cloud methods may enforce $SO(3)$ or $SE(3)$ equivariance, respectively. These methods can be grouped according to where equivariance is incorporated within the overall learning and prediction pipeline. These categories may overlap in practice:

- **Equivariant point convolutions / kernel operators** build $SO(3)$ - or $SE(3)$ -equivariant layers directly on point sets so that the extracted features transform consistently with the input. Representative examples include EPN (Chen et al., 2021a), E2PN (Zhu et al., 2023a), and CSEConv (Kim et al., 2024a).
- **Equivariant attention / transformers** integrate tensor features or group representations into attention mechanisms so that feature aggregation remains consistent under 3D rotations and translations. Representative examples include the $SE(3)$ -Transformer (Fuchs et al., 2020) and BITR for point cloud assembly (Wang & Jörnsten, 2024).

- **Equivariant priors for pose, registration, and fitting** incorporate SE(3)-equivariant representations into downstream transformation-estimation tasks. Representative applications include self-supervised category-level object pose estimation (Li et al., 2021b), articulated human fitting through ArtEq (Feng et al., 2023), sparse and robust point cloud registration through Equi-GSPR and PARE-Net (Kang et al., 2024; Yao et al., 2024), and clothed-human fitting through ETCH (Li et al., 2025a).
- **Equivariant generative models on SE(3)** define diffusion processes over rigid transformations so that the denoising procedure respects both rotational and translational transformations. Such formulations have been applied to point cloud registration and robust 6D object pose estimation (Jiang et al., 2023a), as well as pose denoising and 3D object reassembly (Min et al., 2025).

These categories differ in the level at which equivariance is incorporated and in the resulting computational and modeling trade-offs. Equivariant point operators introduce the geometric constraint directly into the feature extractor and can therefore be reused across multiple point cloud tasks. However, maintaining group features or orientation channels generally increases architectural complexity and computational cost. Equivariant transformers additionally capture long-range geometric interactions, but their attention operations may require more memory than local point operators. Task-specific methods combine equivariant features with stronger assumptions about correspondences, object articulation, or body geometry, which can improve performance in their intended applications but may limit transferability. Equivariant diffusion models can represent uncertainty over multiple plausible transformations, although their iterative denoising process typically results in higher inference cost than direct prediction.

6.3.1 Conclusion, Challenges, and Future Directions of Rotation Invariance and Equivariance

Rotation invariance and rotation equivariance are important properties for robust 3D point cloud learning. Rotation-invariant methods aim to produce the same output when the input point cloud is rotated, while rotation-equivariant methods preserve feature transformations consistently with the rotation of the input. Despite their importance, handling arbitrary rotations in real-world point clouds remains challenging, especially when models must also remain accurate, efficient, and scalable. Some rotation-aware methods introduce additional architectural complexity or computational cost compared with standard point cloud networks. Future research should therefore focus on developing methods that generalize better across rotations, sensor viewpoints, and 3D tasks while maintaining high accuracy, low computational cost, and practical usability in real-world applications.

6.4 Domain Adaptation

Let $\mathcal{D}_s = (x_i^s, y_i^s)$ denote a labeled source domain and $\mathcal{D}_t$ a target domain drawn from a different distribution. Domain adaptation (DA) aims to transfer a model learned on $\mathcal{D}_s$ to $\mathcal{D}_t$. In unsupervised domain adaptation (UDA), a common setting in 3D perception, target samples are available without labels. Source-free DA further removes access to the source samples and retains only the pretrained source model. In 3D data, the shift may arise from sensor configuration, sampling density, occlusion, weather, scene layout, or discrepancies between simulated and real observations. These factors alter both geometric structure and semantic statistics, making direct transfer unreliable.

General DA theory, methods, and detector-specific adaptation are reviewed in (Wilson & Cook, 2020; Redko et al., 2020; Oza et al., 2023), while point-cloud transfer and LiDAR-specific adaptation are surveyed in (Sohail et al., 2025; Triess et al., 2021). Across 3D tasks, existing methods can be grouped into five paradigms. **Feature alignment** reduces discrepancies between source and target representations. **Self-training** converts confident target predictions into pseudo-labels. **Domain mixing and sensor simulation** create intermediate or target-like inputs. **Cross-modal transfer** uses complementary modalities, typically images, to guide 3D predictions. **Reconstruction and generation** reduce geometric mismatch by recovering canonical or missing structure. These paradigms may be combined within one method. Their main assumptions and trade-offs are summarized in Table 14.

Table 14: Major paradigms in 3D domain adaptation, their primary tasks, and principal trade-offs.

Paradigm	Core Mechanism	Representative Methods	Primary Tasks	Advantage	Limitation / Cost
Feature Alignment	Match source and target representations while preserving semantic structure.	PointDAN (Qin et al., 2019a); CounterPC (Yang et al., 2025a).	Classification	Low additional inference cost.	Alignment may be unstable or merge class-conditional distributions.
Self-Training & Pseudo-Labeling	Iteratively train on selected target predictions.	RefRec (Cardace et al., 2021); GAST (Zou et al., 2021); ConUDA (Li et al., 2025f); MLC-Net (Luo et al., 2021b); DPGLA (Li et al., 2025g).	Classification, Segmentation, Detection	Provides target-specific supervision directly from unlabeled predictions.	Repeated training is costly and errors may reinforce themselves.
Domain Mixing & Sensor Simulation	Construct intermediate samples or imitate target sampling patterns.	ConDA (Kong et al., 2021); UMDMix (Nihal et al., 2025); BeyondMix (Chen et al., 2025e); ePointDA (Zhao et al., 2021c).	Segmentation	Little deployment overhead.	Transformations must remain realistic and label-preserving.
Cross-Modal Transfer	Enforce agreement or distill knowledge between paired modalities.	xMUDA (Jaritz et al., 2022); (Wu et al., 2023b).	Segmentation	Combines appearance and geometry.	Requires paired, calibrated sensors and additional encoders.
Reconstruction & Generation	Recover canonical surfaces or generate missing object points.	Complete & Label (Yi et al., 2021); DiffRefine (Shin et al., 2025).	Segmentation, Detection	Directly addresses geometric sparsity.	Adds model cost and may introduce incorrect geometry.

Having established this methodological taxonomy, we next examine how the five paradigms are instantiated in three core 3D perception tasks: point-cloud classification, semantic segmentation, and object detection. This task-level organization is important because each prediction granularity exposes different forms of domain shift and therefore favors different adaptation mechanisms.

6.4.1 Point Cloud Classification

At the object level, point-cloud classification assigns a semantic label to a point set and is sensitive to cross-dataset differences in viewpoint, density, and partial shape. PointDAN established feature alignment at both global and local scales through adversarial learning (Qin et al., 2019a). CounterPC later formulated alignment in a causal latent space and used counterfactual feature generation to separate transferable semantic information from domain-specific geometric variation (Yang et al., 2025a). PointDAN provides a comparatively direct alignment objective, whereas CounterPC introduces stronger structural assumptions and additional training components to preserve class discrimination.

Self-training methods instead exploit the geometry of unlabeled target objects. RefRec (Cardace et al., 2021) refines pseudo-labels using shape descriptors learned through reconstruction. GAST (Zou et al., 2021) combines self-paced self-training with rotation and distortion prediction to regularize geometric representations, while Fan et al. (2022a) models global and local structures and retains target pseudo-labels through source-neighbor voting. These methods reduce reliance on global distribution matching, but their performance depends on the initial target predictions. Reconstruction consistency, geometric pretext tasks, and voting provide complementary mechanisms for controlling pseudo-label noise, with additional training cost.

PC-Adapter (Park et al., 2023b) uses shape- and locality-aware adapters with class-wise pseudo-label correction, avoiding auxiliary self-supervision and adversarial training. BlendCLIP (Khoche et al., 2025) instead applies curriculum-based multimodal pretraining across synthetic and real samples for zero-shot classification. The former requires target-specific optimization; the latter requires image-text supervision and broader training data.

6.4.2 Semantic Segmentation

Semantic segmentation assigns a label to every point and is therefore highly sensitive to local density, occlusion, class imbalance, and scene-layout changes. Cross-modal methods compensate for missing geometric evidence with image features. xMUDA (Jaritz et al., 2022) enforces mutual agreement between 2D and 3D predictions, whereas Wu et al. (2023b) fuses multi-scale features before bidirectional distillation. Feature fusion offers richer interaction than prediction-level agreement but increases memory and computation. Both approaches also assume synchronized and calibrated sensors, limiting their use in LiDAR-only settings.

Domain mixing constructs intermediate samples between source and target distributions. ConDA (Kong et al., 2021) interchanges source and target regions in range-view projections for self-training. UMDMix

(Nihal et al., 2025) adds urban, occlusion-aware multi-domain mixing, while BeyondMix introduces structural priors and long-range dependency modeling (Chen et al., 2025e). ConDA is simpler; UMDMix targets urban layout and occlusion; BeyondMix captures broader context at greater architectural cost. All assume that mixed scenes remain semantically coherent.

Pseudo-labeling and structure-aware alignment methods differ in how they control target-label noise and preserve geometric consistency. ConUDA (Li et al., 2025f) uses confidence-guided probabilistic sampling rather than fixed thresholds. DPGLA (Li et al., 2025g) combines dynamic filtering with prior-guided augmentation and mixing consistency for synthetic-to-real transfer, whereas SADA-3D (Huang et al., 2025d) uses structure-aware feature alignment. Sampling and filtering are lightweight but may underrepresent difficult classes; structural constraints better preserve geometry at additional optimization cost.

Other methods reduce the sensor gap before or during segmentation. Complete & Label reconstructs sparse scans into a canonical surface and then transfers semantic labels across sensors (Yi et al., 2021). ePointDA (Zhao et al., 2021c) combines self-supervised dropout-noise rendering with feature alignment for synthetic-to-real transfer, while adaptive adversarial masking learns which synthetic points should be removed to resemble real scans (Li et al., 2023a). Completion handles sensor-specific sparsity more explicitly but requires an additional network and may propagate reconstruction errors. Noise rendering and masking are less expensive, yet they assume that point dropout captures the dominant source-target discrepancy.

6.4.3 Object Detection

3D object detection must transfer both semantic recognition and localization. Domain shifts in object size, point density, range, and scan pattern consequently affect proposal quality as well as classification. SF-UDA3D (Saltori et al., 2020) considers the source-free setting and adapts a pretrained LiDAR detector using unlabeled target data. By contrast, MLC-Net (Luo et al., 2021b) uses a teacher-student framework with point-, instance-, and neural-statistics-level consistency when source data remain available. Source-free adaptation is preferable when source data cannot be retained, but it depends more strongly on the pretrained detector; multi-level consistency provides richer supervision at higher training and storage cost.

Sensor-oriented approaches directly address LiDAR sampling differences. See Eye to Eye interpolates object geometry and normalizes scan patterns before detection (Tsai et al., 2022). Tanaka et al. (2025) instead uses downstream and partial-layer fine-tuning across sensor configurations. Normalization has lower adaptation cost but cannot recover missing evidence; fine-tuning specializes to each configuration. DiffRefine (Shin et al., 2025) uses conditional diffusion to densify proposal-specific points before box refinement. It handles sparse proposals but increases inference cost and may generate unsupported geometry. Digital twins provide a complementary way to improve simulated LiDAR realism (Shahbaz & Agarwal, 2025), consistent with evidence that sensor modeling remains important even in matched real and simulated scenes (Huch et al., 2023).

6.4.4 Conclusion, Challenges, and Future Directions of Domain Adaptation

In summary, 3D domain adaptation methods address distribution shifts through complementary strategies that differ in their assumptions, computational requirements, and access to source or target data. The appropriate strategy depends on deployment constraints and the dominant source of shift. Alignment, mixing, and sensor normalization are efficient but may fail under class-conditional or geometric shifts. Self-training is flexible but vulnerable to confirmation bias; cross-modal and generative methods offer stronger corrective signals at higher computational and hardware cost. The same principles extend to source-free primitive segmentation and synthetic-to-real bridge-component instance segmentation (Wang et al., 2025j; Xu et al., 2026b). Open challenges identified across 3D scene understanding (Knights, 2025) include compound shifts, calibrated pseudo-label uncertainty, drift-free continual adaptation, and generalization to unseen domains.

Future work should develop uncertainty-aware pseudo-labeling, efficient source-free and continual adaptation, and methods that remain stable under combined sensor, environmental, and geometric shifts. Promising directions also include lightweight cross-modal and generative approaches, improved adaptation across

heterogeneous LiDAR configurations, and standardized evaluation protocols for assessing robustness under realistic and evolving domain changes.

6.5 Anomaly Detection

Anomaly detection plays a key role in ensuring stable performance, preventing errors, and improving the efficiency of industrial systems. Earlier approaches relied on manual inspection, which is labor-intensive and prone to human error. With the advent of deep learning and computer vision, the field has shifted towards automated visual inspection systems, substantially increasing inspection speed and reliability. Recent RGB image-based anomaly detection methods employ several strategies to improve automated inspection, including knowledge distillation in teacher–student frameworks (Rudolph et al., 2023; Gu et al., 2023), adaptation of pretrained vision models for efficient feature transfer (Reiss et al., 2021; Zhou et al., 2023c; Pan et al., 2025b), and synthesis of artificial anomalies for robust optimization (Cao et al., 2023; Zavrtnik et al., 2024a). However, 2D methods are fundamentally constrained by their inability to capture structural and geometric deviations (Hoang et al., 2026). 3D anomaly detection, using data such as point clouds and depth maps, complements 2D methods and enables the identification of structural, geometric, and internal defects that are difficult or impossible to detect with 2D techniques.

Unsupervised methods provide a practical approach that is trained only on normal data. These methods can be broadly organized into several categories based on their core strategy. **Reconstruction-based methods**, such as autoencoders (Masuda et al., 2021), diffusion models (Zhou et al., 2024d), and dual reconstruction architectures (Li et al., 2024i;c; Hoang et al., 2025; Liang et al., 2025b), detect anomalies through high reconstruction error (Sun et al., 2025b; Li et al., 2025j). **Feature-based methods**, including memory banks (Chu et al., 2023; Wang et al., 2023g; Tao et al., 2024; Bhunia et al., 2024), model normal feature distributions and identify anomalies through feature discrepancies (Jing et al., 2025). **Teacher-student architectures** (Rudolph et al., 2023; Gu et al., 2023; Sun et al., 2024b) distill knowledge from a teacher network to detect anomalies via feature discrepancy. **Self-supervised learning** (Tu et al., 2024) and **contrastive learning** (Zhu et al., 2024b) learn representations without labels and identify inconsistencies by comparing samples in latent space. A key challenge across these unsupervised approaches remains the discriminative capacity of features, particularly in 3D, where rotation-invariant and high-resolution point-level representations are critical.

In production environments, **supervised methods** learn from labeled samples, often using limited, domain-specific datasets (Wang et al., 2025c; Kaji et al., 2022; Xu et al., 2024b; Bolourian et al., 2023; Hwang & Kang, 2023), as summarized in Table 15. These methods can effectively detect predefined defect categories, but their performance depends strongly on the diversity and representativeness of the annotated anomalies, limiting their ability to generalize to unseen defect types and increasing the cost of data collection and labeling.

Few-shot and zero-shot methods for 3D point-cloud anomaly detection address limited access to target-category training data and anomalous samples through several key strategies. **Data synthesis methods** generate new anomalous samples (Li et al., 2024b; 2025l; Xiang et al., 2025a). **Zero-shot methods** leverage pretrained vision-language models without in-domain training (Zhou et al., 2024b; 2025b; Wang et al., 2025h). Reconstruction-based approaches such as 3DR_{EM} (Zavrtnik et al., 2024b) can also be adapted to few-shot scenarios. **Prompt learning methods** aim to teach models the concept of anomalous data using textual or visual prompts (Zuo et al., 2024b).

As shown in Table 15, 3D anomaly detection methods span a wide range of methodological strategies, from reconstruction-based and feature-based approaches to zero-shot and few-shot paradigms. The development of these methods has been accompanied by the creation of tailored benchmark datasets, which are generally categorized into pure point cloud, multi-view, and RGB-D formats and other multimodal acquisition settings (Bergmann et al., 2021; Bonfiglioli et al., 2022; Zhou et al., 2023b; Liu et al., 2023c; Wang et al., 2024a; McHard et al., 2025; Li et al., 2025i; Cheng et al., 2026).

Table 15: Categorization of 3D anomaly detection methods by core methodological strategy, with representative approaches and their primary contributions.

Methodological Strategy / Learning Setting	Representative Methods	Key Contribution
Reconstruction-based	3DR-EM (Zavrtanik et al., 2024b), MVR (Sun et al., 2025b), MADFlow (Li et al., 2025j)	Detect anomalies via high reconstruction error; learn normal geometry representation
Feature-based / Memory Bank	F2PAD (Tao et al., 2024), 3DMulti-FFFH (Jing et al., 2025), PIAD (Yang et al., 2025c)	Model normal feature distributions; identify anomalies through feature discrepancies
Zero-shot / Foundation Model-based	PointAD (Zhou et al., 2024b), PointAD+ (Zhou et al., 2025b), PLANE (Wang et al., 2025h)	Leverage pretrained vision-language models; require no in-domain training
Few-shot / Data Synthesis	DAS3D (Li et al., 2024b), Find (Li et al., 2025l), BridgeNet (Xiang et al., 2025a)	Synthesize anomalous samples or adapt features from limited normal data
Teacher-Student / Distillation	Asymmetric Teacher-Student (Rudolph et al., 2023), Remembering (Gu et al., 2023), Memoryless (Sun et al., 2024b)	Distill knowledge from teacher to student; detect anomalies via feature discrepancy
Supervised	(Wang et al., 2025c), (Kaji et al., 2022), (Xu et al., 2024b), (Bolourian et al., 2023), (Hwang & Kang, 2023)	Learn from labeled anomalies; effective for known defects but limited generalization
Self-supervised & Contrastive	(Tu et al., 2024), (Zhu et al., 2024b)	Learn representations without labels; identify inconsistencies via contrastive learning

6.5.1 Conclusion, Challenges, and Future Directions of Anomaly Detection

In summary, 3D anomaly detection complements RGB inspection by capturing geometric and structural deviations. Reconstruction-based methods detect anomalies through deviations from learned normal geometry, while feature-based approaches model normal feature distributions. Teacher-student architectures (Rudolph et al., 2023; Gu et al., 2023; Sun et al., 2024b) distill knowledge to identify feature discrepancies, and self-supervised or contrastive learning (Tu et al., 2024; Zhu et al., 2024b) enables representation learning without labels. Supervised methods are effective for known defects but require representative labeled anomalies, whereas few-shot and zero-shot methods reduce this dependence through feature transfer (Liu et al., 2024b), anomaly synthesis (Li et al., 2024b; 2025l; Xiang et al., 2025a), prompts (Zuo et al., 2024b), and pretrained models (Zhou et al., 2024b; 2025b; Wang et al., 2025h).

Key challenges include generalization to unseen defects and object categories, motivating zero-shot approaches that leverage vision foundation models and vision-language models without in-domain training (Belhadi et al., 2026; Fučka et al., 2026). Further difficulties include computational cost on edge devices and reliable RGB-3D fusion under cross-modal misalignment, sensor noise, and incomplete geometry. Current benchmarks also provide limited coverage of real production variations, including pose, illumination, occlusion, and sensor degradation.

Future work should focus on unified appearance-geometry foundation models, scalable self-supervised pre-training, data-efficient adaptation, uncertainty estimation, explainability, and lightweight deployment. More realistic evaluation protocols are also needed to assess robustness under distribution shifts and evolving defect patterns.

7 Evaluation and Benchmarking

To provide a comprehensive comparison of existing approaches in 3D point cloud learning, this section presents an evaluation framework that highlights the benchmark datasets, evaluation metrics, and performance analyses commonly reported in the literature. In this survey, widely used datasets are first presented, highlighting their associated tasks, data representations, scale, and data sources. Next, the evaluation metrics commonly adopted to measure performance across different 3D tasks are summarized. Finally, a performance analysis of representative deep learning approaches for point cloud understanding is provided, focusing on major tasks such as classification, segmentation, and object detection. This structured evalua-

tion helps clarify benchmarking practices in the literature and facilitates a more consistent comparison of existing methods.

7.1 Datasets

3D point cloud datasets are very diverse. They cover many tasks, such as classification, segmentation, detection, registration, completion, reconstruction, and generation. They also come from different sources, such as synthetic CAD models or real scans captured by sensors. Consequently, they can differ substantially in scale and annotation richness.

Another key difference is the 3D representation used in each dataset, such as meshes, CAD models, point clouds, voxels, or multi-view images. This choice affects how methods are built, how data are labeled, and how results are evaluated. To describe datasets in a clear and consistent way, we summarize each one using the same attributes: task, representation, number of classes, number of samples, type (source). This helps readers quickly compare datasets along several axes: data quality, label richness, synthetic versus real data, and dataset scale. Table 16 lists these fields, which helps compare datasets and choose the right benchmarks for a specific 3D setting.

Table 16: Summary of representative 3D point-cloud datasets across different tasks. Each dataset is characterized by its primary task, underlying representation, number of classes, sample scale, and data source type

Dataset	Task	Representation	#Classes	#Samples	Type
ShapeNet (2015) (Chang et al., 2015)	Classification, Reconstruction	Mesh	55	51,190	Synthetic
ModelNet10 (2015) (Wu et al., 2015)	Classification, Generation, Reconstruction	Mesh	10	4,899	Synthetic
ModelNet40 (2015) (Wu et al., 2015)	Classification, Generation, Reconstruction	Mesh	40	12,311	Synthetic
ScanNet (2017) (Dai et al., 2017)	Classification, Semantic Segmentation, Reconstruction	RGB-D + Mesh	20	1,513 scans, 12,283 samples	Indoor
ScanNet200 (2022) (Rozenberszki et al., 2022)	Semantic Segmentation, Instance Segmentation	RGB-D + Mesh	200	1,513 scans, 12,283 samples	Indoor
Semantic3D (2017) (Hackel et al., 2017)	Semantic Segmentation	LiDAR Point Cloud	8	4B	Outdoor
S3DIS (Area 5) (2017) (Armeni et al., 2016)	Semantic Segmentation	RGB + Point Cloud	13	695.9M	Indoor
SemanticKITTI (2019) (Behley et al., 2019)	Semantic Segmentation	LiDAR Point Cloud	19	4.549B points	Outdoor
nuScenes (2019) (Caesar et al., 2020)	Detection, Semantic Segmentation; Tracking	LiDAR + Cameras	16	1,000 scenes, ~40k keyframes	Outdoor
SemanticPOSS (2020) (Pan et al., 2020)	Semantic Segmentation	LiDAR Point Cloud	14	216M points	Outdoor
BlendedMVS (2020) (Yao et al., 2020)	Foundation Models	RGB-D	-	17,000 images	Synthetic
CO3D (2021) (Reizenstein et al., 2021)	Foundation Models	RGB + Point Cloud	50	1.5 million frames	Real-world
KITTI (2012) (Geiger et al., 2012)	Detection, Tracking	LiDAR + GPS/IMU	9	14,999	-
Waymo (2019) (Sun et al., 2020b)	Detection, Tracking; Foundation Models	LiDAR + cameras + radar	4 detection; 23–28 seg- mentation	390,000 frames	Real-world
H3D (2019) (Patil et al., 2019)	Tracking	Point Cloud	8	27,721	-
Owlii (2017) (Xu et al., 2017)	Compression	Mesh	-	4 sequences × 600 frames	Real-world

Table 16 Continued

Dataset	Task	Representation	#Classes	#Samples	Type
SiVSLF (2018) (d'Eon et al., 2017)	Compression	Voxel	-	1 sequence $\times$ 300 frames + 6 single-frame Point Clouds	Real-world
SiVFB v2 (2017) (d'Eon et al., 2017)	Compression	Voxel	-	4 sequences $\times$ 300 frames	Real-world
MVUB (2016) (Loop et al., 2016)	Compression	Voxel	-	5 subjects; 7–10 s at 30 fps	Real-world
PKU-DPCC (2024) (Xie et al., 2024c)	Compression	Mesh	-	50 sequences $\times$ 250 frames	Real-world
Matterport3D (2017) (Chang et al., 2017)	Generation, Reconstruction	RGB-D + Mesh	-	194,400 RGB-D images	Real-world
Completion3D (2019) (Tchapmi et al., 2019)	Generation	Point Cloud	8	30,974 3D models	Synthetic
VIPC (2021) (Zhang et al., 2021b)	Generation	RGB + Point Cloud	13	~919,872	Synthetic
Google Scanned Objects (2022) (Downs et al., 2022)	Generation	Mesh + SDF	17	1030	Real-world
Sun3D (2013) (Xiao et al., 2013)	Reconstruction	RGB + Point Cloud	-	415 sequences	Real-world
ABC dataset (2019) (Koch et al., 2019)	Reconstruction	Mesh	-	1 million 3D models	Synthetic
3D-FUTURE (2020) (Fu et al., 2021)	Reconstruction	Mesh	-	20,240 images	Synthetic
LineMod (2012) (Hinterstoisser et al., 2012)	6DoF Pose Estimation	RGB-D	15	~1,200 per class	Indoor
LineMod-Occluded (2014) (Brachmann et al., 2014)	6DoF Pose Estimation	RGB-D	15	1214	Indoor
T-less (2017) (Hodan et al., 2017)	6DoF Pose Estimation	RGB-D	30	~1,600 per class	Industrial
YCB-Video (2018) (Xiang et al., 2017)	6DoF Pose Estimation	RGB-D	21	92 videos	Indoor
ScanObjectNN (2019) (Uy et al., 2019)	Robustness	Point Cloud	15	15,000	Real-world
RobustPointSet (2020) (Taghanaki et al., 2020)	Robustness	Point Cloud	40	17,276	-
ModelNet40-C (2022) (Sun et al., 2022b)	Robustness	Point Cloud	40	185,100	Synthetic
Point Cloud-C (2022) (Ren et al., 2022)	Robustness	Point Cloud	40 classification + 16 part labels	186,970	Synthetic
ShapeNet-C (2022) (Ren et al., 2022)	Robustness	Point Cloud	16	16,881	Synthetic
ModelNet40-E (2025) (Alonso et al., 2025)	Robustness	Point Cloud	40	12,311	Synthetic
MVTec 3D-AD (2019) (Bergmann et al., 2019)	Anomaly Detection	RGB + Point Cloud	15	5,354	Industrial
Real3D-AD (2023) (Liu et al., 2023c)	Anomaly Detection	Point Cloud	12	1,254	-
MAD (2023) (Zhou et al., 2023b)	Anomaly Detection	Multi-view RGB	20	4,000	-
Real-IAD (2024) (Wang et al., 2024a)	Anomaly Detection	RGB-D	30	150,000	Industrial
Anomaly-ShapeNet (2024) (Li et al., 2024c)	Anomaly Detection	Point Cloud	40	1,600	Industrial
ScanRefer (2020) (Chen et al., 2020a)	Captioning; Grounding	RGB-D scans	806	11,046	Indoor
ScanQA (2022) (Azuma et al., 2022)	Captioning; Question Answering	Mesh	-	40,000 QA pairs	Indoor
ScanNet++ (2023) (Yeshwanth et al., 2023)	Captioning	DSLR; RGB-D	460	280k DSLR images	Indoor
Multi3DRefer (2023) (Zhang et al., 2023g)	Captioning; Grounding	Point Cloud scenes	265	11,609	Indoor

Table 16 Continued

Dataset	Task	Representation	#Classes	#Samples	Type
ExCap3D (2025) (Yeshwanth et al., 2025)	Captioning	DSLR images	2000	34,700 objects	Indoor
ReferIt3D (2020) (Achlioptas et al., 2020)	Grounding	Point Cloud	1,273 scenes	11,375 object contexts	Indoor
ALFRED (2020) (Shridhar et al., 2020)	Grounding	3D scans (AI2-THOR)	120 scenes; 84 classes	428,322	Indoor
3DVQA (2022) (Etesam et al., 2022)	Question Answering	Point Cloud	-	484,359 questions	Indoor
3DQA (2022) (Ye et al., 2022)	Question Answering	Point Cloud	-	6,000 questions	Indoor
FE-3DGQA (2022) (Zhao et al., 2022)	Question Answering	Point Cloud	-	20,125 QA	Indoor
CLEVR3D (2023) (Yan et al., 2023b)	Question Answering	Point Cloud	-	171,000 QA	Synthetic + Indoor
Space3D-Bench (2024) (Szymańska et al., 2024)	Question Answering	Mesh + Point Cloud + RGB-D	-	1,000 QA	Indoor
3DSRBench (2025) (Ma et al., 2025a)	3D Reasoning	2D Image	-	2,772 QA	-
SURPRISE3D (2025) (Huang et al., 2026b)	3D Reasoning	Mesh + Point Cloud	-	200,000 QA	Indoor

7.2 Metrics

Choosing appropriate evaluation metrics is essential in point cloud learning, because different tasks assess fundamentally different properties. This section is organized into three parts. First, we summarize general evaluation metrics used across point cloud tasks, including classification, segmentation, detection, tracking, compression, generation, reconstruction, registration, 6DoF pose estimation, captioning, grounding, question answering, reasoning, anomaly detection, domain adaptation, rotation invariance and equivariance, robustness to noise and occlusion, and 3D ML pipeline security. Second, we discuss point-set distance measures in greater detail, focusing on their suitability as training losses and evaluation metrics. Finally, we provide comparative conclusions and practical guidelines for selecting and reporting these measures.

7.2.1 General Evaluation Metrics

Classification metrics measure label correctness, segmentation and detection metrics assess semantic or spatial overlap, and geometric tasks such as generation, reconstruction, registration, and pose estimation require measures of point-set similarity, alignment, or transformation error. Table 17 provides a task-oriented summary of commonly used metrics, their mathematical definitions, and their preferred directions.

The numerical value of a metric should not be interpreted independently of its evaluation protocol. Point cardinality, coordinate normalization, sampling strategy, distance norm, threshold, and numerical scaling can substantially affect reported results. This issue is particularly important for point-set distance measures, whose behavior depends not only on geometric proximity but also on point density, outliers, and the correspondence mechanism. A dedicated comparison of these measures is therefore provided in Section 7.2.2.

Table 17: Summary of commonly used evaluation metrics for 3D point-cloud learning tasks, including their formulations, application domains, and optimization directions.

Metric	Task	Formula	Description
Accuracy	Classification, Robustness	$\frac{TP+TN}{TP+TN+FP+FN}$	Fraction of all predictions that are correct. Higher is better.

Table 17 Continued

Metric	Task	Formula	Description
Precision	Classification, Detection, Robustness	$\frac{TP}{TP+FP}$	Fraction of predicted positives that are truly positive. Higher is better.
Recall	Classification, Detection, Reconstruction	$\frac{TP}{TP+FN}$	Fraction of true positives that are successfully retrieved. Higher is better.
F1-Score	Classification, Generation, Reconstruction	$2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$	Harmonic mean of precision and recall, balancing false positives and false negatives. Higher is better.
IoU	Segmentation, Grounding	$\text{IoU}(A, B) = \frac{ A \cap B }{ A \cup B }$	Overlap between predicted and reference regions. Higher is better.
mAP	Detection, Segmentation, Robustness	$\frac{1}{C} \sum_{c=1}^C \sum_n (R_{c,n} - R_{c,n-1}) P_{c,n}$	AP summarizes the precision-recall curve, while mAP averages AP over the evaluated classes. Higher is better.
mIoU	Segmentation, Robustness	$\frac{1}{C} \sum_{c=1}^C \frac{TP_c}{TP_c + FP_c + FN_c}$	Mean intersection-over-union across classes. Higher is better.
Tracking Success	Tracking	$\text{Success}_\tau = \frac{1}{T} \sum_{t=1}^T \mathbb{I} [\text{IoU}(B_t^p, B_t^{gt}) \geq \tau]$	Fraction of frames whose predicted and ground-truth boxes have an IoU above threshold τ . Some benchmarks report the area under the success curve over multiple thresholds. Higher is better.
Tracking Precision	Tracking	$\text{Precision}_\tau = \frac{1}{T} \sum_{t=1}^T \mathbb{I} [\ c_t^p - c_t^{gt}\ _2 \leq \tau]$	Fraction of frames whose predicted target center lies within distance threshold τ of the ground-truth center. Higher is better.
HOTA (Higher Order Tracking Accuracy)	Tracking	$\text{HOTA} = \frac{1}{ \mathcal{A} } \sum_{\alpha \in \mathcal{A}} \sqrt{\text{DetA}_\alpha \text{AssA}_\alpha}$	Balances detection and association accuracy and averages the resulting score over localization thresholds $\alpha \in \mathcal{A}$. Higher is better.
Point-to-Point PSNR	Compression	$\text{D1-PSNR} = 10 \log_{10} \left(\frac{p^2}{\text{MSE}_{\text{D1}}} \right)$	Measures geometry accuracy using nearest-neighbor Point-to-Point error, where p denotes the adopted peak geometry range. Higher is better.
Point-to-plane PSNR	Compression	$\text{D2-PSNR} = 10 \log_{10} \left(\frac{p^2}{\text{MSE}_{\text{D2}}} \right)$	Measures geometry distortion using point-to-plane error and surface normals. Higher is better.
Bits per point	Compression	$\frac{R}{N}$	Average number of coded bits per point. Lower is better at comparable reconstruction quality.
BD-PSNR	Compression	$\frac{1}{x_{\max} - x_{\min}} \left[\frac{\Delta a}{4} x^4 + \frac{\Delta b}{3} x^3 + \frac{\Delta c}{2} x^2 + \Delta d x \right]_{x_{\min}}^{x_{\max}}$	Bjontegaard Delta PSNR summarizes the average quality difference between two rate-distortion curves over a common bitrate range.
Bits per occupied voxel	Compression	$\frac{R}{N_{\text{occ}}}, \quad N_{\text{occ}} = V_{\text{occ}} $	Bitrate normalized by the number of occupied voxels. Lower is better at comparable quality.
Chamfer Distance (CD)	Generation, Reconstruction, Robustness	$\frac{1}{ X } \sum_{x \in X} \min_{y \in Y} \ x - y\ _2^2 + \frac{1}{ Y } \sum_{y \in Y} \min_{x \in X} \ y - x\ _2^2$	Bidirectional mean nearest-neighbor distance. It is efficient and almost everywhere differentiable, but permits many-to-one matches and is sensitive to outliers and non-uniform sampling. Lower is better.
Density-aware Chamfer Distance (DCD) (Wu et al., 2021)	Generation, Reconstruction	$\frac{1}{2} \left[\frac{1}{ X } \sum_{x \in X} \left(1 - \frac{e^{-\alpha \ x - \pi_Y(x)\ _2^2}}{n_{\pi_Y(x)}^\lambda} \right) + \frac{1}{ Y } \sum_{y \in Y} \left(1 - \frac{e^{-\alpha \ y - \pi_X(y)\ _2^2}}{n_{\pi_X(y)}^\lambda} \right) \right]$	A bounded, density-aware extension of CD. Here, $\pi_Y(x)$ denotes the nearest neighbor of x in Y , and $n_{\pi_Y(x)}$ is its query frequency. DCD penalizes repeated assignments and detects density mismatch, although density weighting may emphasize isolated points. Lower is better.

Table 17 Continued

Metric	Task	Formula	Description
Earth Mover's Distance (EMD)	Generation, Reconstruction	$\min_{\phi: X \rightarrow Y} \frac{1}{ X } \sum_{x \in X} \ x - \phi(x)\ _2$	Minimum global transport cost under a bijective or mass-preserving assignment. It provides stronger global correspondence than CD, but exact computation is expensive and generally requires compatible cardinalities or point masses. Lower is better.
Hyperbolic Chamfer Distance (HyperCD) (Lin et al., 2023a)	Generation, Reconstruction	$\frac{1}{ X } \sum_{x \in X} \operatorname{arcosh} \left(1 + \alpha \min_{y \in Y} \ x - y\ _2^2 \right) + \frac{1}{ Y } \sum_{y \in Y} \operatorname{arcosh} \left(1 + \alpha \min_{x \in X} \ y - x\ _2^2 \right), \quad \alpha > 0$	A hyperbolic transformation of CD that assigns greater relative gradient weight to reliable small-distance matches and updates large residuals more gradually. Lower is better.
InfoCD (Lin et al., 2023b)	Generation, Reconstruction	$\begin{aligned} \mathcal{L}_{\text{InfoCD}}(X, Y) &= \ell_{\text{InfoCD}}(X, Y) + \ell_{\text{InfoCD}}(Y, X), \\ \ell_{\text{InfoCD}}(X, Y) &= -\frac{1}{ Y } \sum_{y \in Y} \log \frac{\exp \left[-\tau^{-1} \min_{x \in X} d(x, y) \right]}{\sum_{y' \in Y} \exp \left[-\tau^{-1} \min_{x \in X} d(x, y') \right]} \end{aligned}$	A contrastively regularized Chamfer objective that discourages clustered matches and improves distribution alignment. It is primarily used as a training loss. Lower is better.
Gumbel Prior Similarity (GPS) (Zhang et al., 2025i)	Generation, Reconstruction	$\begin{aligned} \kappa(X, Y) &= \sum_{k=1}^K \sum_{g=1}^G \left[\sum_{x \in X} D_{k,g}(x, Y) e^{-D_{k,g}(x, Y)} + \sum_{y \in Y} D_{k,g}(y, X) e^{-D_{k,g}(y, X)} \right], \\ D_{k,g}(x, Y) &= \alpha_{k,g} \left[d_k(x, Y) + \delta_{k,g} \right]^{\beta_{k,g}} \end{aligned}$	A probabilistic similarity based on mixtures of Gumbel priors fitted to transformed KNN-distance distributions, where $d_k(x, Y)$ is the distance to the k -th nearest neighbor. Higher is better; $-\kappa$ can be minimized as a training loss.
Jensen-Shannon Divergence (JSD)	Generation, Reconstruction	$\frac{1}{2} D_{\text{KL}}(P \ M) + \frac{1}{2} D_{\text{KL}}(Q \ M), \quad M = \frac{1}{2}(P + Q)$	Measures similarity between two estimated point distributions. Lower is better.
Point-cloud Fréchet Inception Distance (P-FID)	Generation	$\ \mu_X - \mu_Y\ _2^2 + \operatorname{Tr} \left(\Sigma_X + \Sigma_Y - 2(\Sigma_X \Sigma_Y)^{1/2} \right)$	FID-style distance computed from point-cloud embeddings produced by a specified feature extractor. Results depend on the selected embedding model and protocol. Lower is better.
Point-cloud Kernel Inception Distance (P-KID)	Generation	$\frac{1}{m(m-1)} \sum_{i \neq j}^m k(u_i, u_j) + \frac{1}{n(n-1)} \sum_{i \neq j}^n k(v_i, v_j) - \frac{2}{mn} \sum_{i=1}^m \sum_{j=1}^n k(u_i, v_j), \quad u_i = f(x_i), \quad v_j = f(y_j)$	Kernel-based distribution discrepancy computed from point-cloud embeddings using maximum mean discrepancy. Lower is better.
Fréchet Pointcloud Distance (FPD)	Generation	$\ m_P - m_Q\ _2^2 + \operatorname{Tr} \left(\Sigma_P + \Sigma_Q - 2(\Sigma_P \Sigma_Q)^{1/2} \right)$	A Fréchet distance convention commonly computed from PointNet feature vectors of real and generated point sets. Although algebraically similar to P-FID, it is distinguished by its feature extractor and evaluation protocol. Lower is better.
Minimum Matching Distance (MMD)	Generation, Reconstruction	$\frac{1}{ Y } \sum_{y \in Y} \min_{x \in X} d(x, y)$	Average distance from each reference sample to its nearest generated sample, where d is usually a point-cloud distance such as CD or EMD. Lower is better.
Hausdorff Distance (HD)	Reconstruction, Robustness	$\max \left\{ \sup_{x \in X} d(x, Y), \sup_{y \in Y} d(y, X) \right\}$	Measures the maximum mismatch between two point sets and is highly sensitive to outliers. Lower is better.
Normal Consistency (NC)	Reconstruction	$\frac{1}{2 \partial M } \int_{\partial \hat{M}} \langle n(p), n(\pi_2(p)) \rangle dp + \frac{1}{2 \partial M } \int_{\partial M} \langle n(\pi_1(q)), n(q) \rangle dq$	Mean absolute agreement between corresponding surface normals. Higher is better.
Light Field Descriptor (LFD)	Reconstruction	$\min_i \sum_{k=1}^{10} d(I_{1k}, I_{2k})$	Measures visual similarity between 3D shapes using descriptors extracted from multiple rendered views. Lower is better.
Average Distance of Distinguishable Model Points (ADD)	6DoF Pose Estimation	$\frac{1}{ M } \sum_{x \in M} \ (Rx + t) - (\hat{R}x + \hat{t})\ _2$	Average distance between model points transformed by predicted and ground-truth poses. Lower is better.

Table 17 Continued

Metric	Task	Formula	Description
Average Distance of Distinguishable Model Points–Symmetric (ADD-S)	6DoF Pose Estimation	$\frac{1}{ M } \sum_{x_1 \in M} \min_{x_2 \in M} \left\ (Rx_1 + t) - (\hat{R}x_2 + \hat{t}) \right\ _2$	Closest-point variant of ADD commonly used for symmetric objects. Lower is better.
Attack Success Rate (ASR)	Robustness	–	Proportion of eligible adversarial examples that successfully fool the model. A higher attack success rate indicates lower robustness.
Defense Accuracy	Robustness	–	Ratio of test samples correctly classified after applying a defense. Higher is better.
Corruption Error (CE)	Robustness	$\frac{1 - \text{OA}_i}{1 - \text{OA}_{\text{baseline}, i}}$	Normalizes the model’s error under corruption i by the corresponding baseline error. Lower is better.
Relative mean CE (RmCE)	Robustness	$\frac{\text{mCE}_{\text{model}}}{\text{mCE}_{\text{baseline}}}$	Measures mean corruption degradation relative to a baseline. Lower is better.
AUROC	Anomaly Detection	$\int R_{\text{TP}} dR_{\text{FP}}$	Area under the receiver operating characteristic curve over all decision thresholds. Higher is better.
AUPRO	Anomaly Detection	$\int \frac{1}{N} \sum_{i=1}^N \frac{ P \cap C_i }{ C_i } dR_{\text{FP}}$	Measures average per-region overlap as a function of the false-positive rate. Higher is better.
BLEU	Captioning	$\text{BP} \cdot \exp \left(\frac{1}{4} \sum_{n=1}^4 \log p_n \right)$	Compares generated captions with references using modified n -gram precision and a brevity penalty. Higher is better.
METEOR	Captioning	$(1 - \text{Penalty}) \cdot F_{\text{mean}}$	Caption metric combining unigram alignment, recall, precision, and ordering penalties. Higher is better.
CIDEr	Captioning	$\frac{1}{m} \sum_j \frac{\mathbf{g}^n(c_i) \cdot \mathbf{g}^n(s_{ij})}{\ \mathbf{g}^n(c_i)\ \ \mathbf{g}^n(s_{ij})\ }$	Consensus-based caption evaluation using TF–IDF-weighted n -gram similarity. Higher is better.

7.2.2 Point-Set Distance Measures

Let $X = \{x_i\}_{i=1}^M$ and $Y = \{y_j\}_{j=1}^N$ denote two point sets. A point-set discrepancy may be used as a *training loss*, an *evaluation metric*, or both. Training losses should be computationally efficient, differentiable, and provide stable gradients, whereas evaluation metrics should reliably reflect geometric quality and sampling artifacts. Therefore, a measure suitable for training may not be sufficient for evaluation. The following paragraphs review several representative point-set distance measures and their respective properties.

Chamfer Distance (CD). Chamfer Distance performs bidirectional nearest-neighbor matching. It is efficient, permutation invariant, and piecewise differentiable, making it a common training loss. However, it allows many-to-one correspondences, does not explicitly model density mismatch, and can produce clustered predictions. Squared CD is also sensitive to outliers because it assigns large penalties to distant points. The squared or non-squared CD formulation should always be reported. Squared CD provides stronger gradients for distant points but is more sensitive to outliers and coordinate scaling.

Earth Mover’s Distance (EMD). EMD estimates a global optimal transport or bijective assignment between two point sets. It provides stronger global correspondence than CD and avoids many-to-one matching. However, exact EMD is computationally expensive and generally requires compatible point cardinalities or masses. Approximate solvers improve scalability but introduce solver-dependent errors and regularization.

Table 18: Comparison of point-set distances and similarity measures. Computational cost refers to a dense pairwise implementation for point sets of cardinalities M and N . Approximate nearest-neighbor methods may reduce the practical cost of CD-based approaches.

Measure	Matching Mechanism	Computational Cost	Differentiability	Outlier Robustness	Density Sensitivity	Correspondence Quality	Recommended Use
CD	Independent bidirectional nearest neighbors	$O(MN)$ for dense pairwise computation	Piecewise differentiable; non-smooth when the nearest neighbor changes	Low, especially for squared CD	Low; density mismatch is not explicitly modeled	Allows many-to-one matches and point clustering	Efficient training loss and baseline metric, but insufficient as the sole evaluation measure
DCD (Wu et al., 2021)	Density-reweighted nearest neighbors with bounded residuals	$O(MN)$ plus query-frequency counting	Piecewise differentiable; nearest-neighbor assignments and counts are piecewise constant	Moderate; bounded residuals help, but isolated outliers may receive relatively large density weights	High; repeated nearest-neighbor assignments are explicitly reweighted	Improves surface-coverage assessment over CD but does not impose a global bijection	Training and evaluation when sampling density and surface coverage are important
EMD	Global transport or bijective assignment	High; exact balanced assignment is commonly $O(n^3)$	Exact assignment is non-smooth; differentiable approximations exist	Low to moderate because all point mass must be transported	Moderate; behavior depends on point masses and transport formulation	Provides strong global correspondence	Global-distribution evaluation or small point sets; generally expensive for routine training
HyperCD (Lin et al., 2023a)	CD matching with a hyperbolic residual transformation	$O(MN)$ plus an element-wise transformation	Piecewise differentiable	Improved relative to CD because large residuals receive lower relative gradient weight	Low to moderate; no explicit density correction	Preserves reliable matches but retains many-to-one assignments	Primarily a robust training loss; use independent evaluation metrics
InfoCD (Lin et al., 2023b)	Contrastively normalized nearest-neighbor matching	$O(MN)$ plus contrastive normalization	Piecewise differentiable	Improved relative to CD, but dependent on the temperature and optimization setting	Moderate; discourages clustered matches but does not explicitly estimate density	Reduces correspondence congestion but does not impose a global bijection	Primarily a training loss; evaluate with complementary geometric and distributional metrics
GPS (Zhang et al., 2025i)	Probabilistic modeling of transformed KNN-distance distributions	$O(MN)$ plus KNN selection and Gumbel-component evaluation	Differentiable with respect to selected distances; neighbor selection is piecewise constant	Improves distributional robustness relative to first-neighbor CD, but does not explicitly reject outliers	Moderate; captures KNN-distance statistics rather than imposing an explicit density constraint	Captures multi-neighbor distributional statistics without a global bijection	Use $-GPS$ for training and GPS as a complementary distributional similarity score

Density-aware Chamfer Distance (DCD). DCD extends CD by reweighting repeated nearest-neighbor assignments and bounding large residuals (Wu et al., 2021). It can therefore identify density mismatch and distinguish point sets with similar CD values but different surface coverage. DCD is less expensive than exact EMD and can be used for both training and evaluation. However, it does not impose a global bijection, depends on its density-weighting parameters, and may assign relatively high importance to isolated points (Lin et al., 2023a).

Hyperbolic Chamfer Distance (HyperCD). HyperCD applies the transformation $\text{arcosh}(1 + \alpha d^2)$, where $\alpha > 0$, to Chamfer residuals (Lin et al., 2023a). This transformation gives greater relative importance to small, reliable matches while reducing the influence of large residuals during optimization. HyperCD has a computational order similar to CD, but it does not explicitly address density mismatch or many-to-one assignments. It is therefore mainly suitable as a robust training loss.

InfoCD. InfoCD applies contrastive normalization to nearest-neighbor distances (Lin et al., 2023b). It encourages matches to spread across the target surface, reducing point clustering and many-to-one correspondence congestion. Its computational order is similar to CD, although it introduces a temperature parameter and additional optimization complexity. InfoCD is primarily a training objective and should be evaluated using complementary metrics.

Gumbel Prior Similarity. GPS models the distributions of K -nearest-neighbor distances using Gumbel priors (Zhang et al., 2025i). Unlike CD, which uses only the nearest neighbor, GPS captures broader local distributional information. GPS is a similarity measure, so higher values indicate greater similarity, while its negative can be used as a training loss. Its cost is dominated by pairwise distance computation, with additional overhead for KNN selection and Gumbel evaluation. Its behavior also depends on K and the Gumbel parameters.

Conclusion and Metric Selection Guidelines. No single measure captures all aspects of point-set quality. CD is efficient but sensitive to outliers, density mismatch, and many-to-one assignments. EMD provides stronger global correspondence but is computationally expensive. DCD is more suitable for density and surface-coverage evaluation, while HyperCD and InfoCD mainly improve training behavior. GPS

Table 19: Quantitative comparison of representative 3D object classification methods on ModelNet40 and ModelNet10 in terms of overall accuracy (OA) and mean class accuracy (mAcc) (* denotes that results are not reported from the original paper).

Representation	Architecture	Method	ModelNet40 OA	ModelNet40 mAcc	ModelNet10 OA	ModelNet10 mAcc
Projection-based/multi-view	-	MVCNN (Su et al., 2015)	90.10	-	-	-
		GVCNN (Feng et al., 2018)	93.10	-	-	-
		MHBN (Yu et al., 2018b)	94.70	93.10	95.00	95.00
		MVTN(20 view) (Hamdi et al., 2021)	93.50	92.20	-	-
		MVACPN (Wang et al., 2022c)	93.64	91.53	-	-
		DTV-CNN (Xia, 2023)	88.00	-	92.00	-
		CLIP2Point (Huang et al., 2023a)	94.20	-	-	-
Spatially discretized	-	VoxNet (Maturana & Scherer, 2015)	-	83.00	-	92.00
		FPNN (Li et al., 2016)	88.40	-	-	-
		OctNet (Riegler et al., 2017)	-	85.50*	-	91.00*
		O-CNN(Voting) (Wang et al., 2017)	89.90	-	-	-
		MRCNN (Ghadai et al., 2019)	-	86.20	-	91.30
Point-based	MLP-based	DeepSets (Zaheer et al., 2017)	82.00	-	-	-
		PointNet (Qi et al., 2017a)	89.20	86.20	-	-
		PointNet++(+Normals) (Qi et al., 2017b)	91.90	-	-	-
		SRN (Duan et al., 2019b)	91.50	-	-	-
		PointWeb (Zhao et al., 2019)	92.30	89.40	-	-
		Momenet (Joseph-Rivlin et al., 2019)	92.40	90.30	-	-
		PointNeXt (Qian et al., 2022)	93.20	90.80	-	-
		PointMLP (Ma et al., 2022)	94.50	91.40	-	-
		PointGL (Li et al., 2024a)	93.00	90.40	-	-
		DualMLP (Paul et al., 2024)	93.70	-	-	-
		PointMIL(PointNet) (De Vries et al., 2025)	90.7	87.1	-	-
	Convolution-based	PointWiseCNN (Hua et al., 2018)	86.10	81.40	-	-
		PointCNN(Unaligned) (Li et al., 2018)	92.20	88.10	-	-
		PCNN (Atzmon et al., 2018)	92.30	-	94.90	-
		A-CNN (Komarichev et al., 2019)	92.60	90.30	95.50	95.30
		KPConv(rigid) (Thomas et al., 2019)	92.90	-	-	-
		ShellNet (Zhang et al., 2019b)	93.10	-	-	-
		DensePoint (Liu et al., 2019c)	93.20	-	96.60	-
		RS-CNN (Liu et al., 2019d)	93.60	-	-	-
		ConvPoint (Boulch, 2020)	91.80	88.50	-	-
		PACConv(Voting) (Xu et al., 2021b)	93.90	-	-	-
		DeltaConv (Wiersma et al., 2022)	93.80	91.20	-	-
		CompositeNets(Conv) (Floris et al., 2024)	91.30	87.10	-	-
	GNN-based	ECC(12 votes) (Simonovsky & Komodakis, 2017)	87.40	83.20	90.80	90.00
		PointGCN (Zhang & Rabbat, 2018)	89.51	86.05	91.91	91.57
		RGCNN (Te et al., 2018)	90.50	87.30	-	-
		KCNet (Shen et al., 2018)	91.00	-	94.40	-
		SpecGCN(+Normals) (Wang et al., 2018a)	91.80	-	-	-
		DGCNN (Wang et al., 2019c)	92.90	90.20	-	-
		3D-GCN (Lin et al., 2020)	92.10	-	-	-
		PointNGCNN (Lu et al., 2020)	92.80*	-	-	-
		PointManifold (Yang & Gao, 2020)	93.00	90.10	-	-
		LDGCNN (Zhang et al., 2021a)	92.90	90.30	-	-
		CurveNet(Voting) (Xiang et al., 2021a)	94.20	-	96.30	-
		PointViG (Zheng et al., 2024b)	94.30	91.20	-	-
		Point-SkipNet (Saeid et al., 2025)	92.29	89.84	-	-
	Transformer-based	PAT(FPS + GSS) (Yang et al., 2019b)	91.70	-	-	-
		Point Transformer (Zhao et al., 2021a)	93.70	90.60	-	-
		PCT (Guo et al., 2021)	93.20	-	-	-
		PTv2 (Wu et al., 2022)	94.20	91.60	-	-
		LCPFormer (Huang et al., 2023b)	93.60	90.70	-	-
		PPT (Wang et al., 2025i)	93.90	91.20	-	-
Hybrid Methods	-	SPT(Q-SDE768) (Wu et al., 2025a)	91.22	88.45	94.76	93.69
		PointView-GCN (Mohammadi et al., 2021)	95.40	-	-	-
		PV-Ada (Zhu et al., 2022)	92.30	-	-	-

provides a complementary distribution-sensitive similarity measure. The selected measure should depend on its intended use. CD remains a useful baseline, EMD is appropriate when global correspondence is important, DCD is suitable for density-aware comparison, and HyperCD and InfoCD are primarily training losses. Models should not be evaluated only with the loss used during training.

For reproducibility, studies should report the CD formulation, point cardinality, sampling and normalization procedures, distance scaling, solver settings, and method-specific hyperparameters.

7.3 Performance Analysis

This section analyzes the performance of representative deep learning methods for point cloud understanding, focusing on classification, segmentation, and object detection. These tasks are selected because they have widely used benchmark datasets and commonly reported evaluation metrics, making task-specific comparison more meaningful. The reported values are taken from the original papers rather than reproduced through re-training. Re-training and testing every method under a unified setting is often impractical due to differences

in implementations, hyperparameter choices, training strategies, hardware requirements, and computational cost. Therefore, numerical results are compared only within compatible settings, such as the same task, dataset, and metric. Sections 7.1 and 7.2 further reviewed the benchmark datasets and evaluation metrics commonly used in point cloud learning.

7.3.1 Classification

Point cloud classification aims to assign a semantic label to an entire 3D object or scene and has served as one of the most fundamental benchmarks for evaluating 3D deep learning models. Table 19 summarizes the performance of representative classification methods on the ModelNet40 (Wu et al., 2015) and ModelNet10 (Wu et al., 2015) datasets using overall accuracy (OA) and mean class accuracy (mAcc). Consistent with the taxonomy introduced earlier, the evaluated methods are grouped into projection-based, volumetric-based, and point-based approaches.

Early projection-based and volumetric approaches applied deep learning to 3D data by converting point clouds into image views or voxel grids. MVCNN (Su et al., 2015) achieved promising early results, but these representations often introduce information loss due to projection or discretization. As shown in Table 19, later projection-based models such as CLIP2Point (Huang et al., 2023a) improved performance on ModelNet40, reaching 94.2% OA.

Point-based methods later became dominant because they operate directly on raw point sets and better preserve geometric information. PointNet (Qi et al., 2017a) and PointNet++ (Qi et al., 2017b) established the foundation for this paradigm, and many subsequent architectures improved performance through better local feature learning. For example, PointView-GCN (Mohammadi et al., 2021) reports 95.4% accuracy. Transformer-based models such as PTV2 (Wu et al., 2022) also achieve competitive results by capturing long-range dependencies.

7.3.2 Segmentation

3D point cloud segmentation aims to partition a three-dimensional environment by assigning a semantic label or instance identifier to every individual point, serving as a critical prerequisite for spatial understanding in navigation, robotics, and augmented reality. To evaluate these capabilities, various models are benchmarked across diverse datasets, such as ScanNet200 (Rozenberszki et al., 2022) and S3DIS (Armeni et al., 2016), utilizing metrics including mean Intersection over Union (mIoU) and Average Precision (AP).

Table 20 indicates that recent 3D segmentation research is increasingly extending beyond fully supervised performance toward open-vocabulary, few-shot, weakly supervised, and domain-adaptive settings. Although fully supervised methods generally achieve stronger benchmark results, limited-supervision and adaptation approaches address more challenging scenarios involving scarce labels, unseen categories, or distribution shifts between training and deployment domains. Since these methods differ in supervision setting, task, and evaluation metric, comparisons should primarily be made among methods evaluated under similar conditions.

7.3.3 Object Detection

3D object detection has been extensively evaluated on benchmarks such as KITTI (Geiger et al., 2012), nuScenes (Caesar et al., 2020), and Waymo Open Dataset (Sun et al., 2020b). While earlier works primarily reported results on KITTI, recent methods increasingly focus on nuScenes and Waymo due to their larger scale, richer annotations, and more diverse driving scenarios. Therefore, in this work, we present comparisons on nuScenes and Waymo to align with contemporary literature.

Since detection performance varies significantly depending on the input modality, we categorize methods into camera-based, LiDAR-based, and multi-modal fusion approaches to enable a more meaningful comparison. As shown in Table 21, the reported results are collected from the original papers, using the test set of nuScenes and the validation set of Waymo (car category), which is the common evaluation protocol adopted by most prior works for fair comparison. Performance is evaluated using standard metrics, including mean Average Precision (mAP) and nuScenes Detection Score (NDS) for nuScenes, and Level 1/Level 2 mAP and mAPH for Waymo, where mAPH additionally accounts for heading accuracy.

Table 20: Segmentation performance comparison across different supervision settings. Methods are grouped by supervision type. Zero-shot, few-shot, weakly supervised, and unsupervised domain adaptation results are **not directly comparable** to fully supervised results. Metric variants are reported per model.

Category	Model	Setting	Metric	Dataset			
				S3DIS	ScanNet	ScanNet200	nuScenes
Fully Supervised							
Semantic	BFANet (Zhao et al., 2025d)	Fully Sup.	mIoU	–	78.0	37.3	–
Semantic	CDSegNet (Qu et al., 2025b)	Fully Sup.	mIoU	–	77.9	36.3	81.2
Semantic	FastAdapter (Sun & Yan, 2025)	Fully Sup.	mIoU	78.2	78.0	–	–
Semantic	GHEA (Zhang et al., 2025c)	Fully Sup.	mIoU	79.5	76.6	–	–
Semantic	PointNet-KAN-MLP (Kashefi, 2025)	Fully Sup.	mIoU	51.5	–	–	–
Semantic	VDG-Uni3DSeg (Han et al., 2025b)	Fully Sup.	mIoU	73.2	71.5	29.7	–
Zero-Shot / Open-Vocabulary							
Semantic	SMAP (Wei et al., 2025b)	Zero-shot	mIoU	–	40.5	14.6	36.2
Semantic	Mosaic3D (Lee et al., 2025b)	Zero-shot	f-mIoU	–	84.4	28.3	–
Semantic	SAS (Li et al., 2025o)	Zero-shot	mIoU	–	61.9	–	47.5
Semantic	AIDE (Wang & Czarnecki, 2025)	Open-Vocab	hIoU	42.2	72.8	–	62.2
Instance	Any3DIS (Nguyen et al., 2025b)	Open-Vocab	AP	–	–	25.8	–
Instance	Any3DIS (Nguyen et al., 2025b)	Open-Ended	AP	–	–	19.1	–
Instance	SAM2Object (Zhao et al., 2025b)	Zero-shot	mAP	–	34.0	13.3	–
Few-Shot							
Semantic	GFS-VL (An et al., 2025b)	5-shot	hIoU	–	61.9	43.1	–
Semantic	LARM (Li et al., 2025n)	5-shot	hIoU	63.1	–	–	–
Semantic	TaylorSeg-PN (Wang et al., 2025b)	5-shot (2-way)	mIoU	71.3	70.0	–	–
Semantic	DyPolySeg (Wang et al., 2025a)	5-shot (2-way)	mIoU	75.6	72.4	–	–
Semantic	VIP-Seg (Wang et al., 2026a)	5-shot (2-way)	mIoU	77.0	72.6	–	–
Semantic	EDS-Net (Wang et al., 2026b)	5-shot (2-way)	mIoU	76.7	73.4	–	–
Other Limited Supervision							
Instance	Sketchy-3DIS (Deng et al., 2025b)	Weakly Sup.	AP ₅₀	64.6	65.8	–	–
Semantic	CACE (Chen et al., 2025b)	Unsup. DA	mIoU	57.1	58.2	–	–
Semantic	D3CTTA (Zhao et al., 2025a)	Test-time Adapt.	mIoU	–	–	–	24.1
Semantic	LogoSP (Zhang et al., 2025h)	Unsup. DA	mIoU	46.3	35.8	–	20.1
Cross-Modal / Foundation Models							
Semantic	SAM and Alpha-CLIP (Jung et al., 2025b)	Zero-shot	mAP	22.6	–	25.8	–
Semantic	VDG-Uni3DSeg (Han et al., 2025b)	Cross-modal	mAP	63.7	59.3	29.5	–

From the results, camera-based methods generally achieve lower performance compared to LiDAR-based approaches, primarily due to the lack of accurate depth information. LiDAR-based methods demonstrate strong and consistent performance, benefiting from precise 3D geometric cues, with voxel- and point-based architectures achieving competitive results across both datasets. Fusion-based methods further improve performance by leveraging complementary information from both camera and LiDAR modalities, often achieving state-of-the-art results. However, these gains come at the cost of increased model complexity and computational requirements. Overall, the comparison highlights the trade-off between accuracy and sensor dependency, with multi-modal fusion providing the best performance when multiple sensors are available.

8 Overall Conclusion, Open Challenges, Future Directions, and Broader Impact

This section summarizes the main insights of the survey, discusses the remaining challenges in deep point cloud learning, outlines future research directions, and examines the broader impacts, ethical considerations, and limitations associated with the development and deployment of 3D point cloud models. The overall conclusion first highlights the scope, organization, and main findings of the survey. The challenges section then identifies common limitations across tasks, representations, architectures, and deployment settings, followed by future directions toward more robust, efficient, generalizable, and trustworthy 3D systems. Finally, the broader-impact discussion addresses the consequences of perception failures in safety-critical applications, the dual-use nature of adversarial techniques, privacy and surveillance concerns associated

Table 21: Quantitative comparison of representative 3D object detection methods on the nuScenes (test set) and Waymo (validation set) benchmarks. Methods are grouped by modality (camera, LiDAR, and fusion).

Method	Category	nuScenes		Waymo (car)			
		mAP	NDS	L1 mAP	L1 mAPH	L2 mAP	L2 mAPH
CenterNet (Duan et al., 2019a)	Camera-Monocular	33.80	40.00	–	–	–	–
DETR3D (Wang et al., 2022e)	Camera-Multi-View	41.20	47.90	–	–	–	–
PolarFormer (Jiang et al., 2023b)	Camera-Multi-Camera	49.30	57.20	–	–	–	–
BEVDepth (Li et al., 2023j)	Camera-Depth	52.00	60.90	–	–	–	–
CAPE (Xiong et al., 2023)	Camera-Multi-View	52.50	61.00	–	–	–	–
PV-RCNN (Shi et al., 2020a)	LiDAR-Point-Voxel	–	–	77.51	76.89	68.98	68.41
PV-RCNN++ (Shi et al., 2023)	LiDAR-Point-Voxel	–	–	80.17	79.70	72.17	71.70
PVTransformer (Leng et al., 2024)	LiDAR-Point-Voxel	–	–	82.20	81.70	74.30	73.90
RSN (Sun et al., 2021b)	LiDAR-Range Image	–	–	78.40	78.10	69.50	69.10
PillarNet (Shi et al., 2022a)	LiDAR-Pillars	66.00	71.40	79.09	78.59	70.92	70.46
BiProDet (Zhang et al., 2023f)	LiDAR-BEV	–	–	78.36	77.91	69.45	69.04
HEDNet (Zhang et al., 2023c)	LiDAR-BEV	67.70	72.00	81.10	80.60	73.20	72.70
MGTANet (Koh et al., 2023)	LiDAR-BEV	67.50	72.70	–	–	–	–
BEVDilation (Zhang et al., 2026a)	LiDAR-BEV	75.40	73.10	–	–	–	–
Li3DeTr (Erabati & Araujo, 2023)	LiDAR-Point	61.30	67.60	–	–	–	–
SEED (Liu et al., 2024f)	LiDAR-Point	–	–	79.80	79.30	71.90	71.50
Lion (Liu et al., 2024e)	LiDAR-Point	69.80	79.30	80.30	79.90	72.00	71.60
GeoFormer (Jin et al., 2025b)	LiDAR-Point	69.80	73.70	80.48	79.98	72.21	71.75
CenterPoint (Yin et al., 2021)	LiDAR-Voxel	–	–	76.70	76.20	68.80	68.30
SST (Fan et al., 2022b)	LiDAR-Voxel	–	–	77.04	76.56	68.50	68.08
UVTR (Li et al., 2022c)	LiDAR-Voxel	63.90	69.70	–	–	–	–
FocalConvs (Chen et al., 2022e)	LiDAR-Voxel	70.10	73.60	–	–	–	–
MSSVT++ (Li et al., 2023b)	LiDAR-Voxel	–	–	79.96	79.43	71.30	70.86
LargeKernel3D (Chen et al., 2023d)	LiDAR-Voxel	65.40	70.60	78.07	77.61	69.81	69.38
VoxelNeXt (Chen et al., 2023e)	LiDAR-Voxel	66.20	71.40	78.20	77.70	69.90	69.40
OCTR (Zhou et al., 2023a)	LiDAR-Voxel	–	–	78.12	77.63	69.79	69.34
DSVT (Wang et al., 2023a)	LiDAR-Voxel	68.40	72.70	82.10	81.60	74.50	74.10
VoxelMamba (Zhang et al., 2024c)	LiDAR-Voxel	69.00	73.00	80.80	80.30	72.60	72.20
FSHNet (Liu et al., 2025e)	LiDAR-Voxel	71.70	68.10	82.20	81.70	74.50	74.00
DSTR (Cai et al., 2025)	LiDAR-Voxel	72.30	74.40	–	–	–	–
WinMamba (Zheng et al., 2026b)	LiDAR-Voxel	–	–	78.00	77.50	69.60	69.10
MapFusion (Fang et al., 2021)	Fusion (multi-view + point cloud)	60.61	67.97	–	–	–	–
PointAugmenting (Wang et al., 2021a)	Fusion (monocular + point cloud)	66.80	71.00	67.41	–	62.70	–
EPNet++ (Liu et al., 2022)	Fusion (monocular + point cloud)	–	–	76.57	76.10	68.29	67.86
MSMDFusion (Jiao et al., 2023)	Fusion (multi-view + point cloud)	71.50	74.00	–	–	–	–
LoGoNet (Li et al., 2023i)	Fusion (multi-view + point cloud)	–	–	83.21	82.72	75.84	75.38
SAFDNet (Zhang et al., 2024b)	Fusion (multi-view + point cloud)	68.30	72.30	80.60	80.10	72.70	72.30
IS-Fusion (Yin et al., 2024)	Fusion (multi-view + point cloud)	73.00	75.20	–	–	–	–
MambaFusion (Wang et al., 2025f)	Fusion (multi-view + point cloud)	72.70	75.90	–	–	–	–
UniMamba (Jin et al., 2025a)	Fusion (multi-view + point cloud)	70.20	74.00	80.60	80.06	72.28	71.77

with 3D data, related ethical responsibilities, and considerations that affect the scope and interpretation of this survey.

8.1 Overall Conclusion

This survey reviewed deep point cloud models across a broad range of 3D tasks, with particular emphasis on how representation choices and architectural paradigms shape model design and performance. The literature was organized into coherent task families, including core tasks, geometric modeling, alignment and pose estimation, foundation models and scene understanding, and robustness. The survey also discussed benchmark datasets, evaluation metrics, and performance analysis with emphasis on comparable task-specific settings.

8.2 Overall Challenges

Despite this progress, several general challenges remain unresolved. A major limitation is robustness under real-world sensing conditions. Many models achieve strong performance on clean or synthetic benchmarks but

degrade under noise, occlusion, sparsity, density variation, clutter, missing regions, and motion distortion. This issue affects nearly all 3D tasks. In addition, generalization across domains remains a significant challenge. Point cloud data can differ substantially across sensors, environments, viewpoints, sampling densities, and acquisition settings, often causing performance degradation when models are deployed outside their training distribution. Differences between synthetic and real-world data, as well as between indoor RGB-D scans, outdoor LiDAR data, and industrial sensing systems, continue to pose challenges for reliable deployment.

Another major challenge is the trade-off between representation power and efficiency. Transformer-based, graph-based, diffusion-based, and multimodal foundation models often provide stronger reasoning or generation capabilities, but they also introduce high computational cost, memory consumption, and latency. This is problematic where real-time processing is required.

A further limitation is the lack of realistic and standardized evaluation protocols. Many benchmarks focus on specific tasks, clean datasets, or simplified metrics that do not fully reflect real-world performance. Common metrics may overlook factors such as robustness, uncertainty, spatial reasoning, and physical consistency. In addition, fragmented benchmarks make fair comparison across tasks, datasets, sensors, and methods difficult.

Security and trustworthiness are also becoming increasingly important. As 3D models are deployed in safety-critical applications, they become vulnerable to adversarial perturbations, sensor spoofing, poisoning, backdoor attacks, model extraction, and cross-modal attacks. Existing robustness and security studies are often limited to specific datasets, architectures, or attack assumptions, while physically realizable and system-level threats remain underexplored. Reliable deployment therefore requires robustness, interpretability, uncertainty estimation, provenance verification, and secure training and inference pipelines.

8.3 Overall Future Directions

Future research should move toward robust, efficient, generalizable, and trustworthy 3D intelligence. A key direction is developing models that remain reliable under noise, sparsity, occlusion, and domain shifts through realistic training conditions, adaptation techniques, and multimodal fusion of geometry, vision, and language. Another important direction is the design of unified and lightweight architectures. Efficient transformers, sparse computation, distillation, and parameter-efficient adaptation can help bridge the gap between powerful 3D foundation models and real-time deployment while supporting multiple 3D tasks within a single framework.

The field also needs better evaluation frameworks. Future benchmarks should include realistic noise, occlusions, dynamic scenes, domain shifts, adversarial settings, and open-world scenarios. Evaluation should go beyond task-specific metrics to measure robustness, uncertainty, efficiency, and real-world performance. Standardized benchmarks and protocols would enable fairer comparisons across methods.

Finally, future research should move toward open-world and embodied 3D understanding. Next-generation models must reason about unseen objects, changing environments, physical interactions, and human instructions while remaining robust and efficient. Advances in foundation models, multimodal learning, self-supervised training, and secure deployment will be key to building reliable and scalable 3D perception systems.

8.4 Broader Impact, Ethical Considerations, and Limitations

Deep learning on 3D point clouds has broad societal relevance because it supports applications ranging from autonomous driving, robotics, and intelligent infrastructure to industrial inspection, healthcare, digital reconstruction, and large-scale scene understanding. Advances in classification, segmentation, detection, tracking, reconstruction, registration, generation, and multimodal reasoning can improve automation, environmental understanding, and decision-making in complex physical environments. However, the impact of these systems cannot be assessed solely through benchmark accuracy. Their deployment also depends on how reliably they operate under realistic conditions, how the collected 3D data are handled, and whether knowledge of system vulnerabilities can be misused. These considerations are particularly important for point-cloud systems because they directly model the geometry of people, objects, and environments and are increasingly

integrated into safety-critical and privacy-sensitive applications. Accordingly, three broader-impact concerns are particularly relevant to the methods reviewed in this survey:

- **Safety-Critical Consequences of Perception Failures.** Point-cloud perception failures can have direct physical consequences when models are deployed in autonomous vehicles, industrial systems, robotics, or intelligent infrastructure. Errors may arise from sparse or incomplete observations, sensor noise, occlusion, calibration errors, domain shifts, unusual environmental conditions, or deliberate adversarial manipulation. Such failures may cause missed or incorrectly localized objects, unsafe navigation, erroneous robotic actions, unreliable spatial measurements, or delayed responses to hazardous situations. Strong performance on clean benchmarks therefore does not by itself establish deployment readiness. Safety-critical systems should additionally be evaluated under realistic corruptions, compound degradations, distribution shifts, and adversarial conditions, while incorporating uncertainty estimation, failure detection, redundant sensing, fail-safe mechanisms, and appropriate human oversight.
- **Dual-Use Nature of Adversarial and Attack Techniques.** The security and adversarial techniques discussed in this survey have an inherently dual-use character. Analyzing sensor manipulation, evasion attacks, data poisoning, backdoors, privacy attacks, and model-extraction techniques are necessary for exposing weaknesses, constructing realistic threat models, and developing stronger defenses and evaluation protocols. At the same time, detailed knowledge of these vulnerabilities could be misused to evade detection, manipulate 3D observations, compromise deployed perception systems, or interfere with safety-critical decisions. Security research on point-cloud learning should therefore maintain a clear defensive motivation and adopt responsible practices concerning disclosure, experimental containment, release of implementation details, and evaluation on real-world systems. The objective of cataloging these attacks should be to support robustness assessment, red-teaming, standardized security benchmarks, and the design of resilient 3D perception pipelines rather than to facilitate operational misuse.
- **Privacy and Surveillance Implications of 3D Data.** Point clouds collected by LiDAR sensors, RGB-D cameras, body scanners, and related depth-acquisition systems may contain sensitive geometric information about individuals and their surroundings. Human-body shape, facial geometry, movement patterns, occupancy information, and the layouts of private indoor spaces may enable identification, behavioral monitoring, biometric inference, or unauthorized surveillance. Moreover, reconstruction, completion, and generative methods may infer missing geometric details that were not explicitly visible in the original observation, potentially increasing the amount of sensitive information recoverable from partial scans. Responsible development therefore requires informed consent where appropriate, purpose-limited data collection, data minimization, removal or obfuscation of identifiable geometry, secure storage and transmission, access control, retention limits, and compliance with applicable privacy regulations. Dataset creators and model developers should also document how human-centered and privately acquired 3D data are collected, processed, shared, and reused.

These broader-impact concerns imply shared ethical responsibilities for dataset creators, model developers, system integrators, and end users. Point-cloud systems should be developed and evaluated with attention to informed consent, privacy protection, transparency, accountability, equitable performance across sensors and deployment environments, and the prevention of foreseeable harm. Developers should clearly document data provenance, evaluation conditions, known failure modes, and intended uses, while avoiding claims of reliability that are not supported by realistic testing. In high-risk applications, automated predictions should remain subject to appropriate human oversight, and responsibility for failures should not be obscured by the complexity of the underlying models or processing pipelines.

Despite our efforts to provide a comprehensive review through systematic literature search and repeated verification, several considerations should be acknowledged when interpreting the scope and findings of this survey. First, given the breadth of the field, some relevant studies may still have been omitted, particularly when their terminology or scope differs from the search and selection criteria. Second, the rapid development

of 3D machine learning means that new studies may appear after the literature-search cutoff. Moreover, the boundaries among tasks, representations, architectural paradigms, modalities, and learning settings are not always strict, and some methods could reasonably be assigned to multiple categories. Finally, direct comparison across the reviewed studies remains difficult because they often use different datasets, evaluation metrics, supervision settings, preprocessing pipelines, experimental protocols, model architectures, and threat models. Consequently, the proposed taxonomies should be regarded as structured organizational guidance, and reported results should primarily be interpreted under comparable evaluation conditions rather than as a unified ranking of methods.

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